



Resilience Social and Cultural Needs

DR MEGAN BOSTON

ENGCV370

Learning Objectives

- ▶ Understand how resilience strategies impact different demographics of people
- ▶ Recognise the importance of incorporating social and cultural requirements into resilience frameworks

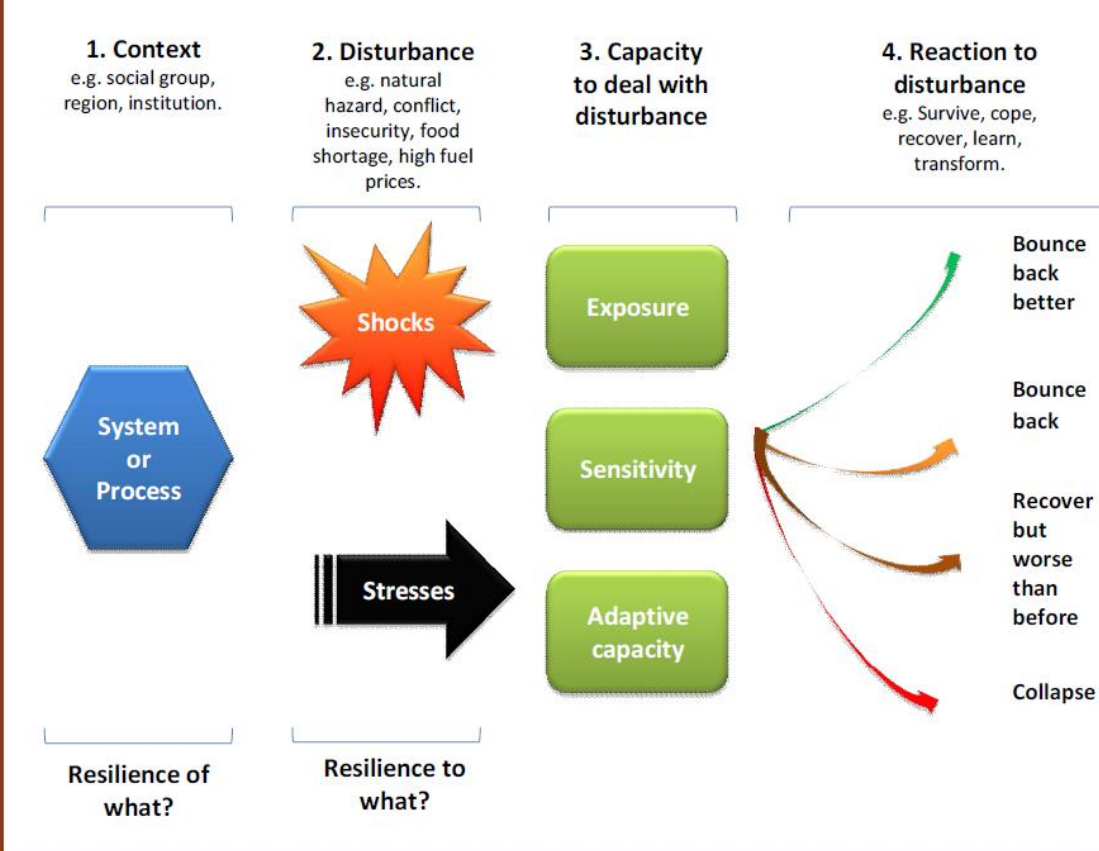
- ▶ I feel that I am generally aware of the impacts of engineering projects on society and the economy?
- ▶ Most large engineering projects have little to no impact on society of communities
- ▶ Most large engineering projects have little to no impact on the economy beyond those employed in design and construction

For the two scenarios presented, briefly respond to the following questions

- A. A large dam is to be constructed in rural area of a developing nation to create a large lake to provide water for irrigation and for generating electricity
 - 1. How do you think such a dam would help the developing nation's economy?
 - 2. How might such a project hurt the economy?
 - 3. How do you think building this large project would benefit the social landscape of the developing nation?
 - 4. How do you think building this large project would (could) hurt the social landscape of the developing nation?
- B. A highway bypass project is proposed around Hamilton
 - 1. How do you think a highway bypass project would help the local economy?
 - 2. How do you think a highway bypass project would hurt the local economy?
 - 3. How do you think a highway bypass project would benefit the social landscape in the community?
 - 4. How do you think a highway bypass project would (could) hurt the social landscape in the community?

Resilience Frameworks – Challenges

The four elements of a resilience framework



Challenges

- ▶ Resilient systems are better able to cope, recover and adapt
- ▶ Fails to account for power relations and complexity of social dynamics
- ▶ Marginalised communities bear the brunt of impacts from natural disasters
- ▶ Lack resources and capacity for successful adaption and mitigation

Key elements for equitable resilience

- ▶ **Recognizing subjectivities**
- ▶ **Inclusion and representation**
- ▶ **Scale**
- ▶ **Transformation**

Recognizing subjectivities

- ▶ How social contexts, power relations and categorization of people determine social and economic entitlements

Inclusion and representation

- ▶ Avoiding processes that disenfranchise some groups

Scale

- ▶ Attending to multiple levels of governance as well as geographical and temporal aspects of scale

Transformation

- ▶ Promoting system(s) **transformation** when existing arrangements degrade well-being or increase risks for certain sections of society

Solutions:

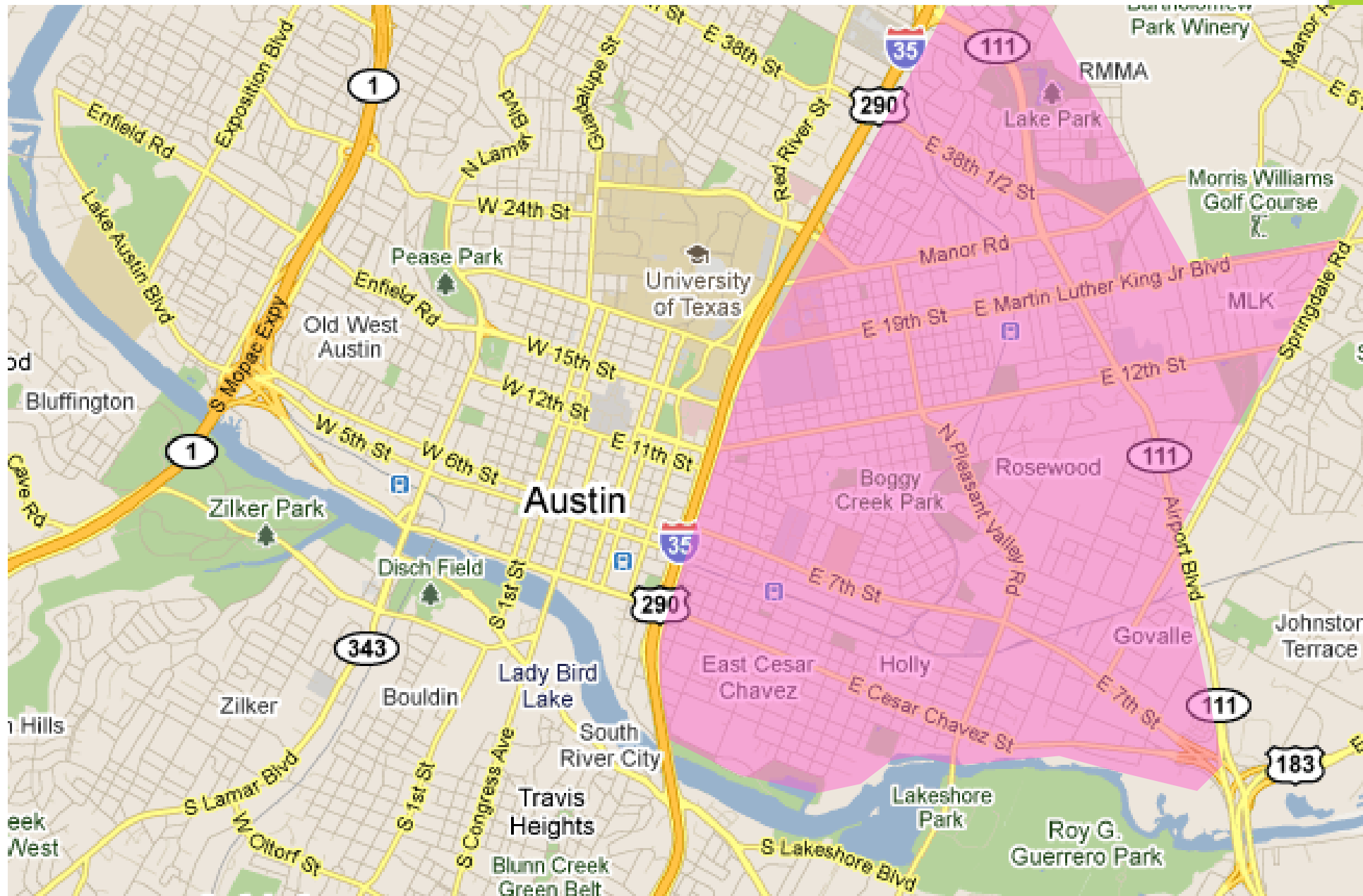
- ▶ Reframing – baseline for disaster readiness
 - ▶ Reinvent – communications
 - ▶ Reinforce – Repetition
 - ▶ Revamp – outreach
-
- ▶ Understand the issues and challenges.



Merowe Dam
in Sudan (Nile River)



Austin, TX

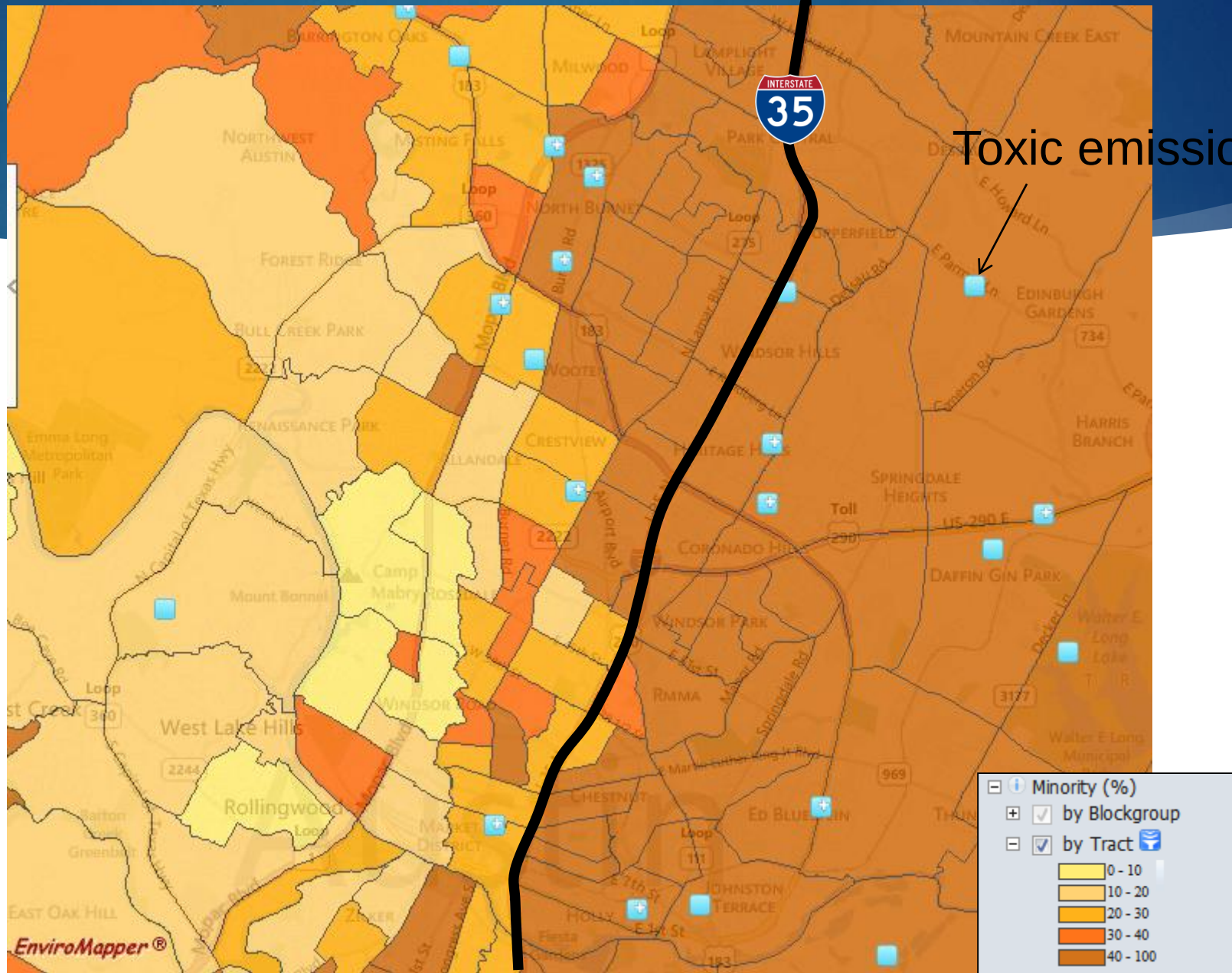






Austin, TX (I-35 segregation)

17









The Study



❖ Cultural Practices

The Study

*Kia Niwha Te Ngakau Ki Nga mahi
Tongikoro, Whakatauki, Purakau*

*Naku te rourou nau te rourou
ka ora ai te iwi*

*“Hutia te rito o te harakeke Kei whea to kōmako e kō? Ki mai ki
ahau He aha te mea nui o te Ao? Maku e kī atu, he tāngata, he
tāngata, he tāngata...”*

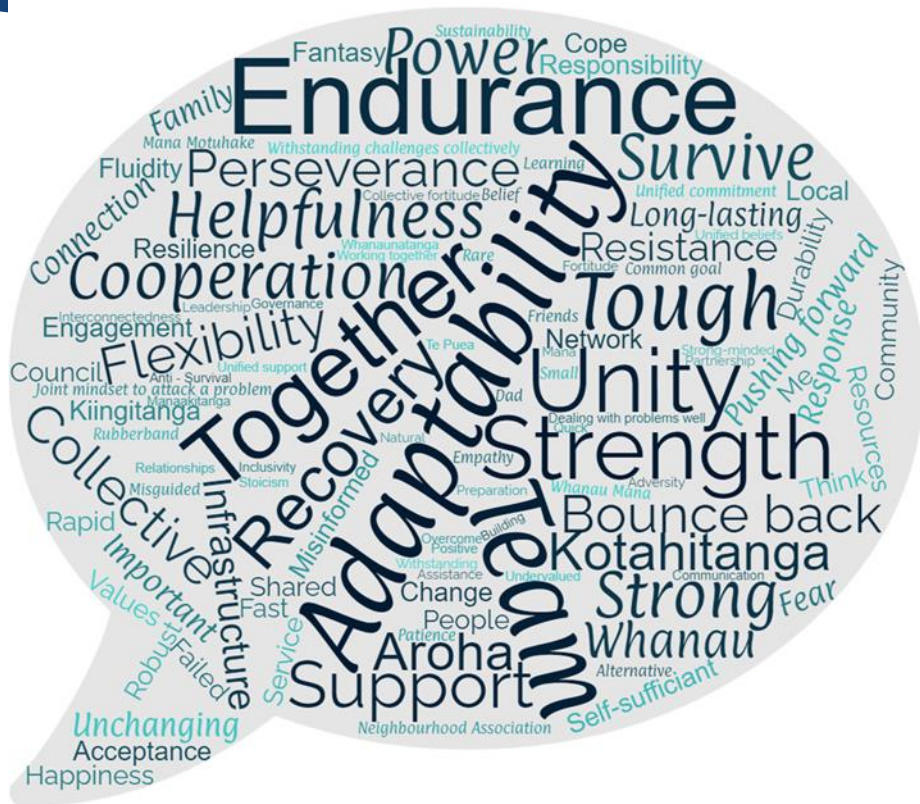
❖ Case Study – 2004 Manawatu Whanganui Flood

The Study



❖ Survey & Interviews

The Study

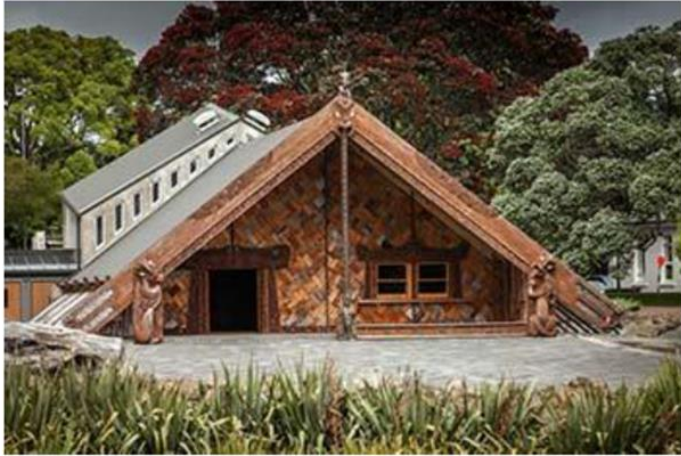


100



❖ Survey & Interviews

The Study



Findings

Mātauranga

“Learn and strengthen resources to become resilient. Reflect on appropriate kōrero tuku iho for tīpuna”

Mātauranga
(knowledge)



“Because the wood used to build pā are whole trees, whereas the modern style of wood used is already cut up pieces of a tree, not the whole thing. ‘Together we stand, divided we fall.’”

Findings

Kotahitanga

"Community is central to recovery
post-disaster"

"there is a sense
of community"

Kotahitanga
(unity)



"the wharehenui is not just a mere
bunch of wood and nails
hammered together, it has a
whole spiritual whare inside it. So
double the strength"

Findings

Kaitiakitanga

Iwi work to ensure the well-being of Māori and all others in the region during a disaster – Carter 2018

“It’s got to be the people. Ensuring the safety of the people – look after the whakapapa – check-in with the whakapapa.”

“I feel like our ancestors were more involved and active in each other’s lives. They were less occupied with the world but also they were more of a community so if something happened then they would help each other immediately. “

“Built to resist attacks, famine, disaster in mind”

Kaitiakitanga
(guardianship)



Findings

Whakawhanaungatanga

Whakawhanaunga
(relationships)



“I feel like our ancestors were more involved and actively engaged in each other’s lives. They were less occupied with the world, but also they were more of a community, so if something happened, then they would help each other immediately.”

“Whanau/hapu would come together and work together to reinstate the whare”

Findings

Manākitanga

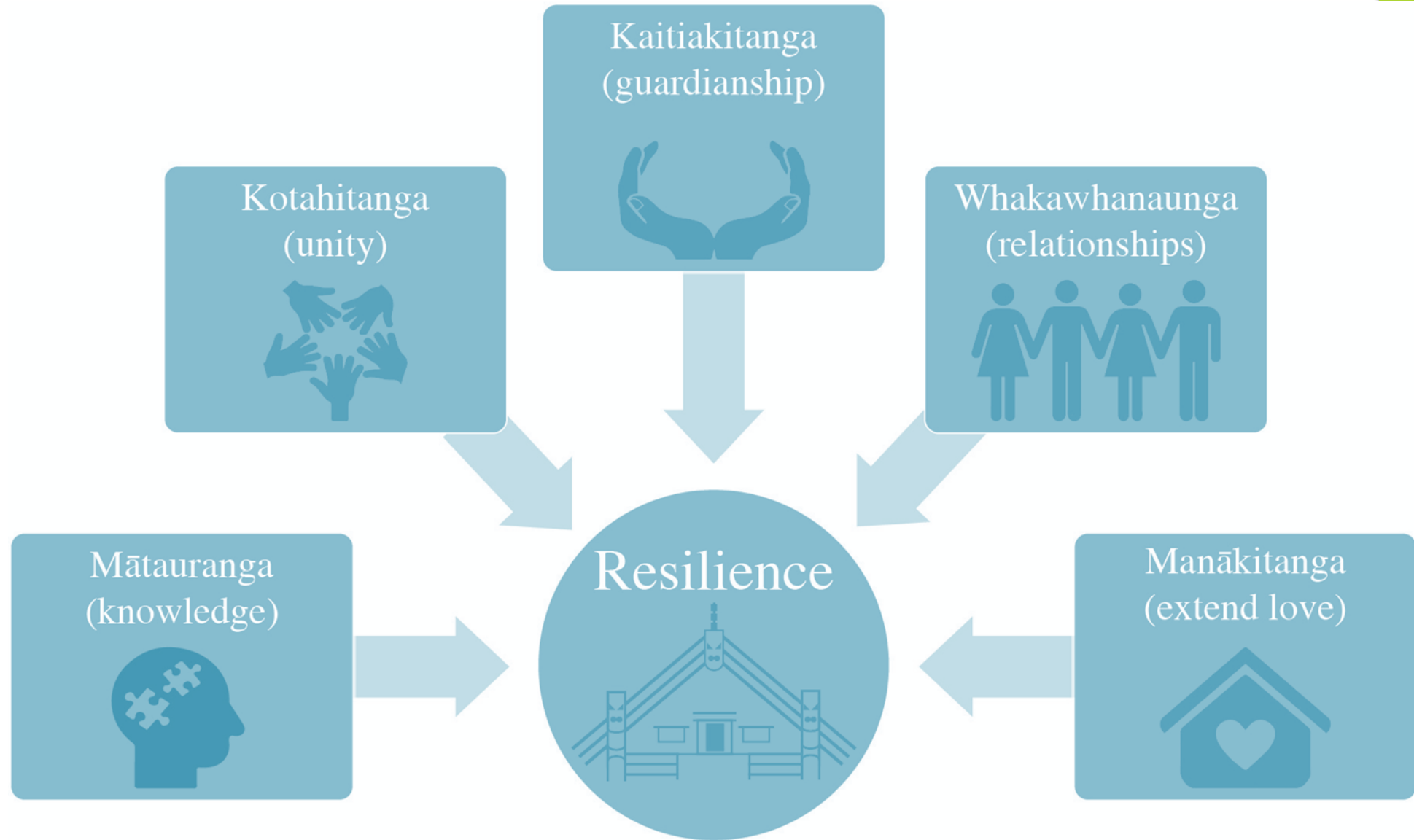
“they would
know one
another and look
after each
other”

Manākitanga
(extend love)



“whanau hapu help each other out,
support each other help raise and
care for the young and old”

“Whare were bare minimum. Communal living means people
were well in touch with each other and their environment.
Connected through whakapapa more often than not -
something more to fight and care for.”



Takeaways



Nepal 2015 Earthquake
PC: Sonny Inbaraj.

Takeaways



Photo: Taneesha Mohan, SEI

Takeaways



Photo: DANIEL BEREHULAK/THE NEW YORK TIMES/REDUX

Social Vulnerabilities

Resilience Practice

ENGCV370

Dr Megan Boston



Learning Objectives

- Practice applying Resilience
 - Robustness
 - Redundancy
 - Resourcefulness
 - Rapidity

Topics Covered

- Resilience
 - Definition of Resilience
 - Resilience Triangle
 - Dimensions of resilience

Resilience

- After the 2016 Kaikoura Earthquake, severe ground slips and landslides destroyed major sections of State Highway 1 between Christchurch and Picton. Initially, communities along the coast were isolated while repairs were made to State Highway 1 and other roads. An abbreviated timeline of the repairs is given.
 - 14 November 2016: Kaikoura Earthquake, State Highway 1 between Picton and Christchurch is closed. The inland road between Kaikoura and Christchurch is closed.
 - 19 December 2016: The inland road reopens between Kaikoura and Christchurch
 - 21 December 2016: State Highway 1 between Kaikoura and Christchurch is reopened
 - 15 December 2017: State Highway 1 reopens from Kaikoura to Picton
- Reopening the different sections of road around Kaikoura increased the ability for people to travel easily between Christchurch and Picton and out of Kaikoura.



Resilience curves

- The Waiho Bridge in Franz Josef is an important link on SH6 providing access across the Waiho River connecting communities along the west coast of the South Island. The bridge was washed away and destroyed on 26 March 2019 during a massive rainstorm. Without the bridge, travel time between Franz Josef and Fox Glacier increased from 25 minutes to 14 hours with vehicles required to travel around Aoraki/Mt Cook. If we assume the following:
 - Prior to the bridge collapse the number of vehicles that travel between Franz Josef and Fox Glacier across the Waiho bridge on SH6 was 1250 each day,
 - During the bridge repair (26 March to 12 April 2019) the number of vehicles that travel across is 0 per day,
 - After repairs are complete the number of vehicles crossing the bridge returns to the original numbers of vehicles per day,
- What does the resilience curve look like that best describes the number of cars able to travel between Franz Josef and Fox Glacier on SH 6?





Milford Rd State Highway 94. Photo: NZTA

SOUTHLAND FLOODS 2020

1100 mm of rain over 60 hr period

~200 tourists in Milford Sound

~200 staff in Milford Sound

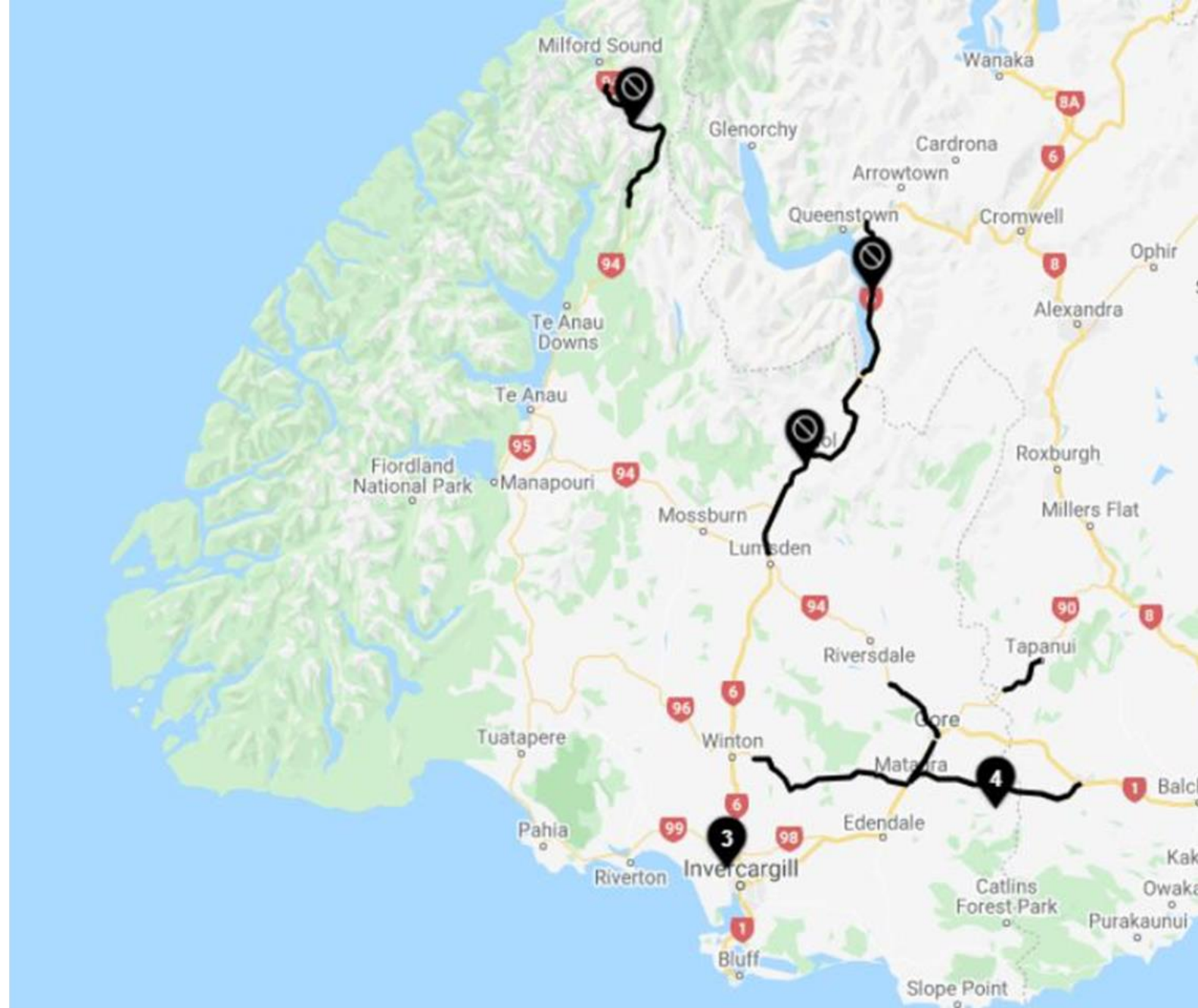
~70 people stranded along Milford road

100+ people trapped in Great Walk and backcountry Huts

6000 people evacuated from Gore, Wyndham, and Mataura

Milford Road closed for weeks, partial reopening to convoys only

SOUTHLAND FLOODS 2020



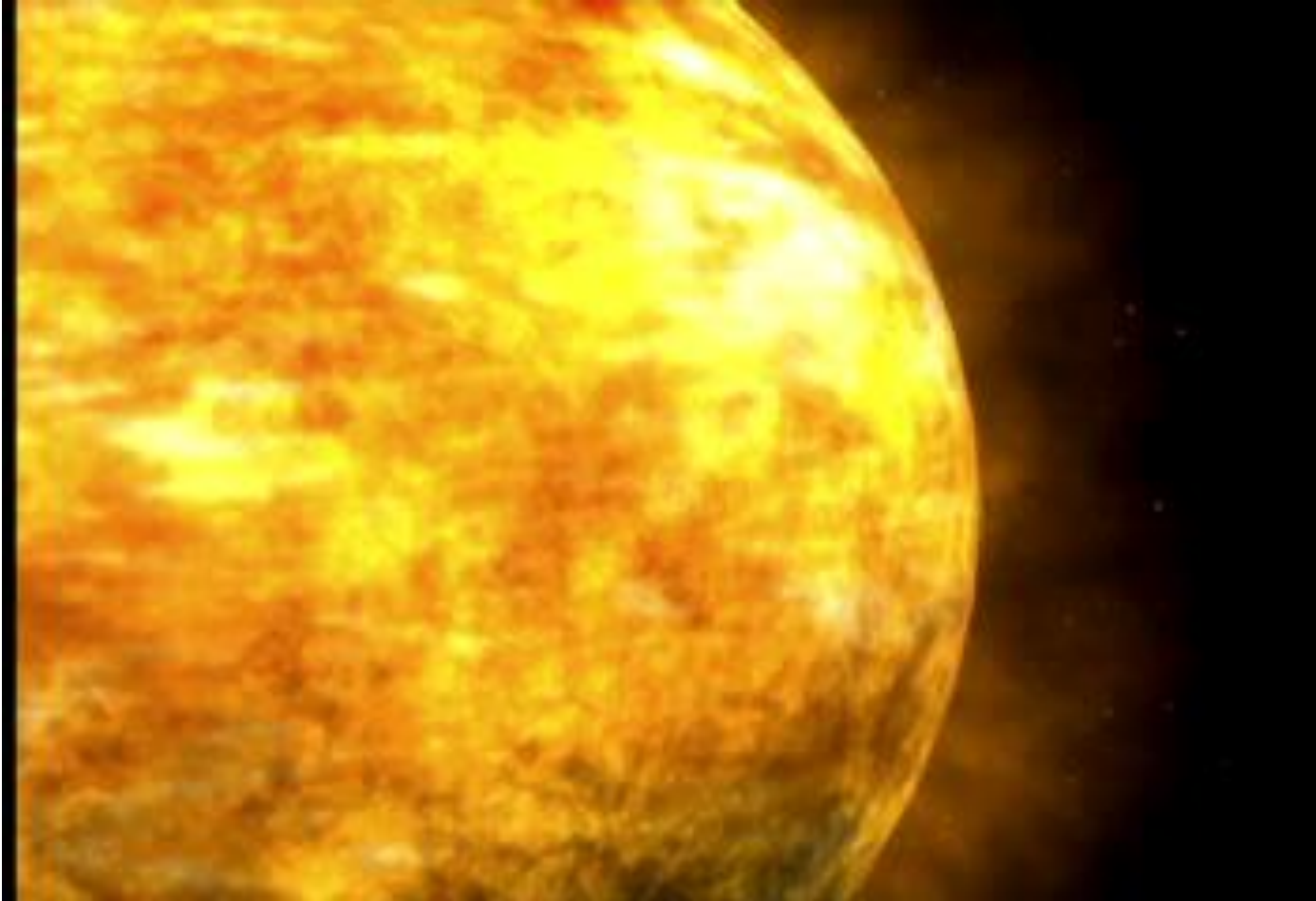
Road Closures due to Southland Flooding, Photo:NZTA

Carrington Solar Flare

August 28-September 2, 1859

- 
- Those who happened to be out late on Thursday night had an opportunity of witnessing another magnificent display of the auroral lights. The phenomenon was very similar to the display on Sunday night, though at times the light was, if possible, more brilliant, and the prismatic hues more varied and gorgeous. The light appeared to cover the whole firmament, apparently like a luminous cloud, through which the stars of the larger magnitude indistinctly shone. The light was greater than that of the moon at its full, but had an indescribable softness and delicacy that seemed to envelop everything upon which it rested. Between 12 and 1 o'clock, when the display was at its full brilliancy, the quiet streets of the city resting under this strange light, presented a beautiful as well as singular appearance.
 - [Baltimore American and Commercial Advertiser](#)
 - 3 September 1859

Carrington Solar Flare

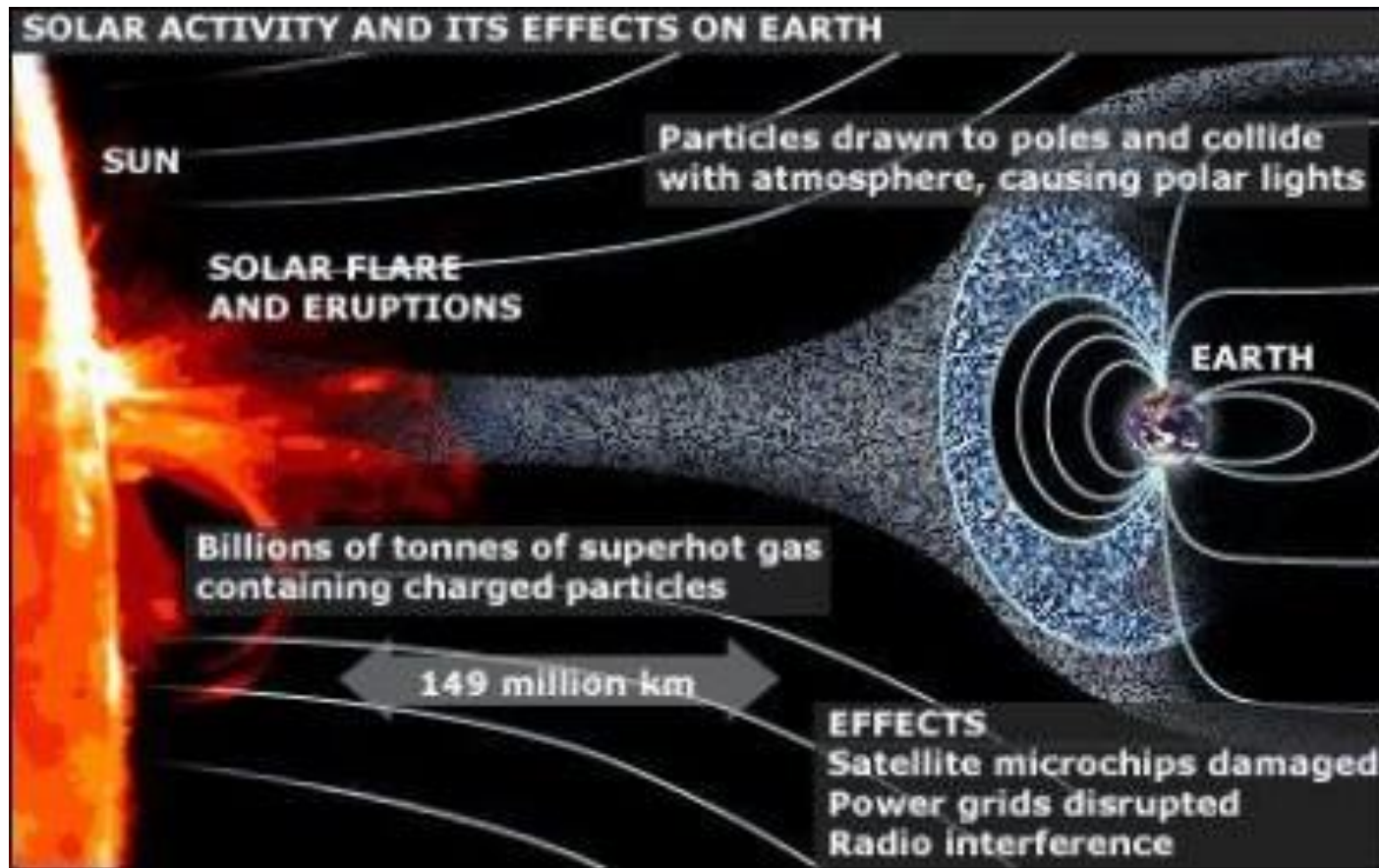


Coronal Mass Ejection

NASA/ Walt Feimer

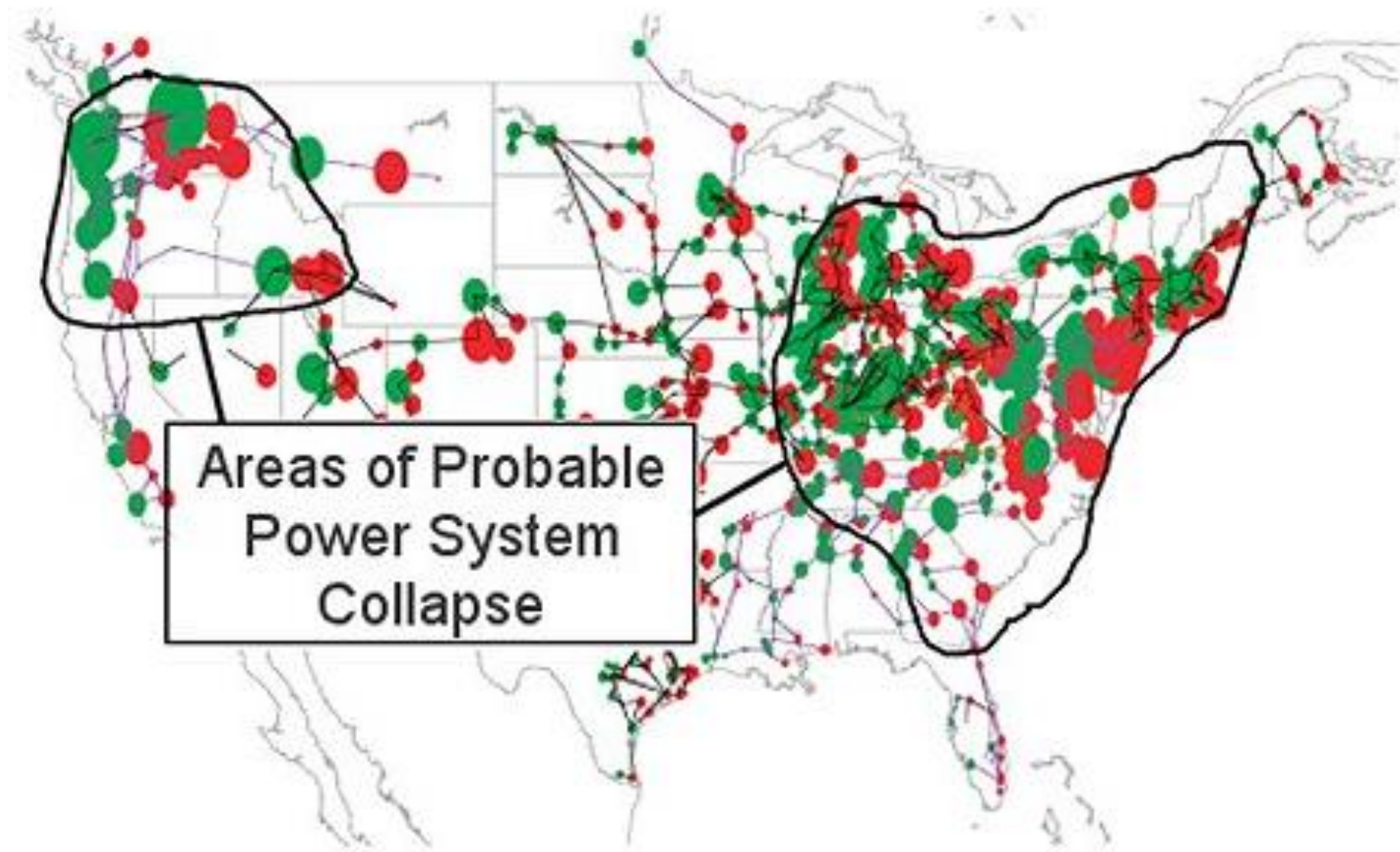
- Geometric solar storm
- Knocked out telegraphs

If it happened today...



- *"We are estimating that 20 to 40 million people might be without power from anywhere up to one, even two years. That has to do with the critical issue of replacement transformers. That number of people without power could result in an economic cost somewhere between \$.6 trillion to \$2.6 trillion."*

-Neil Smith, Exposure Management



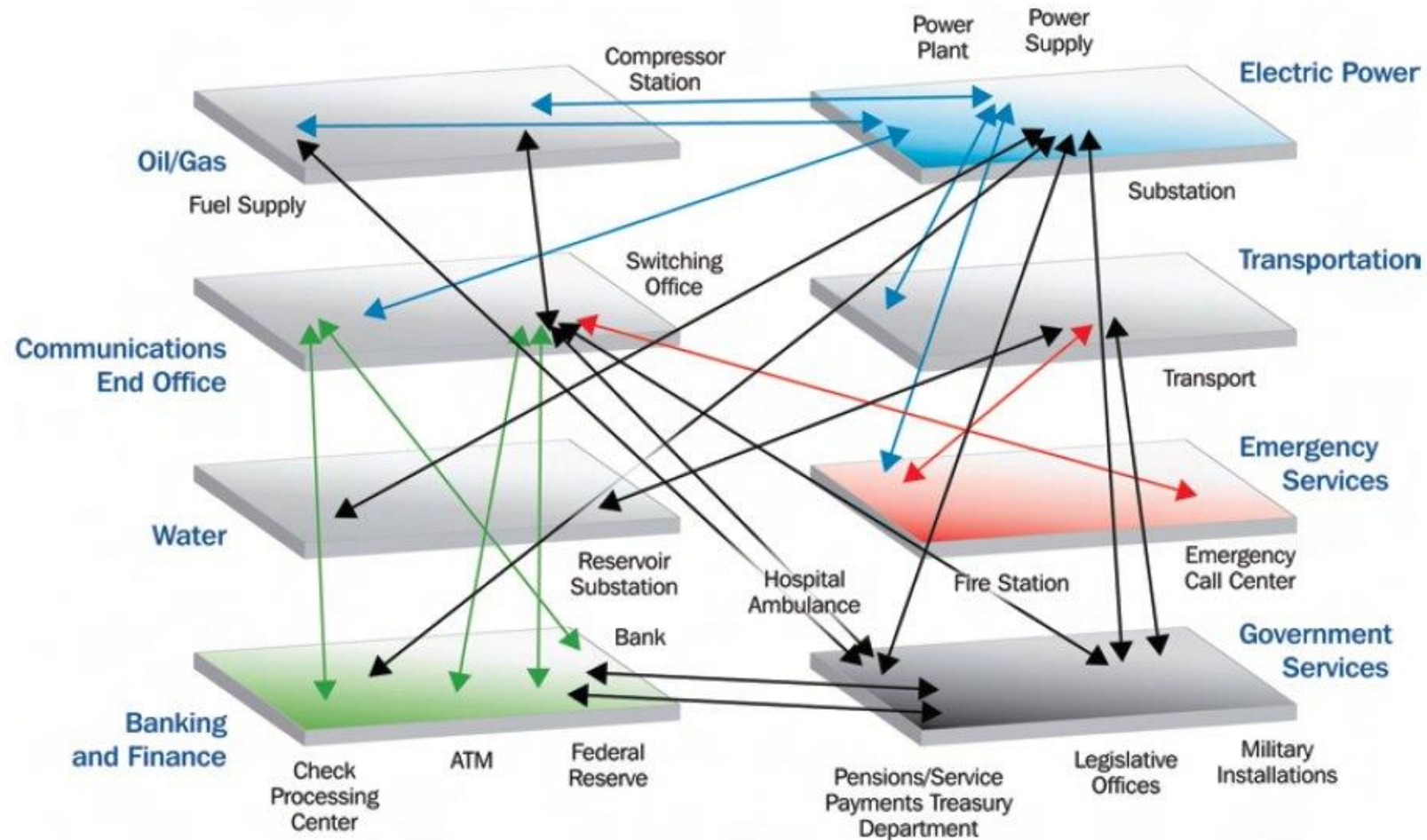


FIGURE 3.1 Connections and interdependencies across the economy. Schematic showing the interconnected infrastructures and their qualitative dependencies and interdependencies. SOURCE: Department of Homeland Security, National Infrastructure Protection Plan, available at http://www.dhs.gov/xprevprot/programs/editorial_0827.shtm.

Similar
Storms 1989
Quebec



Parker Solar Probe



Bhopal Disaster

2 December 1984

- 
- It felt like somebody had filled our bodies up with red chillies, our eyes had tears coming out, noses were watering, we had froth in our mouths. The coughing was so bad that people were writhing in pain. Some people just got up and ran in whatever they were wearing or even if they were wearing nothing at all. Somebody was running this way and somebody was running that way, some people were just running in their underclothes. People were only concerned as to how they would save their lives so they just ran.

- Champa Devi Shukla

A Survivor of Bhopal

WARNING: graphic photo on next slide

What Happened

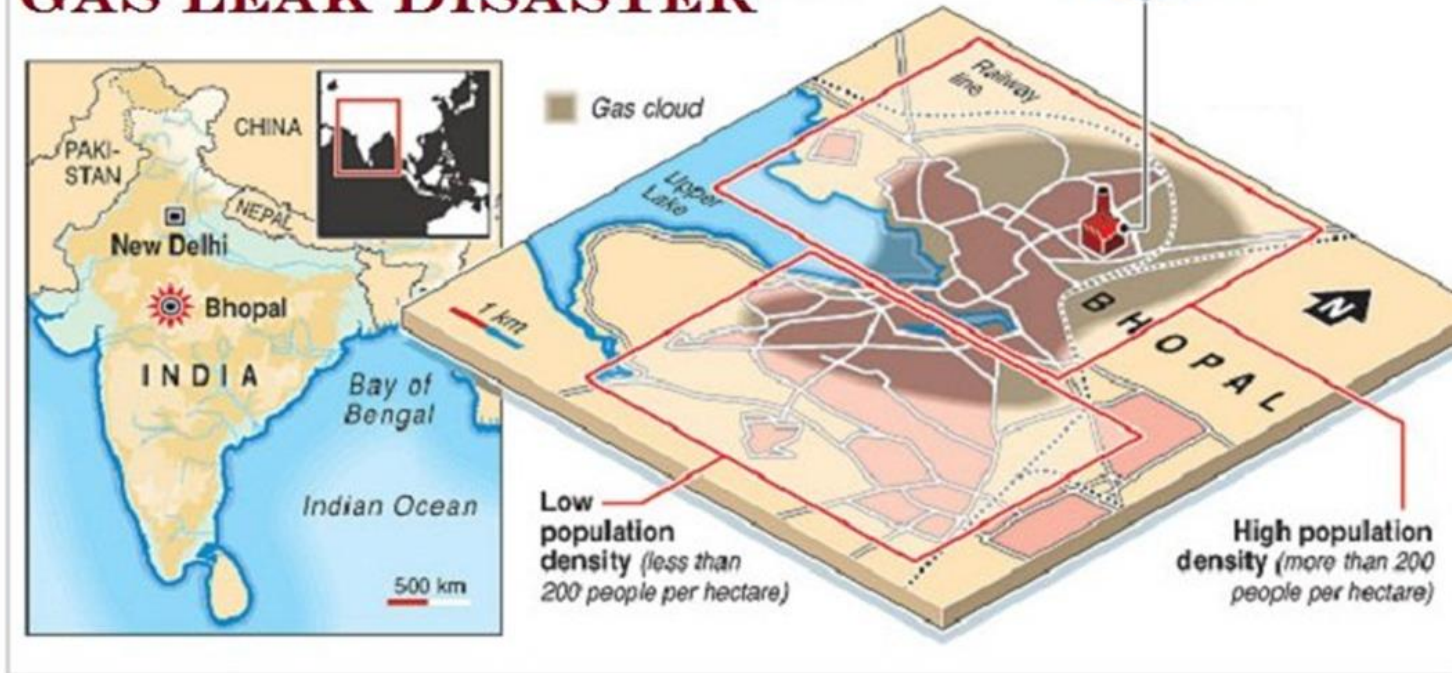
- Gas leak at the Union Carbide plant
- At the time of the leak, safety systems were not working, pipes and valves were in poor condition
- Indication that there was a leak at 11 pm
- Alarm system was triggered at 12:50 am
 - Two part alarm (plant and Public)
 - Public Alarm was turned off
 - Plant Employees evacuated upwind (out of the gas cloud)



Photo Credit: Raghu

BHOPAL

GAS LEAK DISASTER



- 42,000 kilograms of Methyl isocyanate and other gases
- Chemicals hugged the ground causing children and shorter people to be more effected

Gas Leak



Impacts on Society

- Immediate Death toll
 - 3783 official
 - 16,000+ claimed
- Over 500,000 non-fatal injuries
- Those exposed have 10x the cancer risk
- 43% of pregnancies did not survive
 - 14 of babies died within 30 days
 - Many that survived had birth defects

Long term impacts



- Ground water tests
 - Mercury levels 20, 000 and 6 millions times the expected levels
 - Chemicals known to cause cancer, brain damage, and birth defects were found at levels 50 times higher than the EPA safety limits
- Breast milk
 - Dangerous chemical transfer into the milk of mothers
- Continued Exposure to future generations

Implementing Resilience

What resilience strategies can be implemented to prevent this from happening again?





From left to right: David Brown, Rick Husband, Laurel Clark, Kalpana Chawla, Michael Anderson, William McCool, Ilan Ramon

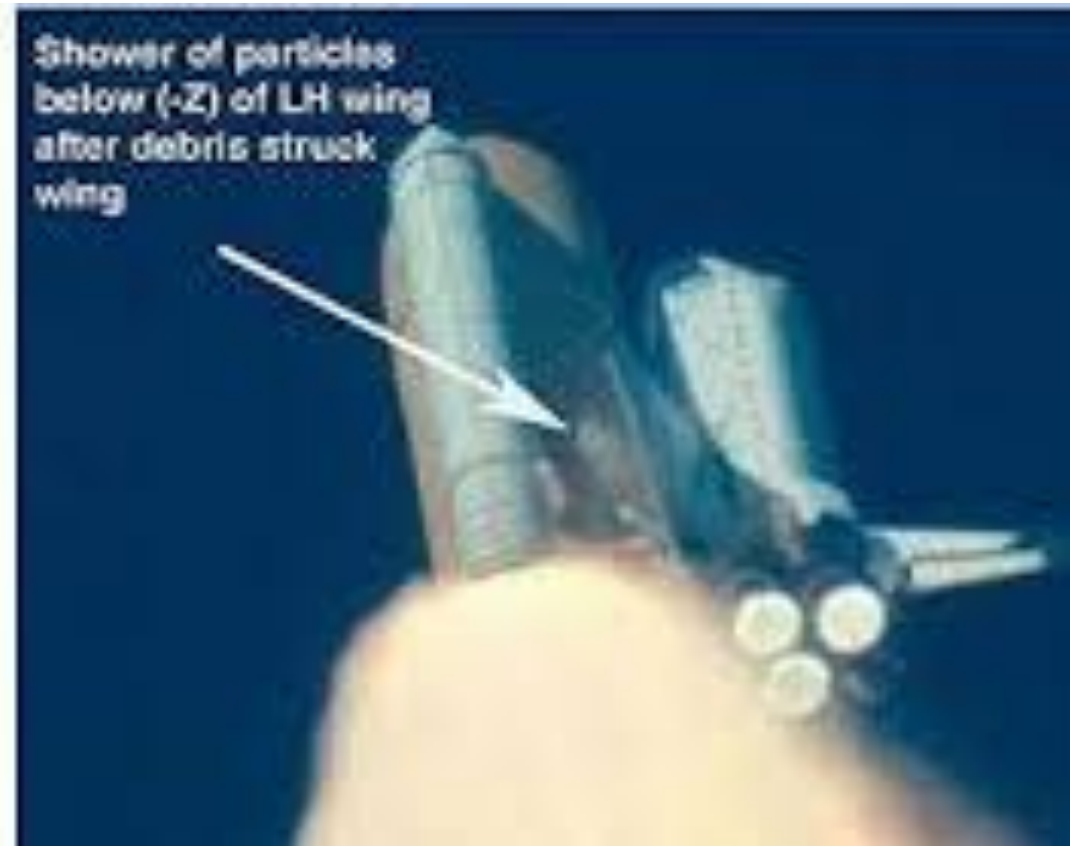
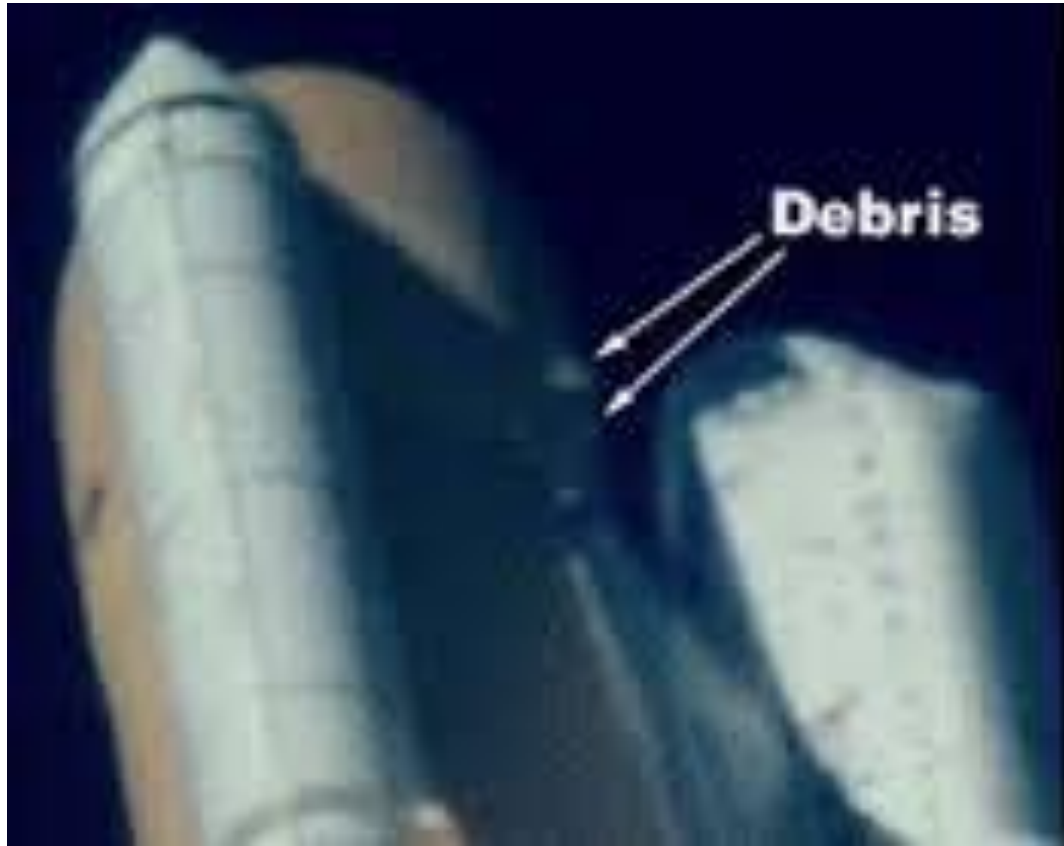
Photo: NASA

Space Shuttle Columbia

What happened

- During launch, a piece of foam broke off of the external fuel tank and hit the left wing of the shuttle
- Engineers suspected that the damage was serious but limited the investigation because they didn't think there was anything the astronauts could do about it
- During re-entry, the damage allowed hot gas to enter and destroy the internal wing structure
- The orbiter became unstable and started to break apart

Cause of the accident



Example scenario – State Highway 94

- Hazards for the region?
- Vulnerabilities for the road?
- Methods to mitigate the hazards?
- In terms of the 4Rs of resilience?
 - Robustness
 - Redundancy
 - Resourcefulness
 - Rapidity



Machinery like this have been kept busy this winter clearing snow from Alpine sections of the Milford Road

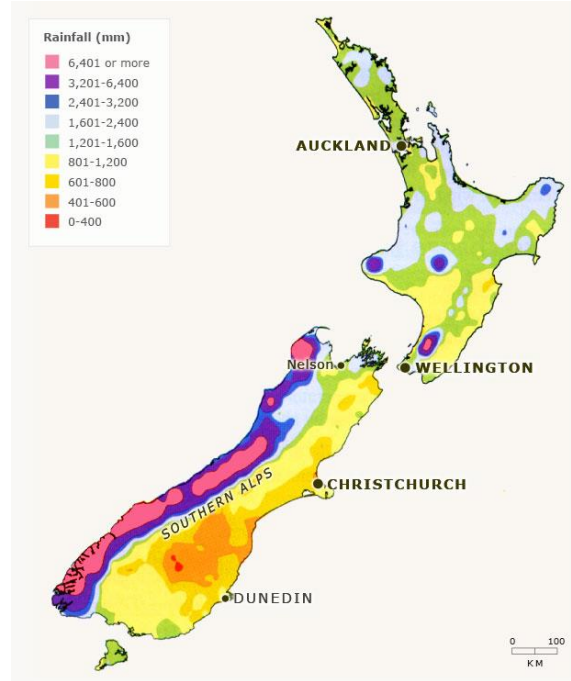
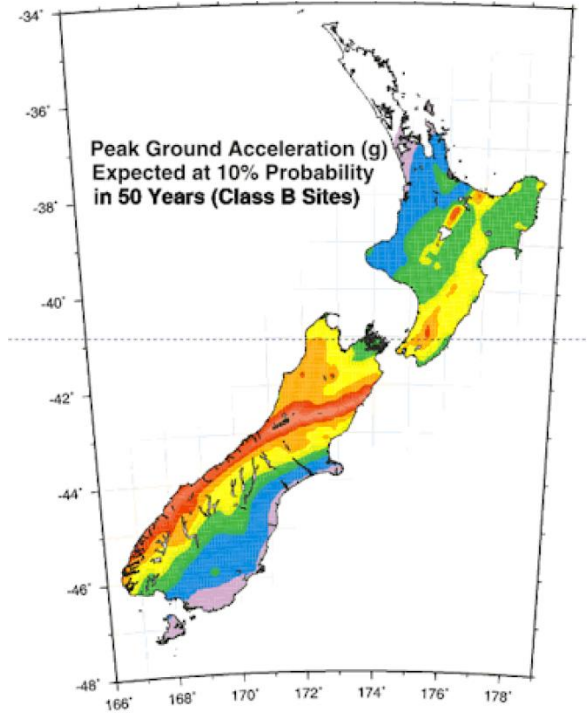


Heavy rain can scour loose rock and bring down slips like this on to Milford Road.



Weather station maintenance underway on a station located high above Milford Sound.

Example scenario – State Highway 94



Heavy rain can scour loose rock and bring down slips like this on to Milford Road.



Machinery like this have been kept busy this winter clearing snow from Alpine sections of the Milford Road



Weather station maintenance underway on a station located high above Milford Sound.

Example scenario – State Highway 94

Connections with other
infrastructure

Connections with the
community

Importance – local,
regional, and national

Community Resilience

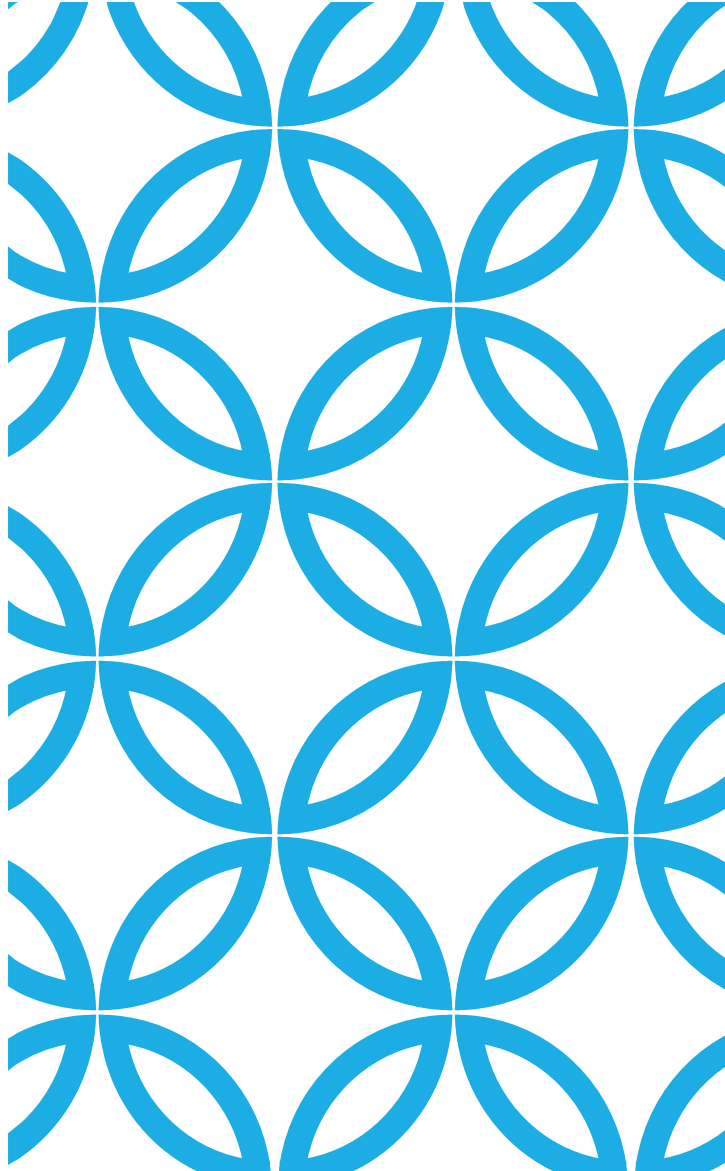


- University of Waikato
 - What are the different components needed for a university?
- What can happen to the different components?
- What will happen if we lose a critical component?
- How can we prevent this?

Infrastructure Interconnectedness



- University of Waikato
 - What infrastructure systems does the university rely on for its day to day operations?
 - How does the university positively or negatively impact the infrastructure that it is tied into?
 - If the university were to expand, what considerations would need to be considered in terms of infrastructure?



RESILIENCE

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ENGEV370

LEARNING OBJECTIVES

01

Define Resilience

02

Relate resilience
curves to system
functionality

03

Quantify resilience

RESILIENCE

What is Resilience?

- How does Resilience relate to Sustainability?

DEFINITIONS

Risk – Risk Triplet

1. What can go wrong? (scenarios combining hazards and outcomes)
2. How likely is it to go wrong? (probability of the scenario)
3. What are the consequences if it goes wrong?

Vulnerability – the diminished capacity of an individual or group to anticipate, cope with, resist and recover from the impact of a natural or man-made hazard

Resilience – The capacity to absorb or mitigate the impact of hazard events while maintaining and restoring essential services

WELLINGTON NEW ZEALAND



DIMENSIONS OF RESILIENCE

Technical: The ability of physical systems to perform to acceptable or desired levels

Organizational: The capacity of organizations to make decisions and take actions that contribute to resilience

Social: Measures specifically designed to lessen the extent to which disaster stricken communities and governmental jurisdictions suffer negative consequence due to loss of critical services

Economical: The capacity to reduce both direct and indirect economic losses resulting from disaster

DIMENSIONS OF RESILIENCE

Technical

- The ability of physical systems to perform to acceptable or desired levels

Organizational

- The capacity of organizations to make decisions and take actions that contribute to resilience

Social

- Measures specifically designed to lessen the extent to which disaster-stricken communities and governmental jurisdictions suffer negative consequence due to loss of critical services

Economical

- The capacity to reduce both direct and indirect economic losses resulting from disaster

RESILIENCE CASE STUDIES

Questions to consider:

1. What are the short, medium, and long term impacts of this disaster on the community?
2. What can be learned to improve community resilience for future disasters?
3. What are the technical, social, organizational, and economic impacts of these disasters?



Great Hanshin (“Kobe”) Earthquake of 1995

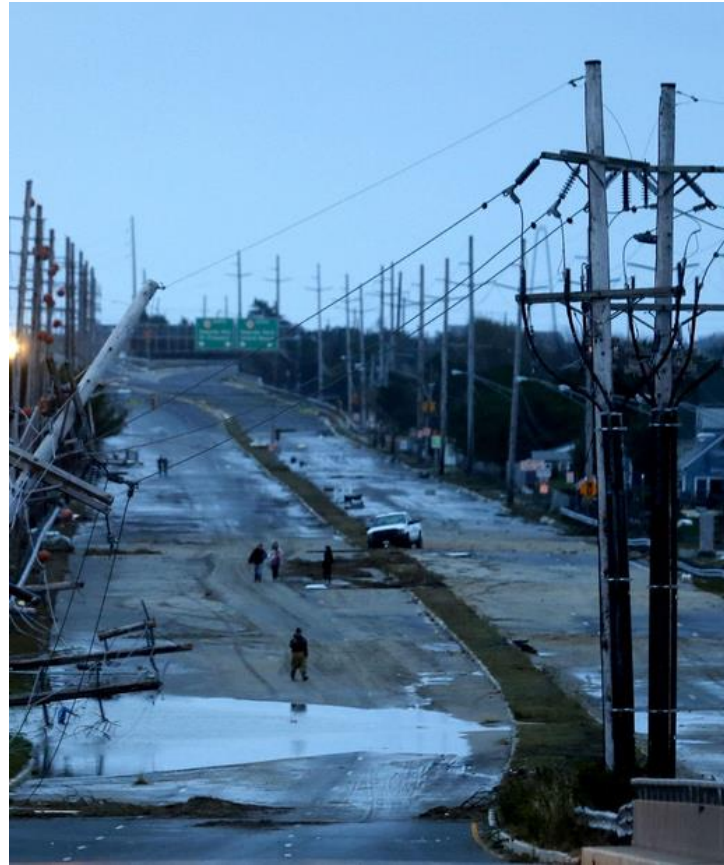
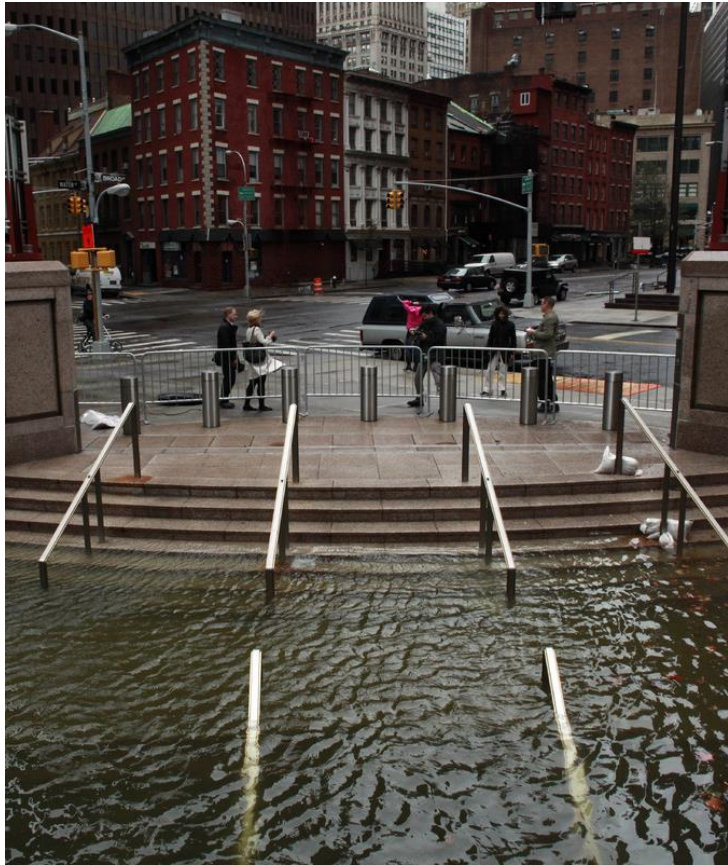
- 6,434 Fatalities
- 400,000 Buildings Destroyed
- 300 Fires



Northeastern Blackout of 2003

- 55 Million people affected
- 8 US States and parts of Canada
- 1 day to 1 week to restore power
- Many infrastructure systems disrupted

<https://www.youtube.com/watch?v=nd3teNgUq8E>



Hurricane Sandy, 29 October 2012

- \$75 Billion in Damage
- 233 Fatalities
- 15 states

Subways
Roads
EMT/Fire
Hospitals
Power
Airports

Water/wastewater
Data centers
Homes (displacement)
Gas rationing



CHRISTCHURCH 2011



Christchurch Earthquake (magnitude 6.2)

22 February 2011

185 Fatalities

Over 100,000 houses damaged



Flooding around Gore. Photo: High Country Hslicopster



Milford Rd State Highway 94. Photo: NZTA



Lake Howden Hut, Photo: DOC

SOUTHLAND FLOOD 2020



Milford Rd State Highway 94. Photo: NZTA

SOUTHLAND FLOODS 2020

1100 mm of rain over 60 hr period

~200 tourists in Milford Sound

~200 staff in Milford Sound

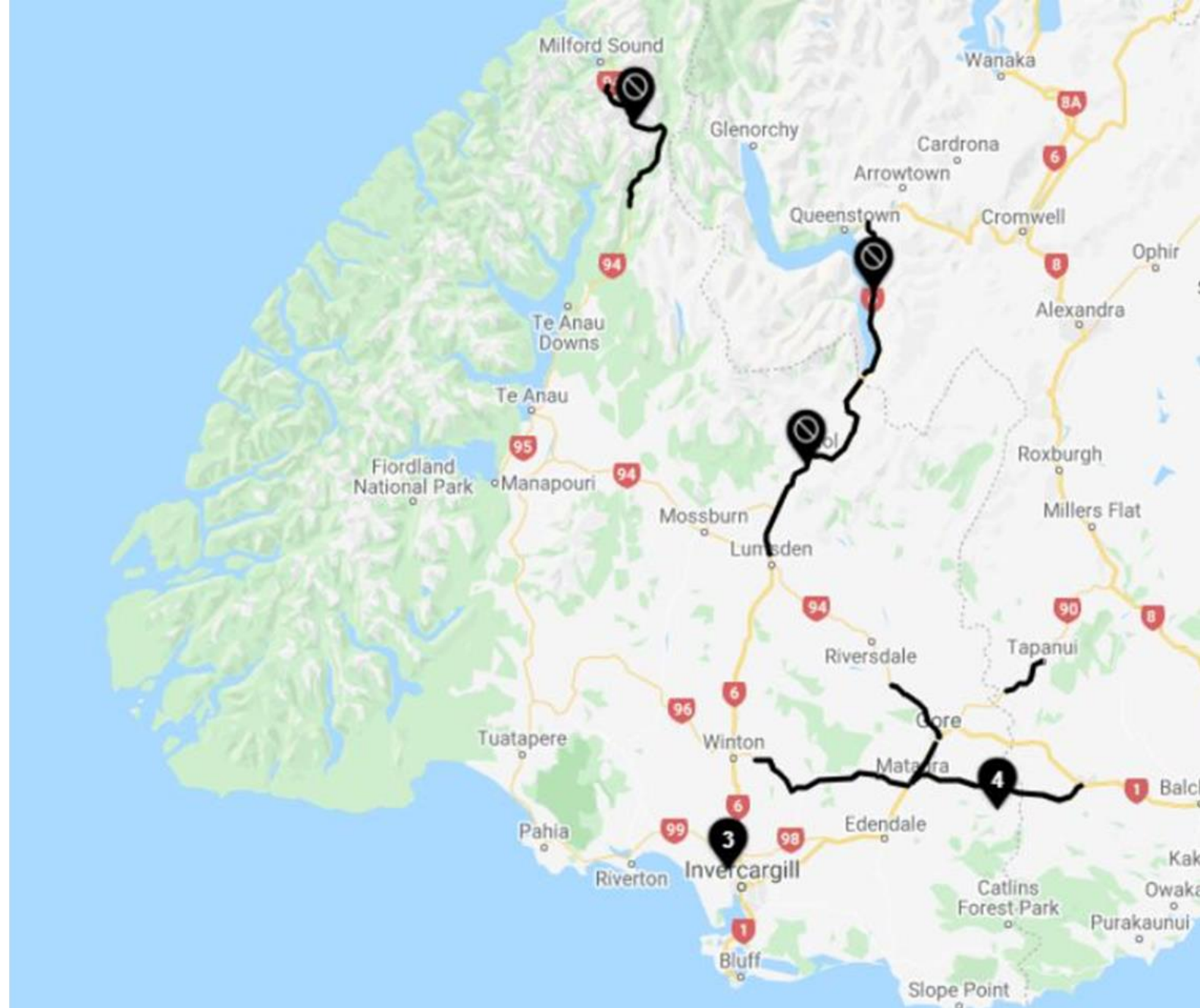
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SOUTHLAND FLOODS 2020



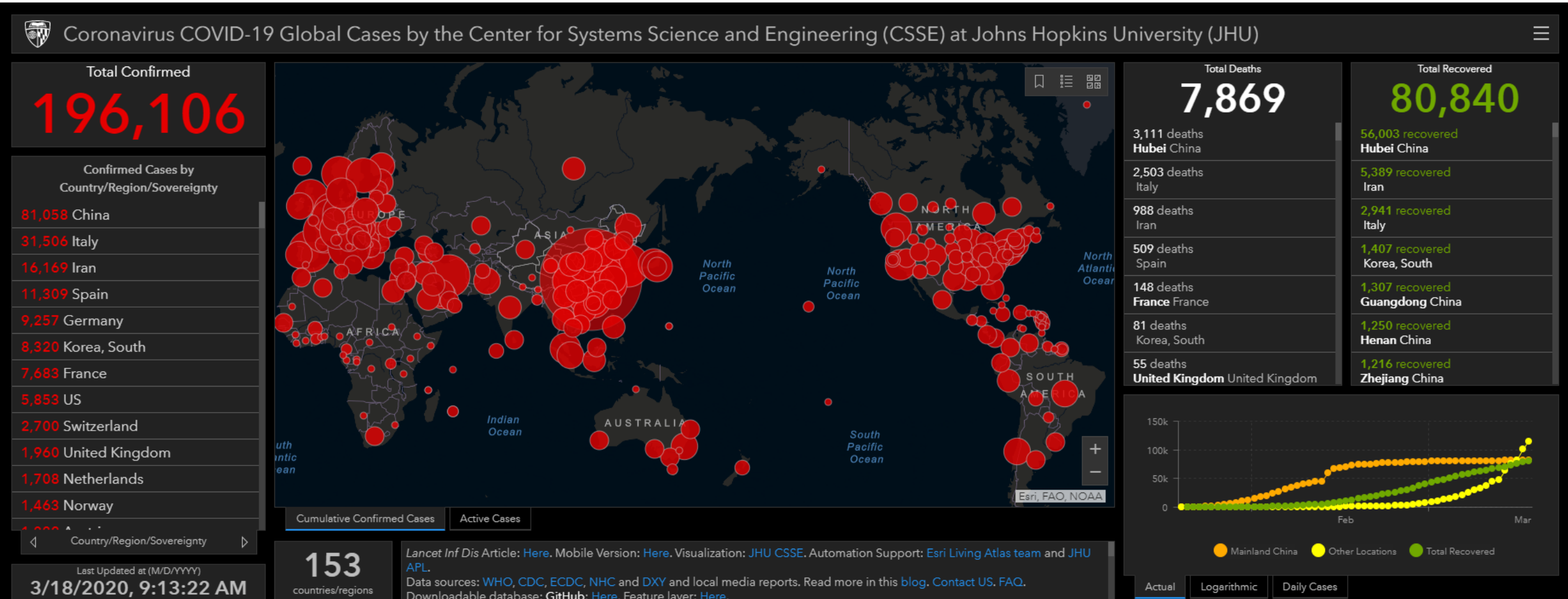
Road Closures due to Southland Flooding, Photo:NZTA

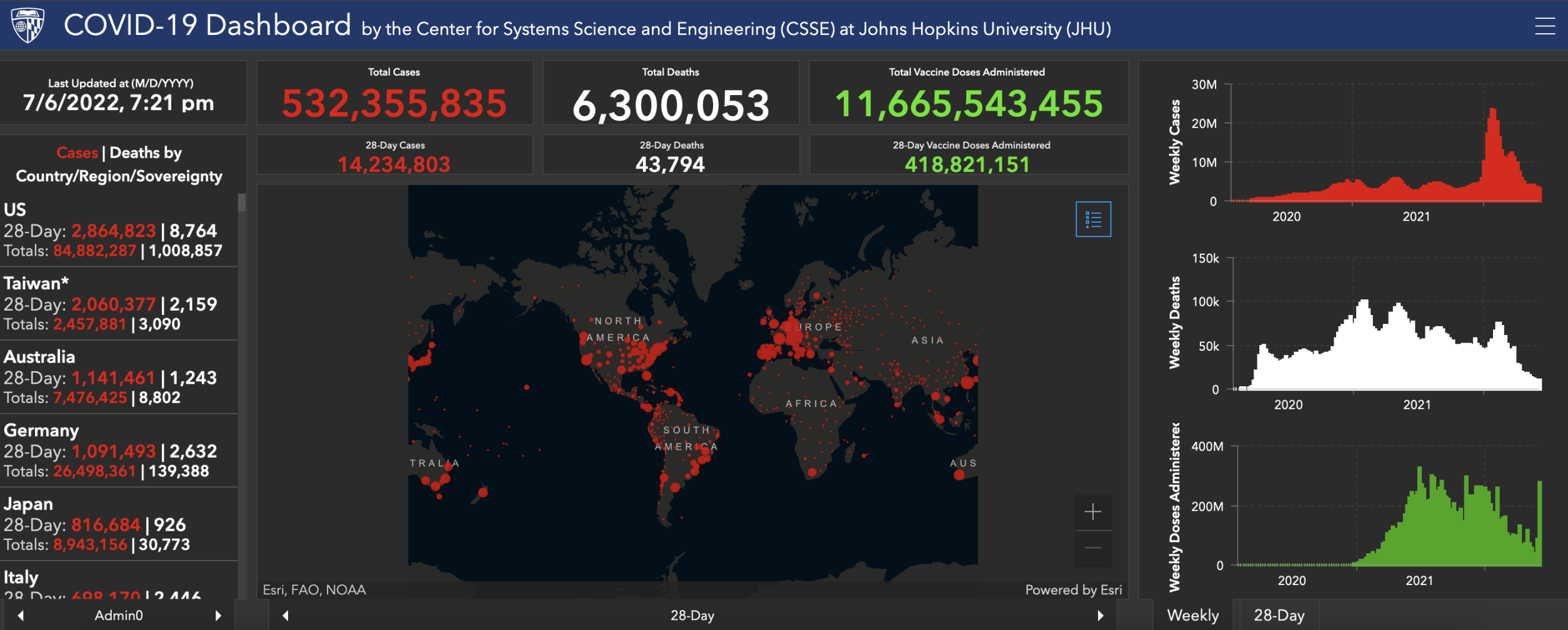


COVID-19

Travel Restrictions
Social Distancing

2020





IN SUMMARY - NATURAL (AND UNNATURAL) DISASTERS:

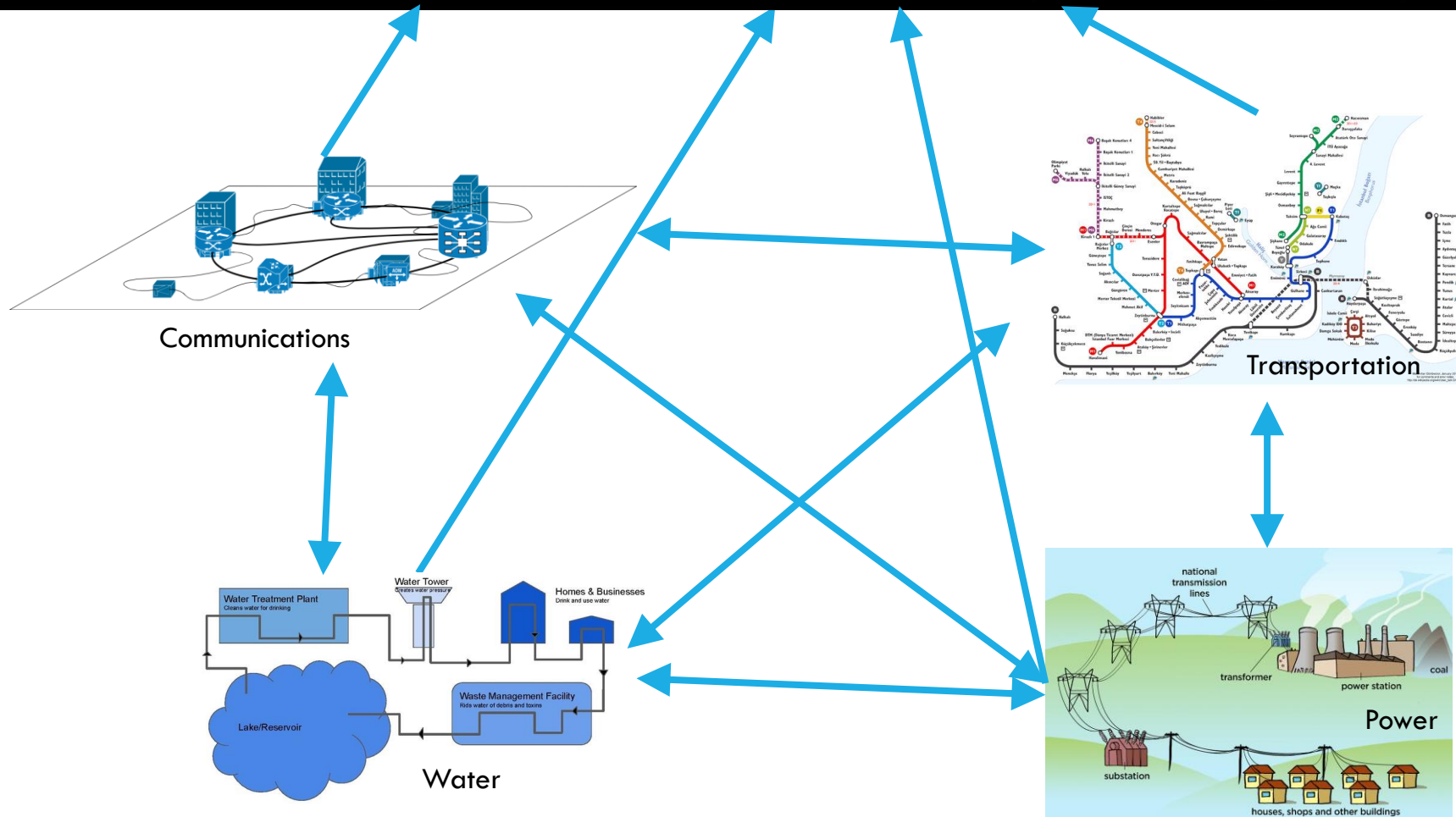
1. Widespread impacts across multiple sectors of infrastructure and society
2. Emergency response efforts often impeded by infrastructure failures
3. Cannot simply think in terms of infrastructure components, but must consider system aspects
4. How do you decide the levels of risk you are willing to tolerate?

RESILIENCE

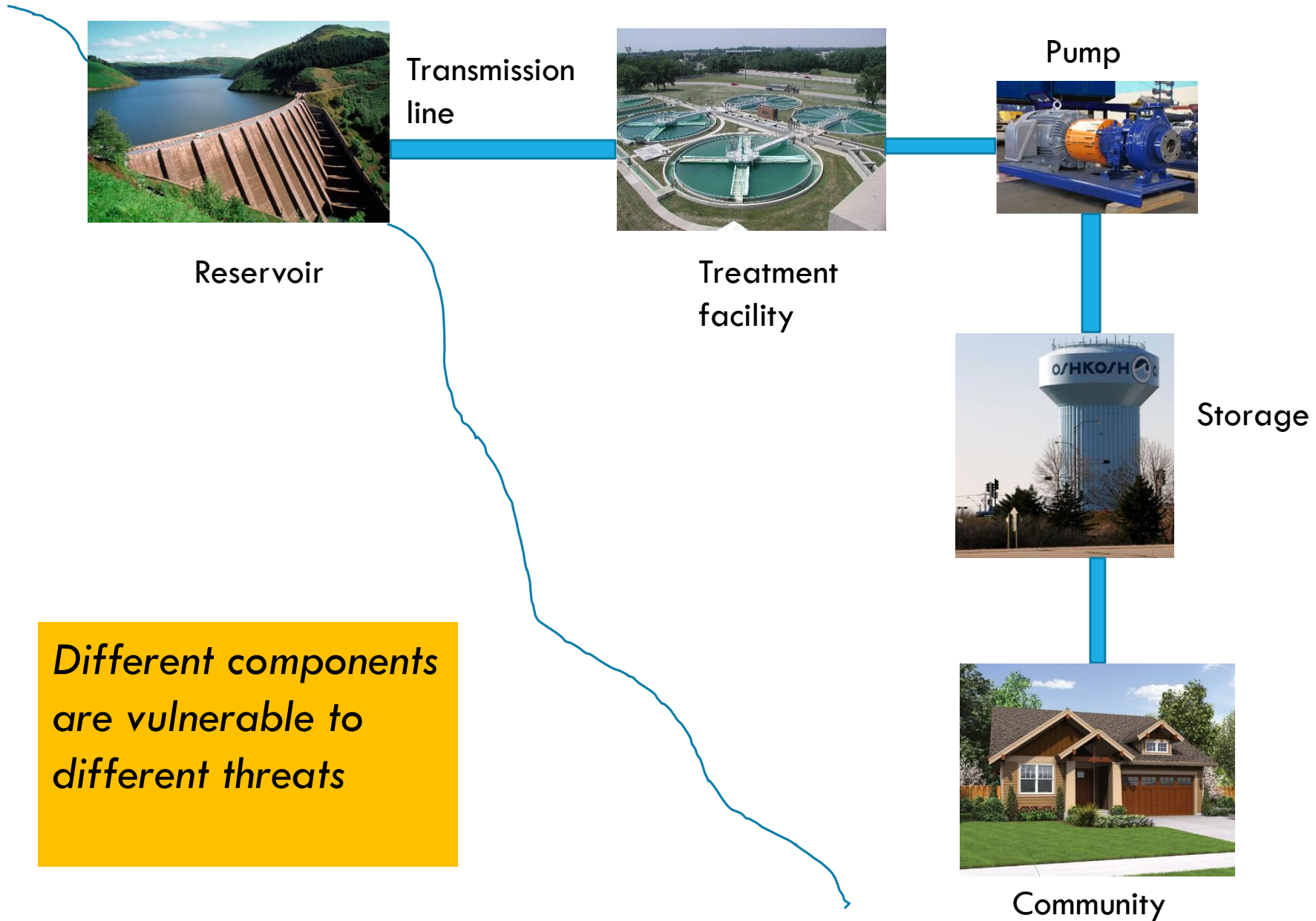
The ability to either withstand a hazard event or recover quickly with minimal disruption of service.

Resilience is inherently a systems aspect of infrastructure because delivering an infrastructure service requires interaction of many components and sectors.

How do you design for resilience?

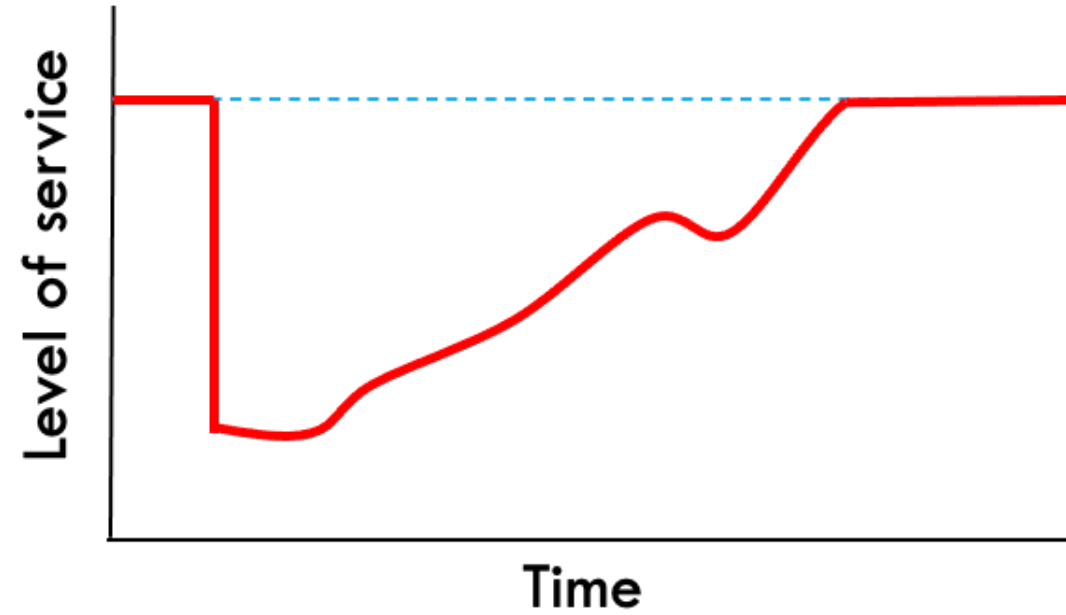


Water Supply Example



The 4 R's of Resilience



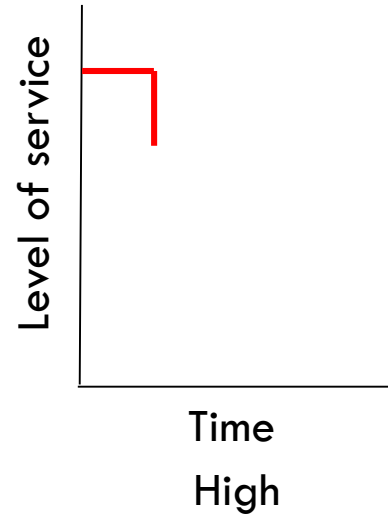
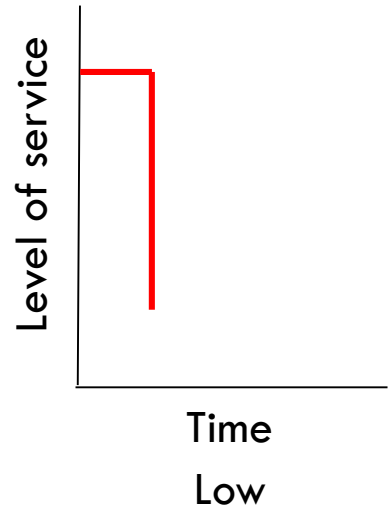


From Bruneau et al. (2003)

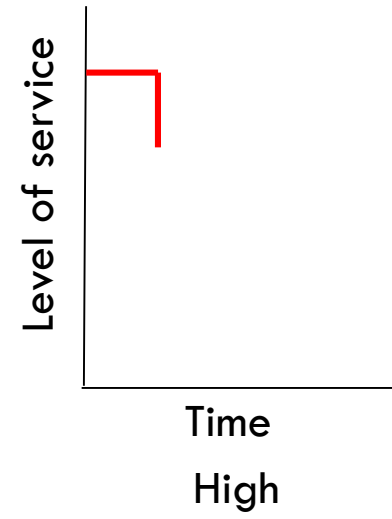
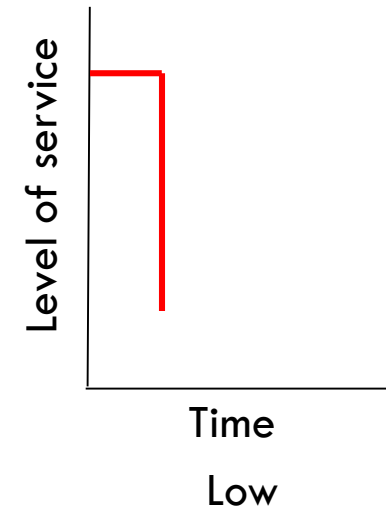


RESILIENCE TRIANGLE

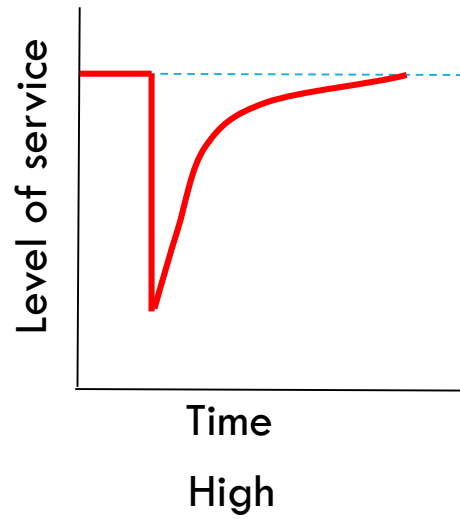
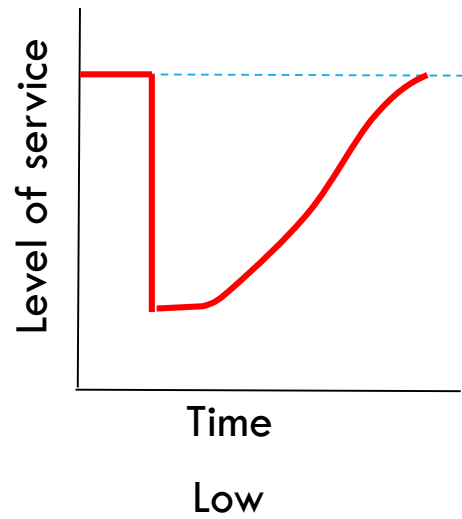
Robustness



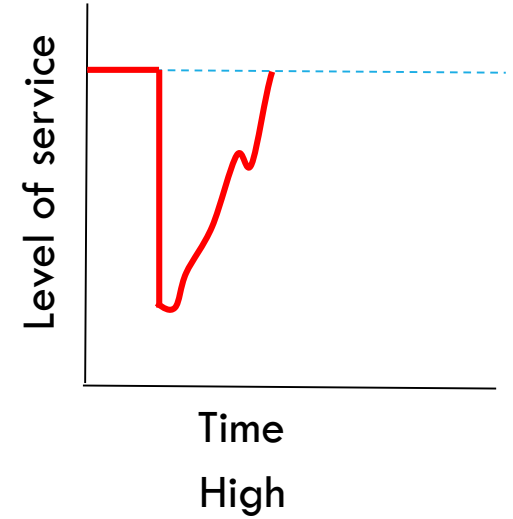
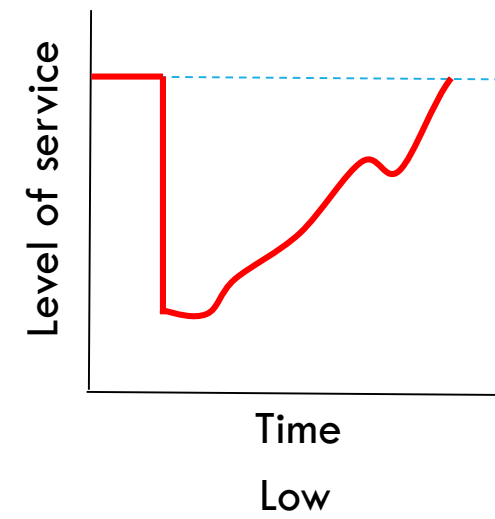
Redundancies



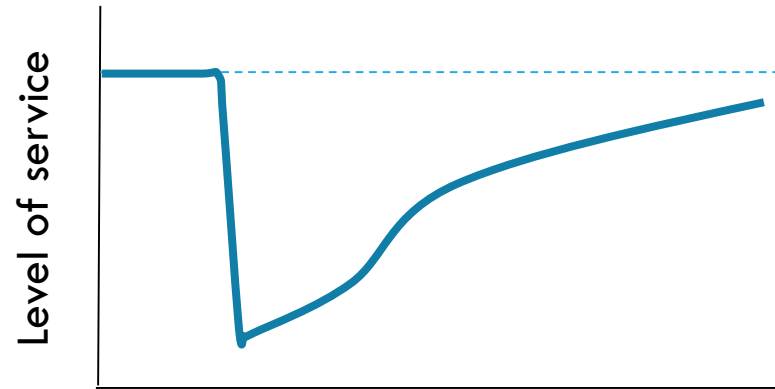
Resourcefulness



Rapidity

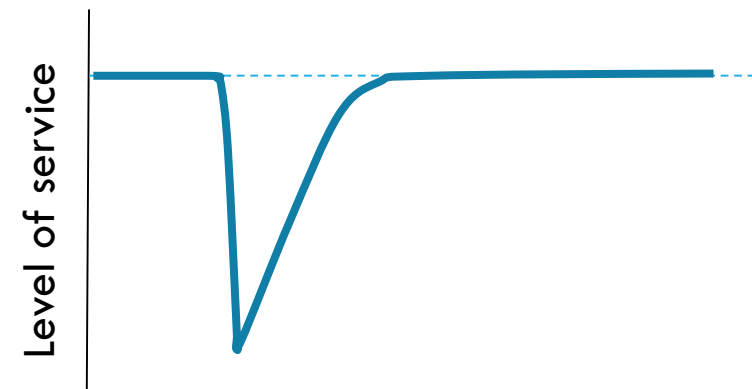


Response of four communities to the same disruption



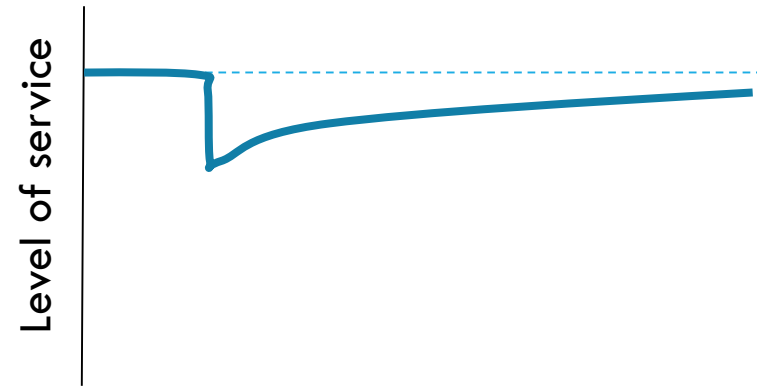
Time

Robustness	☐ High	☒ Low
Redundancy	☐ High	☒ Low
Rapidity	☐ High	☒ Low
Resourcefulness	☐ High	☒ Low



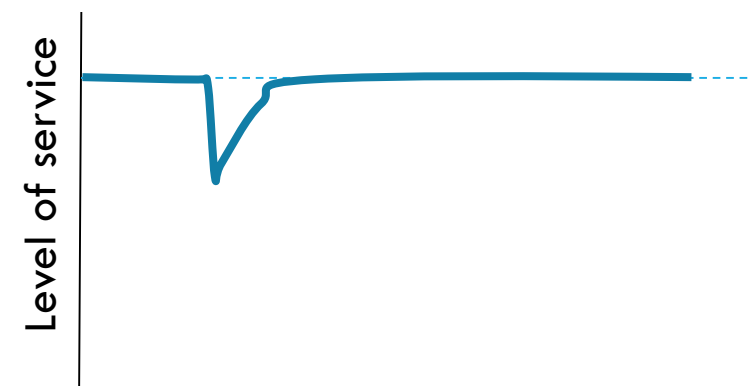
Time

Robustness	☐ High	☒ Low
Redundancy	☐ High	☒ Low
Rapidity	☒ High	☐ Low
Resourcefulness	☒ High	☐ Low



Time

Robustness	☒ High	☐ Low
Redundancy	☒ High	☐ Low
Rapidity	☐ High	☒ Low
Resourcefulness	☐ High	☒ Low



Time

Robustness	☒ High	☐ Low
Redundancy	☒ High	☐ Low
Rapidity	☒ High	☐ Low
Resourcefulness	☒ High	☐ Low

RESILIENCE PRACTICE

For your project topic:

- What are the potential hazards?
- What are the risks associated with the project?

How does your project interface with the 4 dimension of resilience?

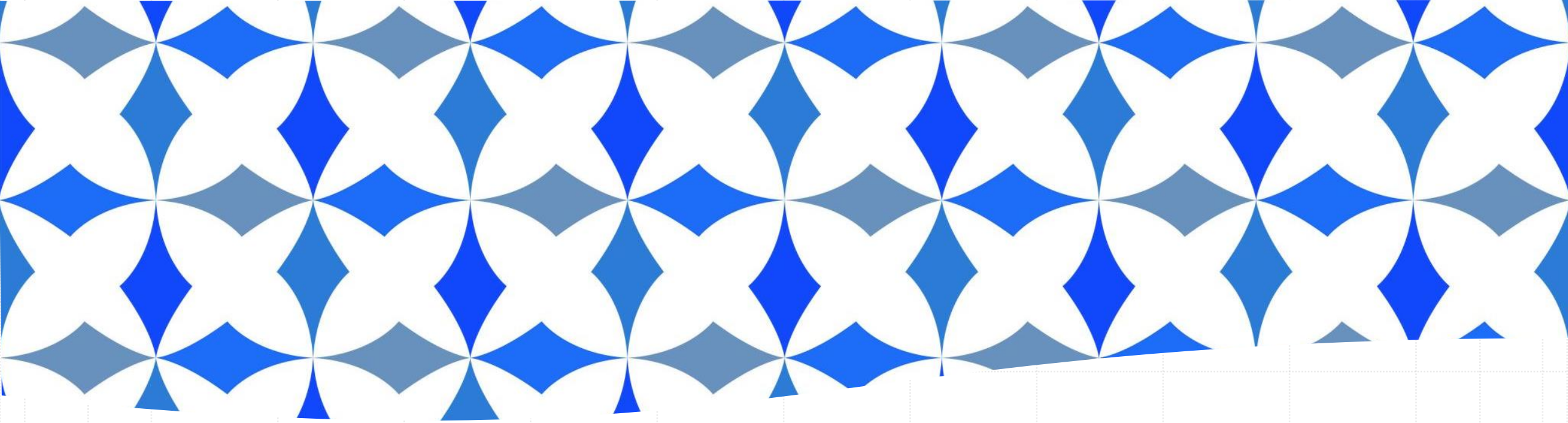
- Technological
- Organizational
- Social
- Economical

How can you use the 4 properties of resilience to decrease the risk?

- Robustness
- Redundancy
- Resourcefulness
- Rapidity



**WHAT DOES RESILIENCE
HAVE TO DO WITH
SUSTAINABILITY?**



Risk Analysis

ENGCV370

Dr Megan Boston



Defining
Risk

Types of
Risk

Safety in
Design

Quantifying
Risk

Risk

Safety in Design

- Easiest to manage risk earlier in a project
- Managing risk helps to improve safety
- Managing risk improves overall project outcomes

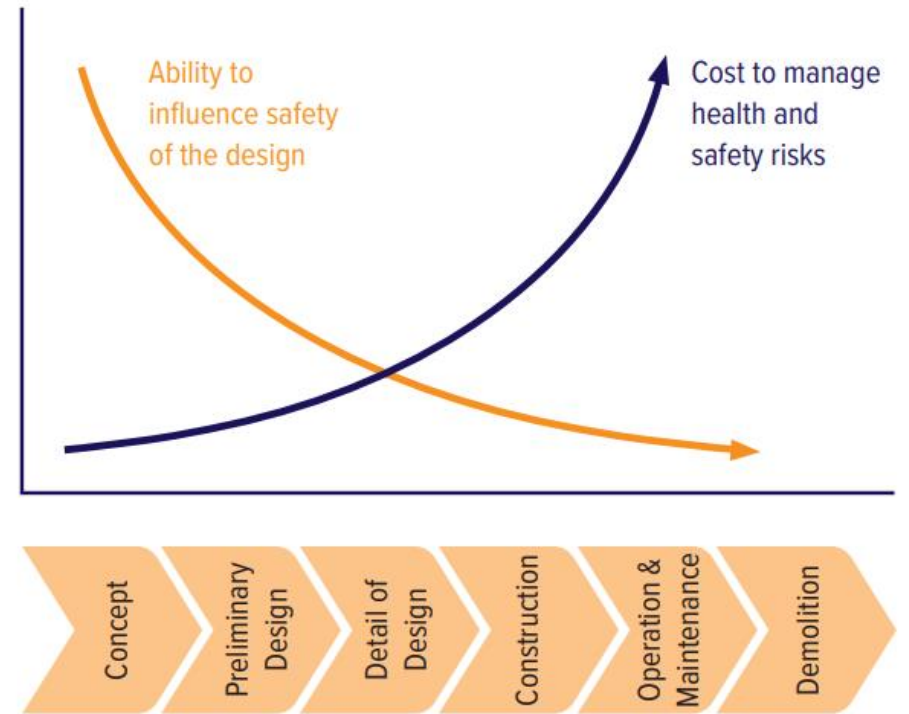
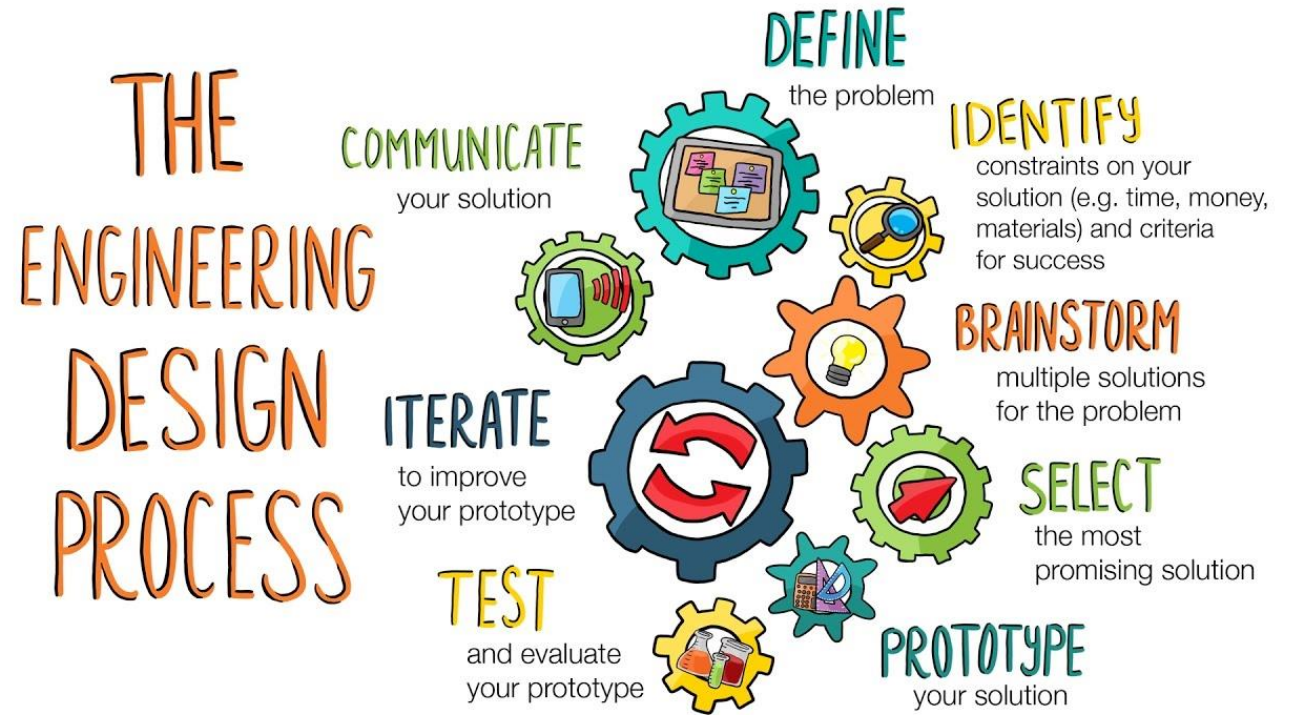


Figure 1: Symberszki chart of influence over a product's life cycle (adapted from Symberszki, R, (1997), Construction Project Safety Planning. TAPPI Journal, 80 (11), 69–74)

Safety in Design

- Elimination of potential hazards at the beginning of the project
- More efficient and effective risk management





Defining Risks

Understanding the final use of the product

- What will the product be used for?
- Who will use it?

Engage with the client and end users

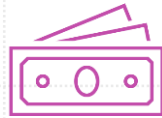
- How will it be used?
- Where will it be located?
- What are all the possible things that can go wrong?
 - Extreme events
 - Failures

Types of Risk



Environmental Risk

Human Health
Ecological
Global Climate Change



Business Risk

Financial
Schedule
Strategic



Technical Risk

Reliability
Catastrophic Failures



Political Risk

Public Acceptance
Public Perception



Classify Risk

Being the lead contractor on a \$600 million-dollar industrial construction project with a fixed-price contract and stiff penalties for completion

Deciding how to design a 47-storey building in an earthquake zone

Deciding where to locate a hazardous waste facility

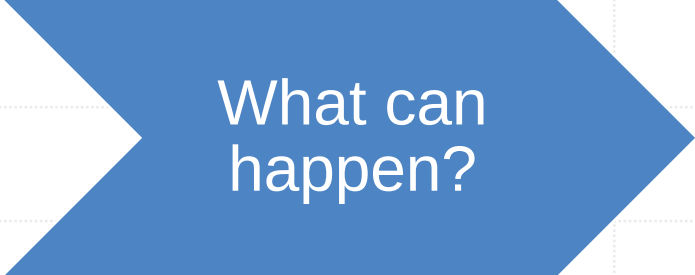
Deciding how to allocated money to protect New Zealand infrastructure

Being the lead project manager of a multi-million-dollar satellite project with a short launch window

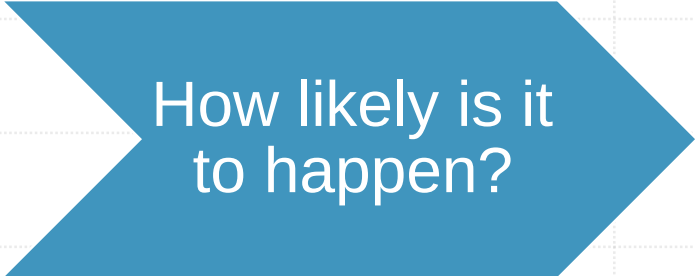


Quantify the risks

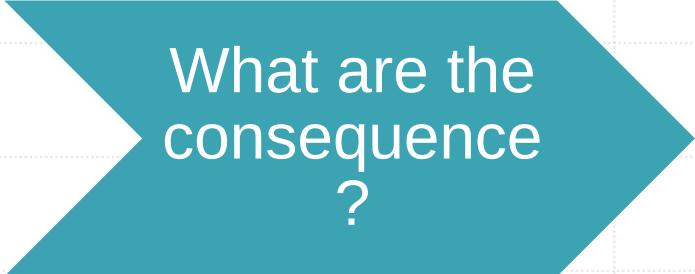
- Scenarios
 - What can go wrong?
- Likelihood (qualitative or quantitative) of the scenario happening
- Consequence (qualitative or quantitative severity of performance degradation) if the scenario happens



What can happen?



How likely is it to happen?



What are the consequence ?

How to Quantify Risk?



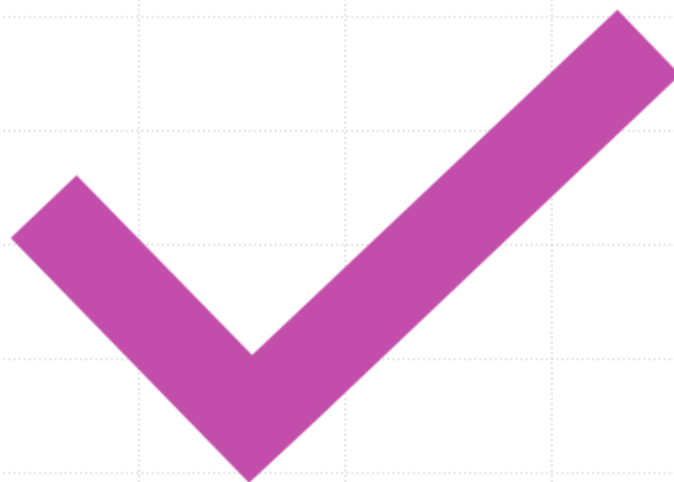
Assessing Risk: Probabilistic Risk Assessment

- Helps to determine risk when there is not a lot of data available
- Steps:
 1. Functional Analysis
 2. Effects of External Events
 3. Probability Calculations
 4. Use Results to Make Decisions



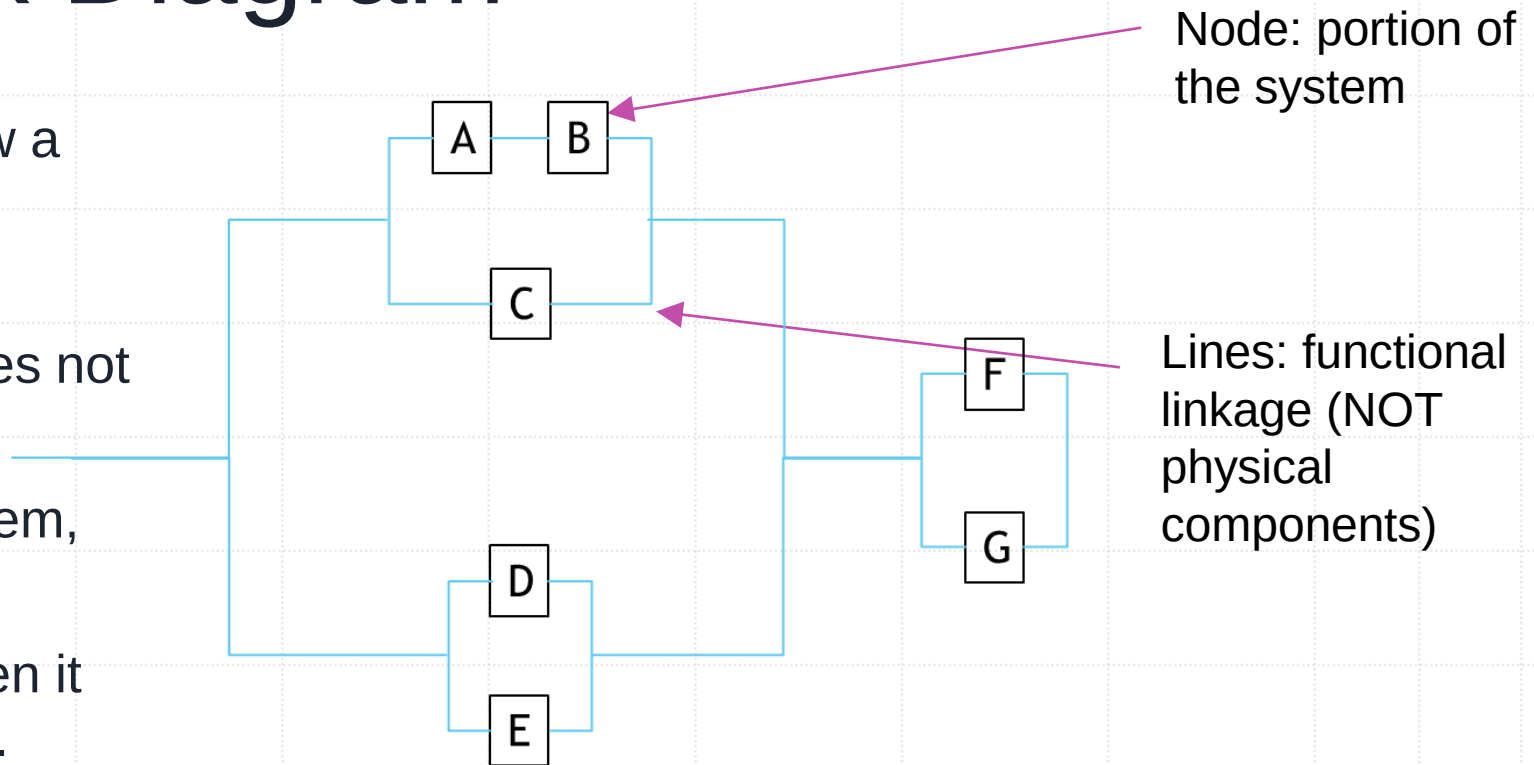


Functional Analysis



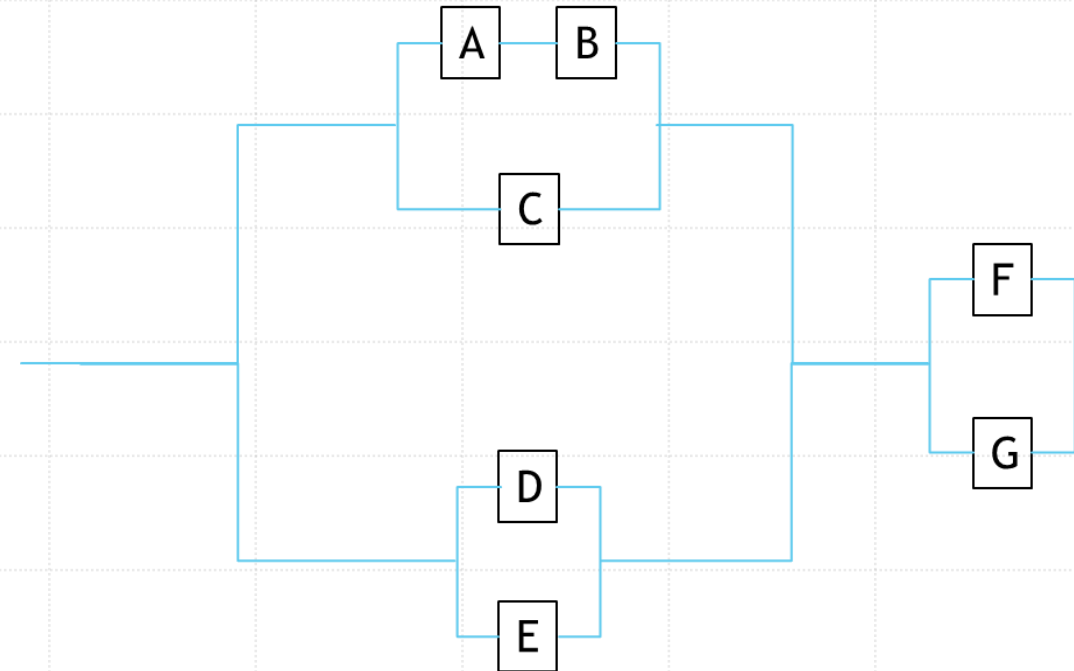
Functional Block Diagram

- Method to graphically display how a system functions
 - System logic diagram
- Binary: the system works or it does not work
- If there is a path through the system, then the system will function
- If a portion of the system fails, then it blocks the path through that area.



Functional Block Diagram

- Tips for Creating a Functional Block Diagram
 1. If two or more components need to work they are in series
 2. If only one of two or more components need to work they are in parallel

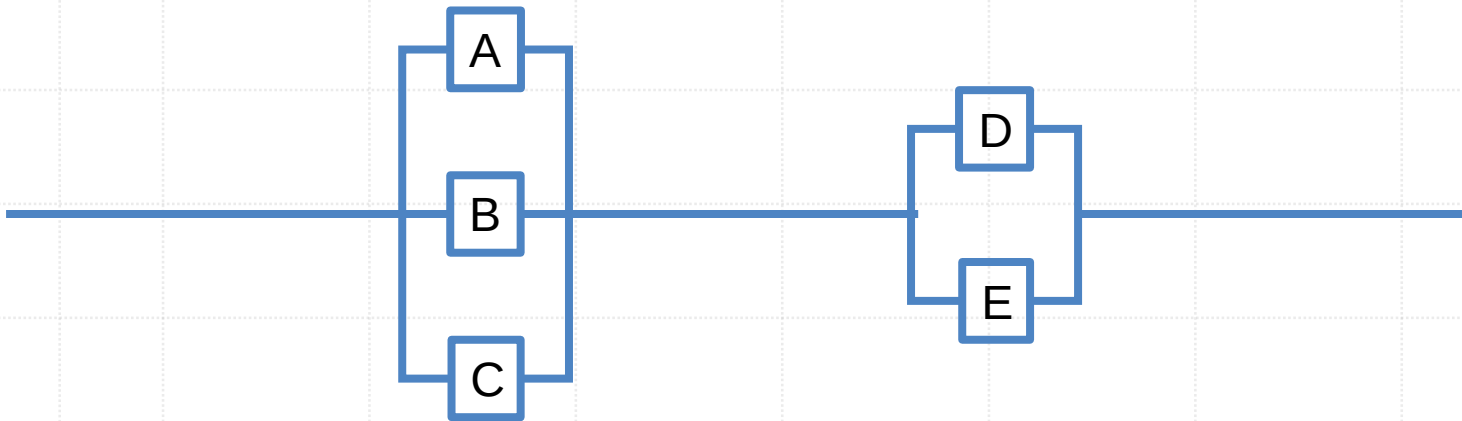




Create functional block diagrams

- A system works if one of A, B and C works AND either D or E works

A system works if one of A, B and C works
AND either D or E works



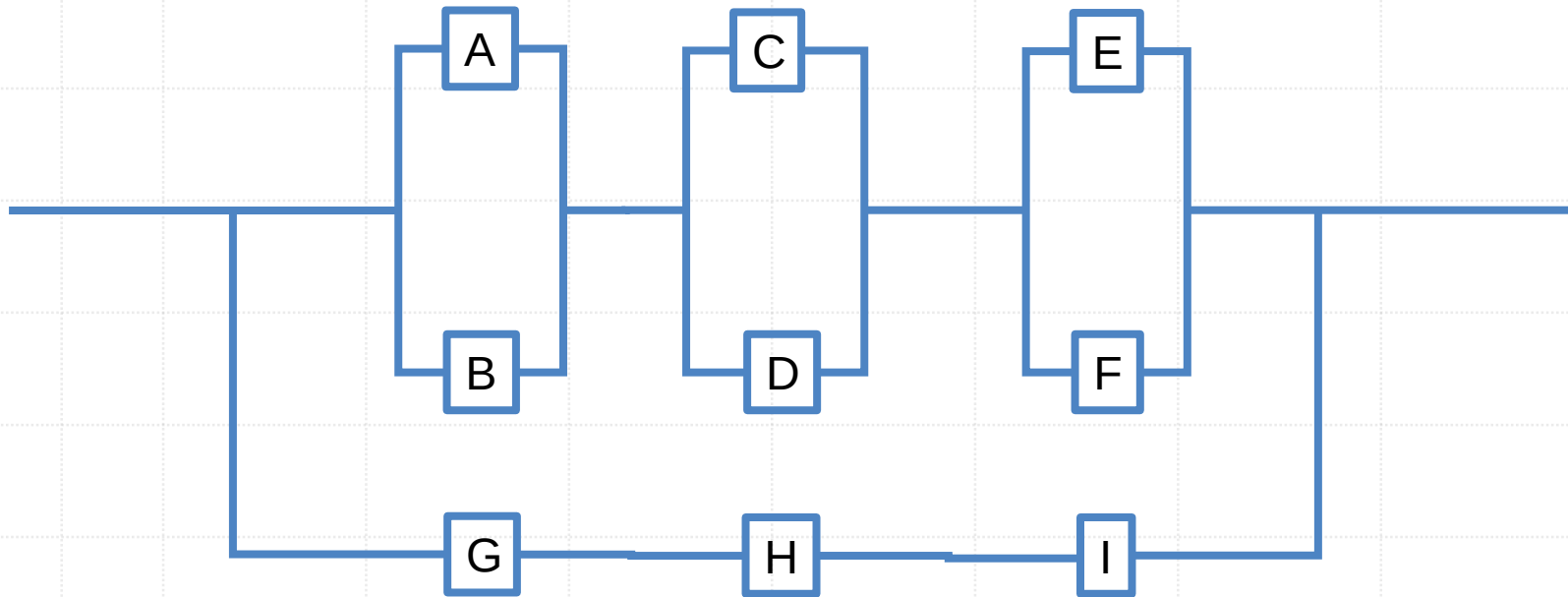


Create functional block diagrams

- A system works if $[(A \text{ or } B) \text{ AND } (C \text{ or } D) \text{ AND } (E \text{ or } F)] \text{ or } (G \text{ and } H \text{ and } I)$

Create functional block diagrams

- A system works if [(A or B) AND (C or D) AND (E or F)] or (G and H and I)





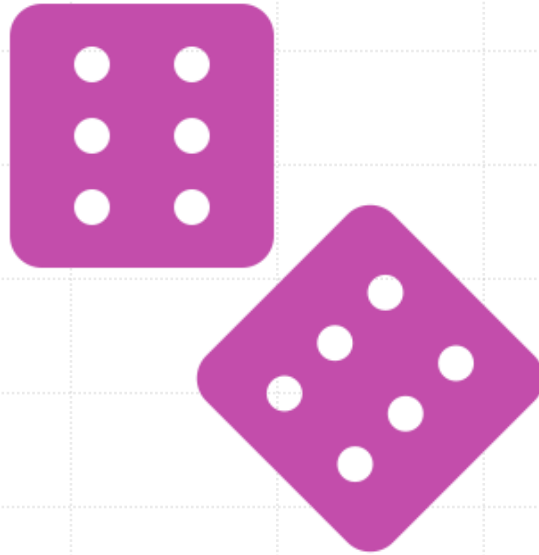
Functional Block Diagrams

- Functional Block Diagrams:
 - Represent functionality of a system or process
 - They are logic diagrams (AND and OR logic)
 - DO NOT provide the framework needed for probability calculations
- Need to convert to a system that takes probabilities
 - Failure Modes and Effects Analysis
 - Event Tree Analysis
 - Fault Tree Analysis



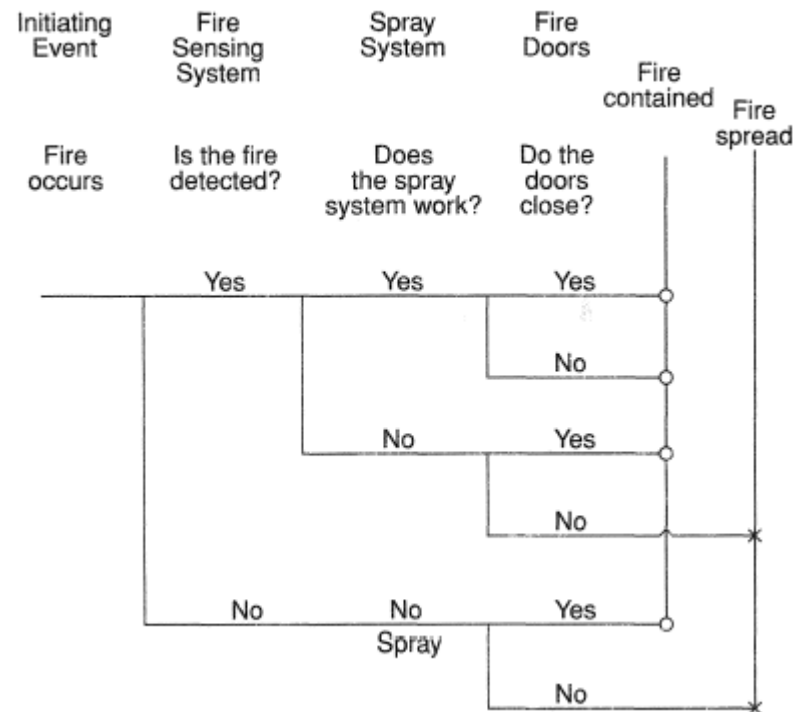
Effects of External Events

Probability of Failure



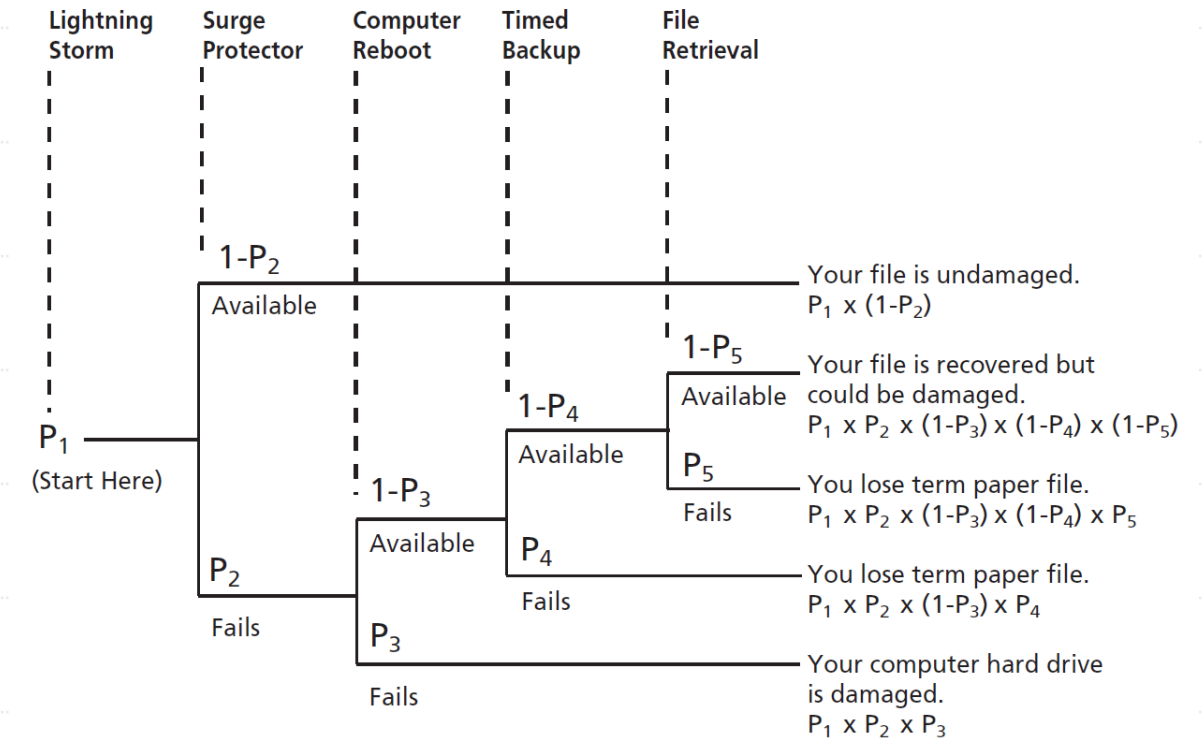
Event Tree Analysis

- Illustrates all possible sequences events after an initial event

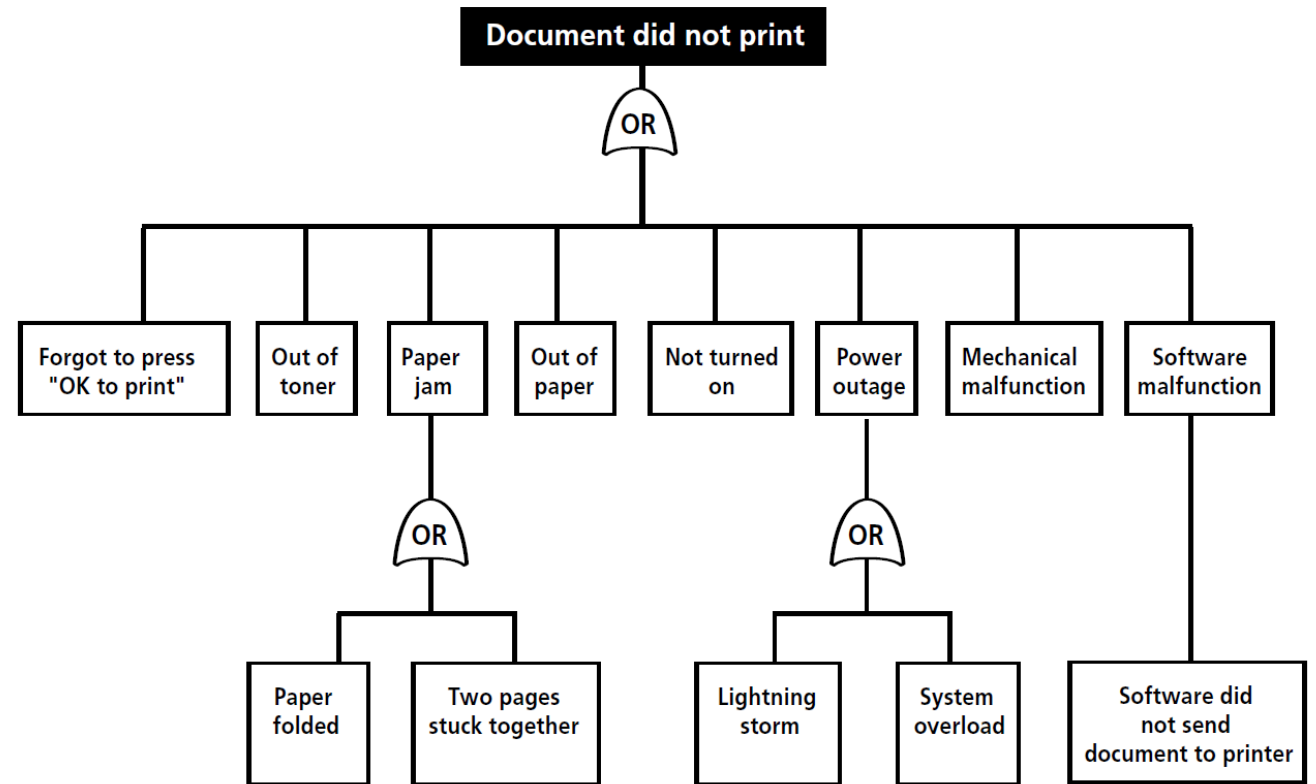


Risk Analysis Methods – Event Trees

*Losing Document due to
Lightning Storm*



Fault Trees





Fault Trees

- Top Down approach to understanding failures
- Identifies all possible contributors (Basic Events) that contribute to a specific outcome, the top level event
- Using Boolean logic to propagate failures (or probability of failure) from the basic level events to the top level events



Fault Trees

Relate the performance of a complex system to the performance of its individual components

Logic flows from bottom (components) to middle (sub-systems) to top (system)

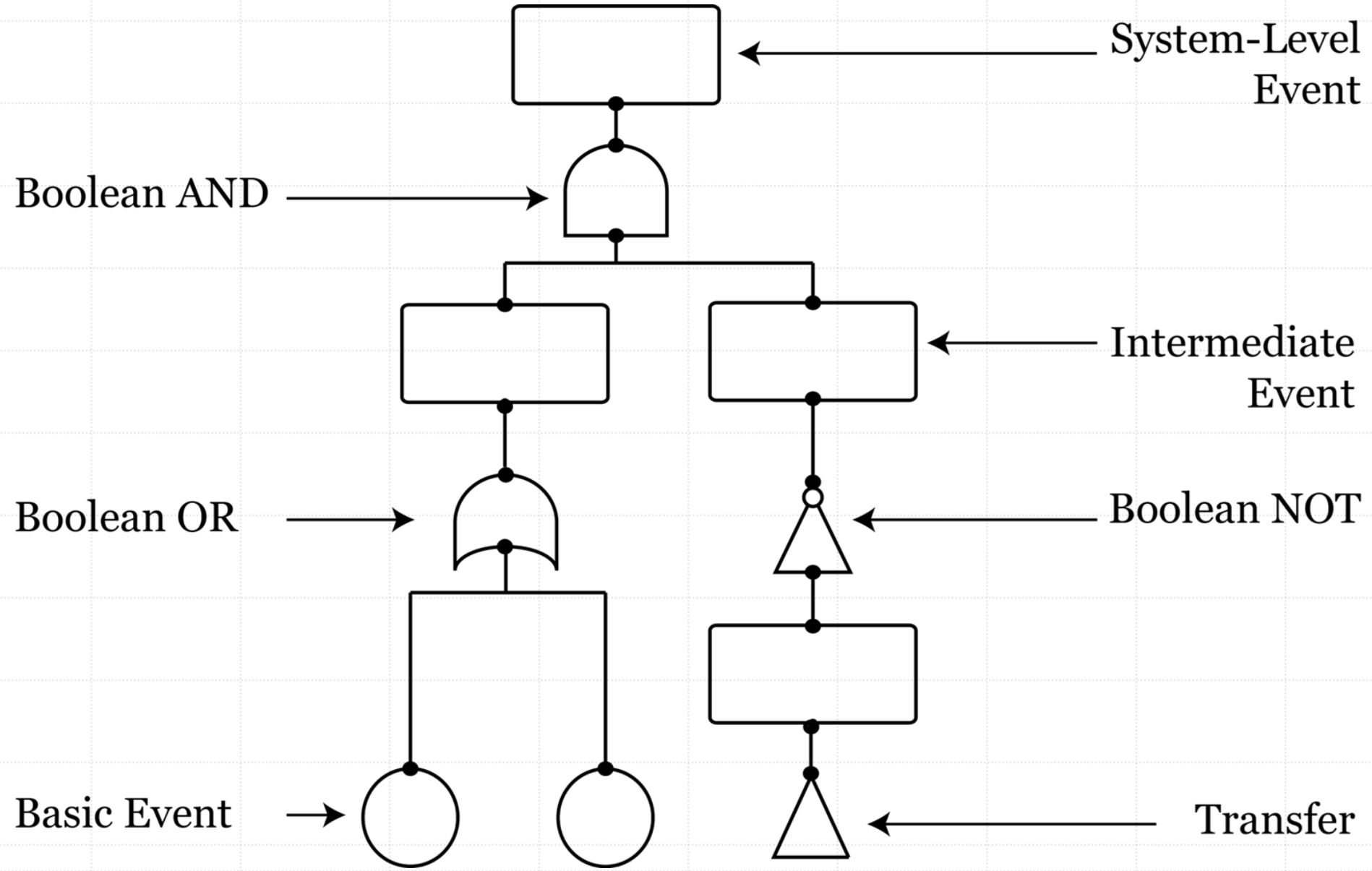
Related objects are connected by lines

Top-Down approach to understanding failures

Identifies all possible contributors (Basic Events) that contribute to a specific outcome, the top-level event

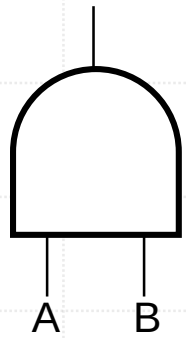
Using Boolean logic to propagate failures (or probability of failure) from the basic level events to the top-level events

Fault Tree Symbols

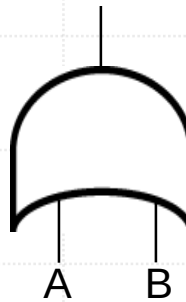


Fault Tree Symbols

Deterministic

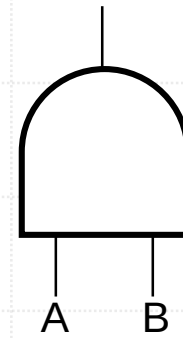


AND Gate: TRUE if **all** the events below the gate are TRUE

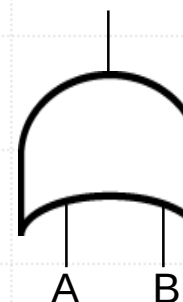


OR Gate: TRUE if **any** of the events below the gate are TRUE

Probabilistic



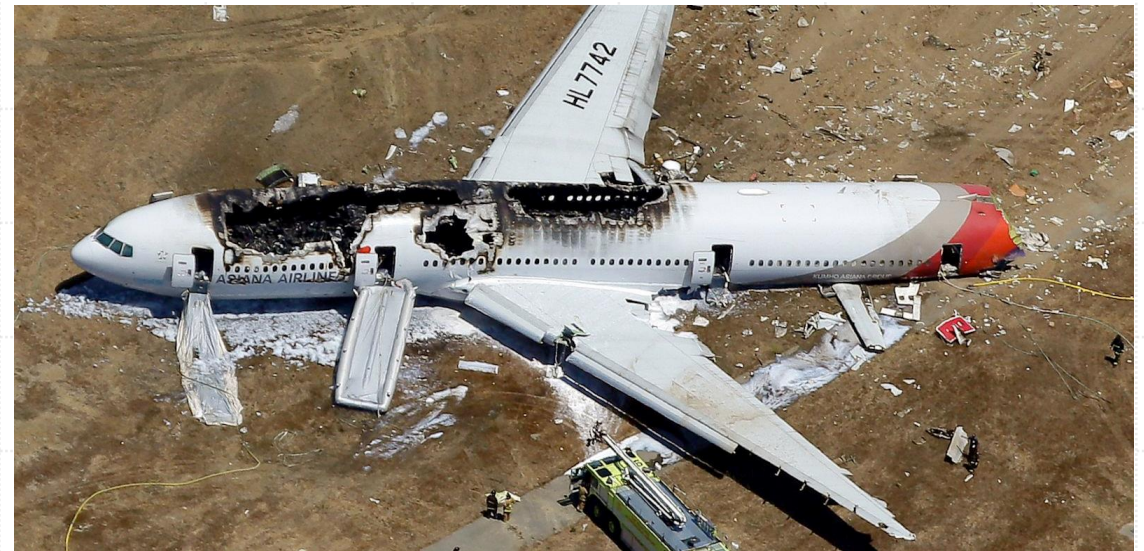
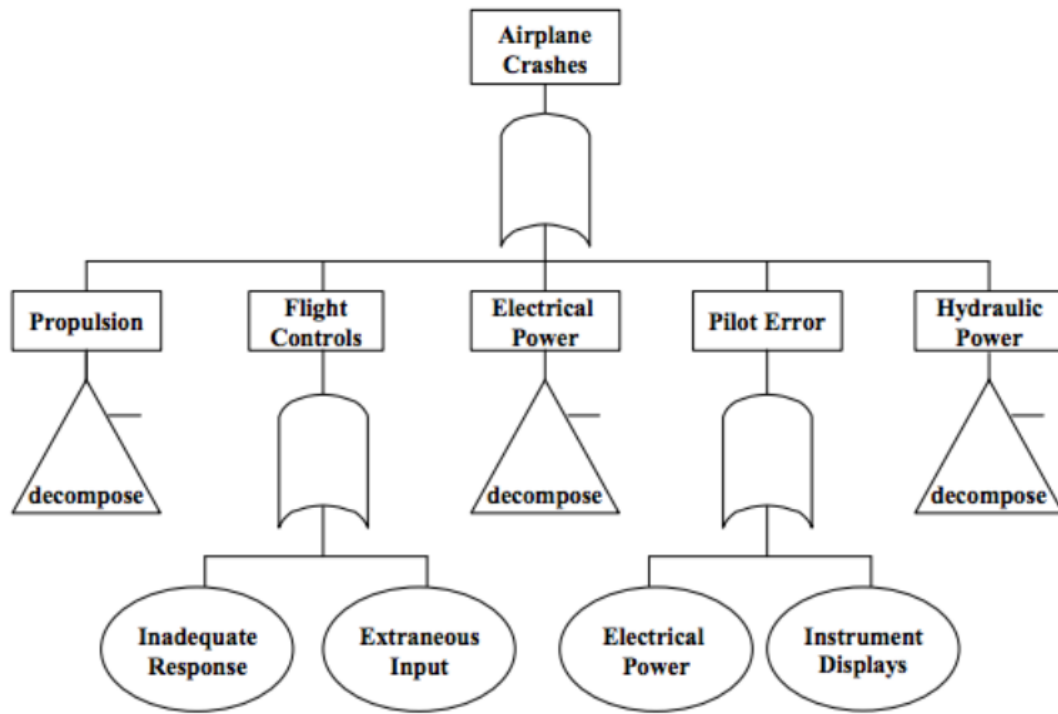
AND Gate: Probability of A and B
 $P(A \text{ and } B) = P(A)P(B)$



OR Gate: Probability of A or B
 $P(A \text{ or } B) = P(A) + P(B) - P(A)P(B)$

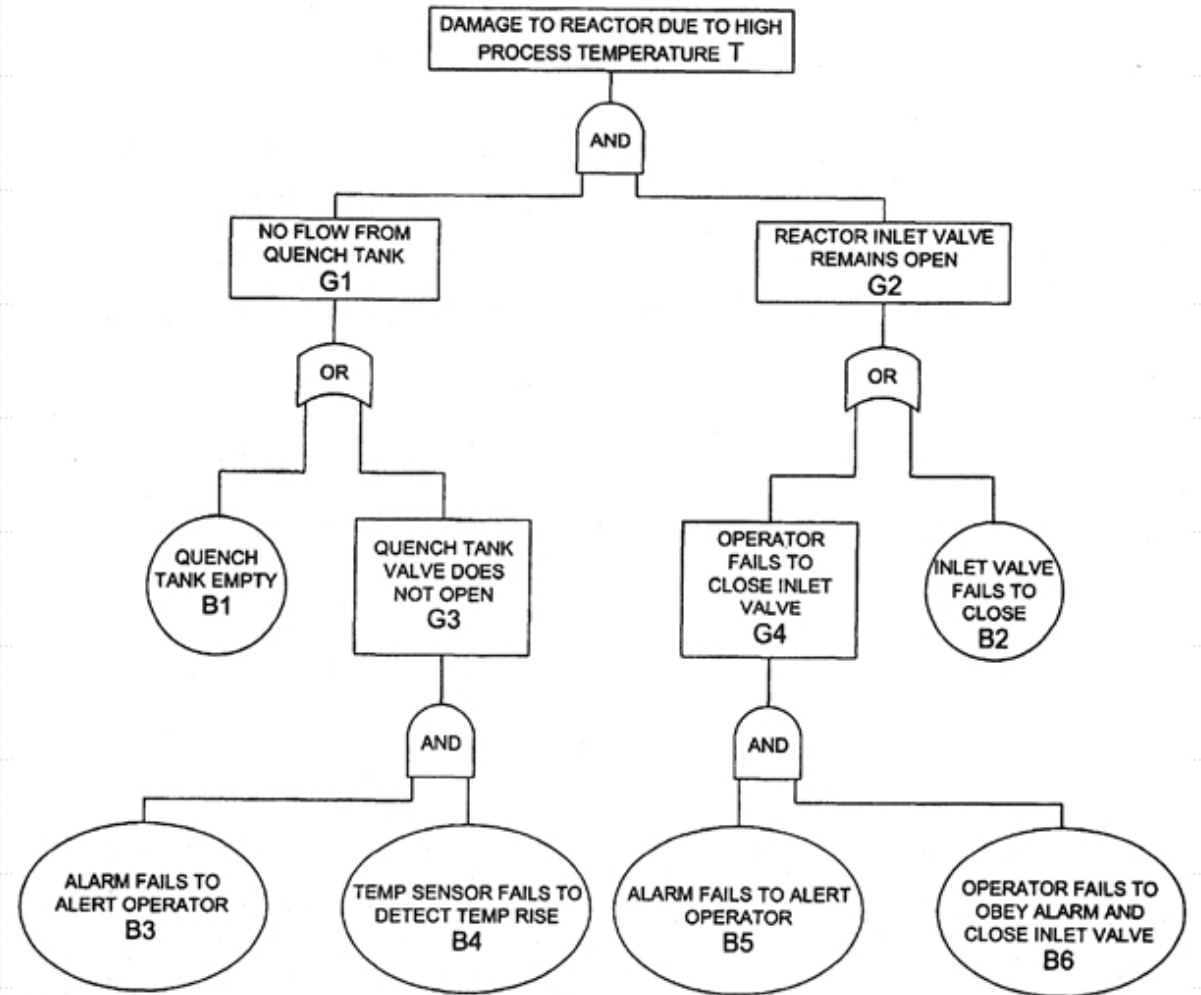
Fault Tree Examples

Airplane Crash



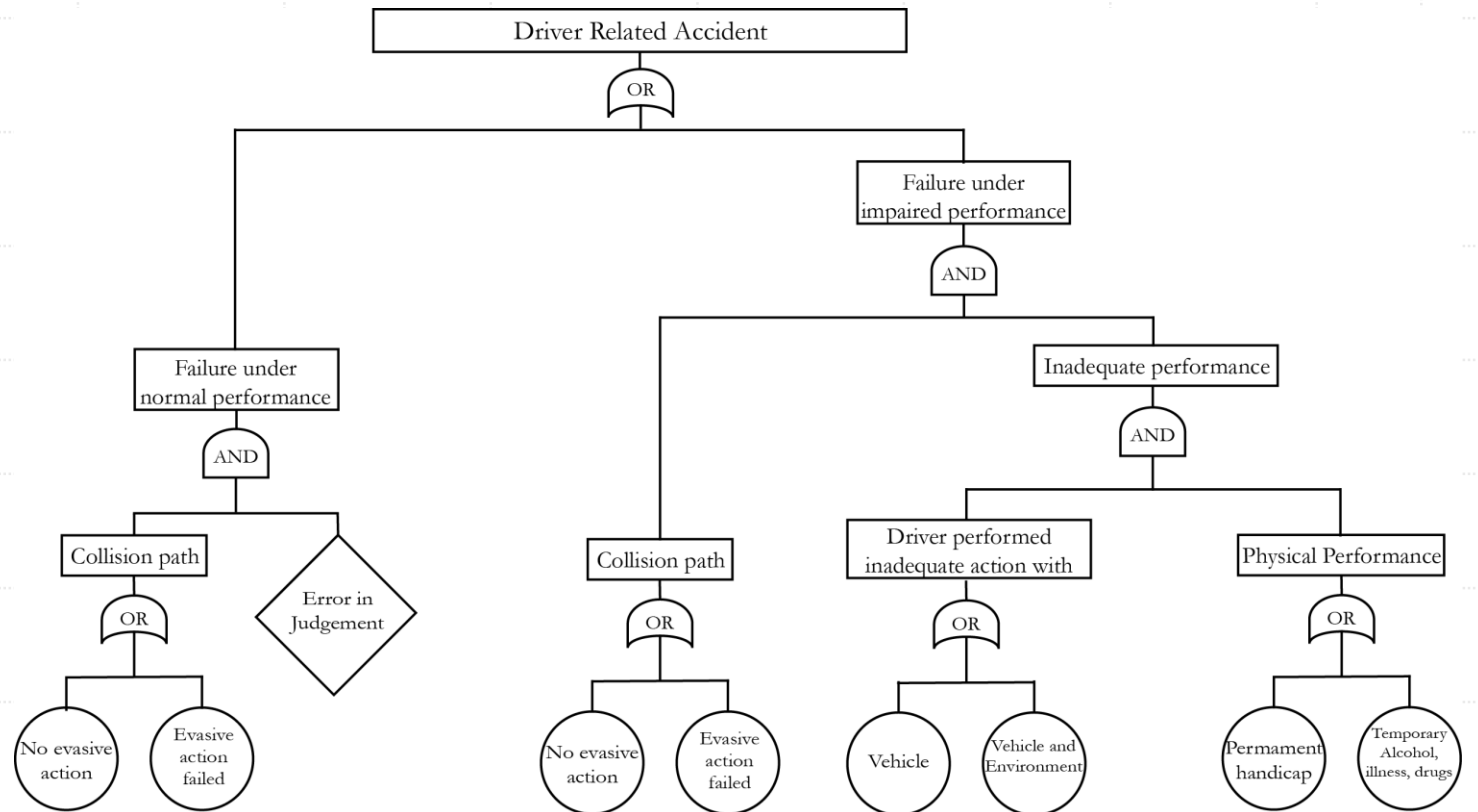
Fault tree examples

Nuclear Reactor Failure



Fault tree examples

Car Accident

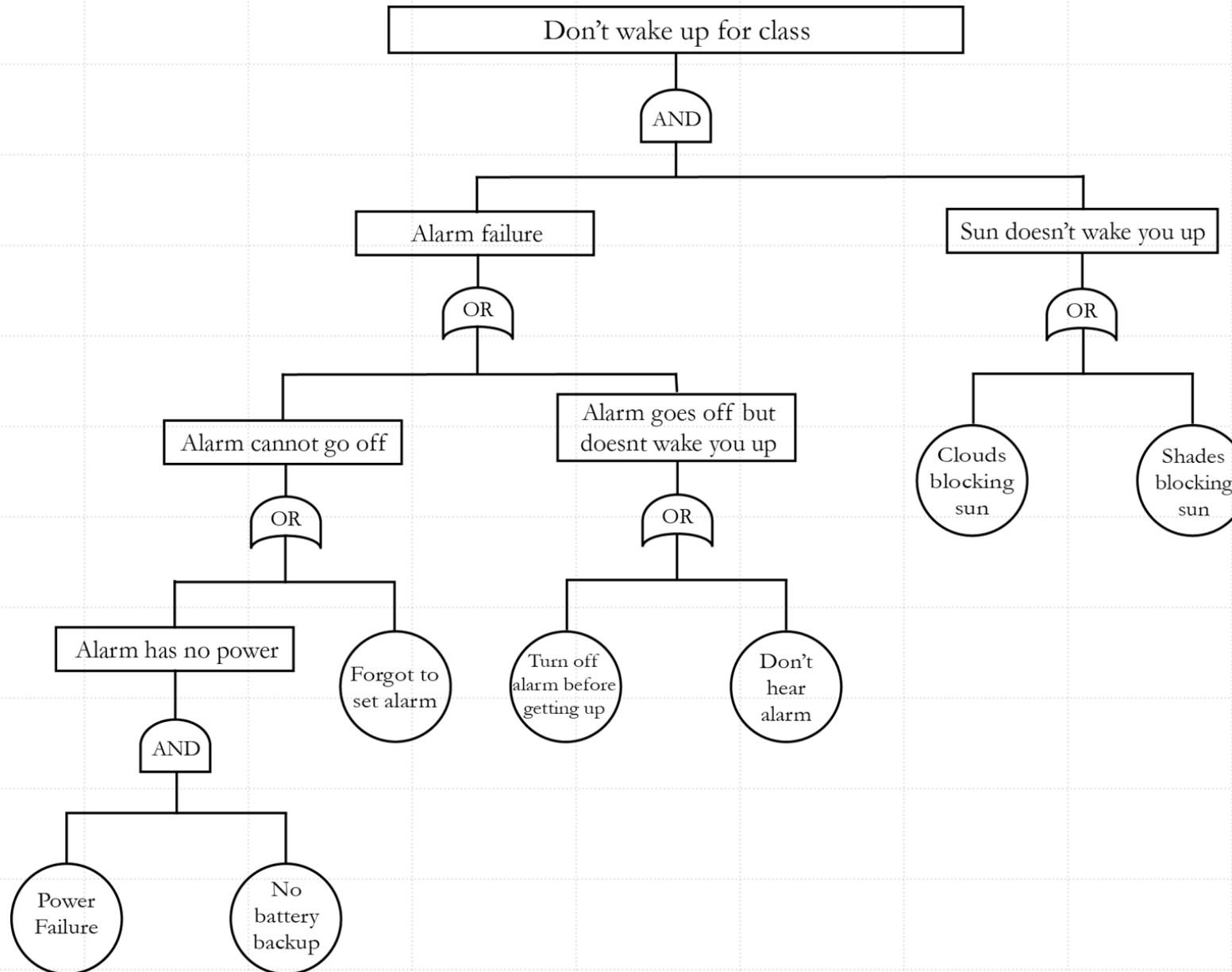




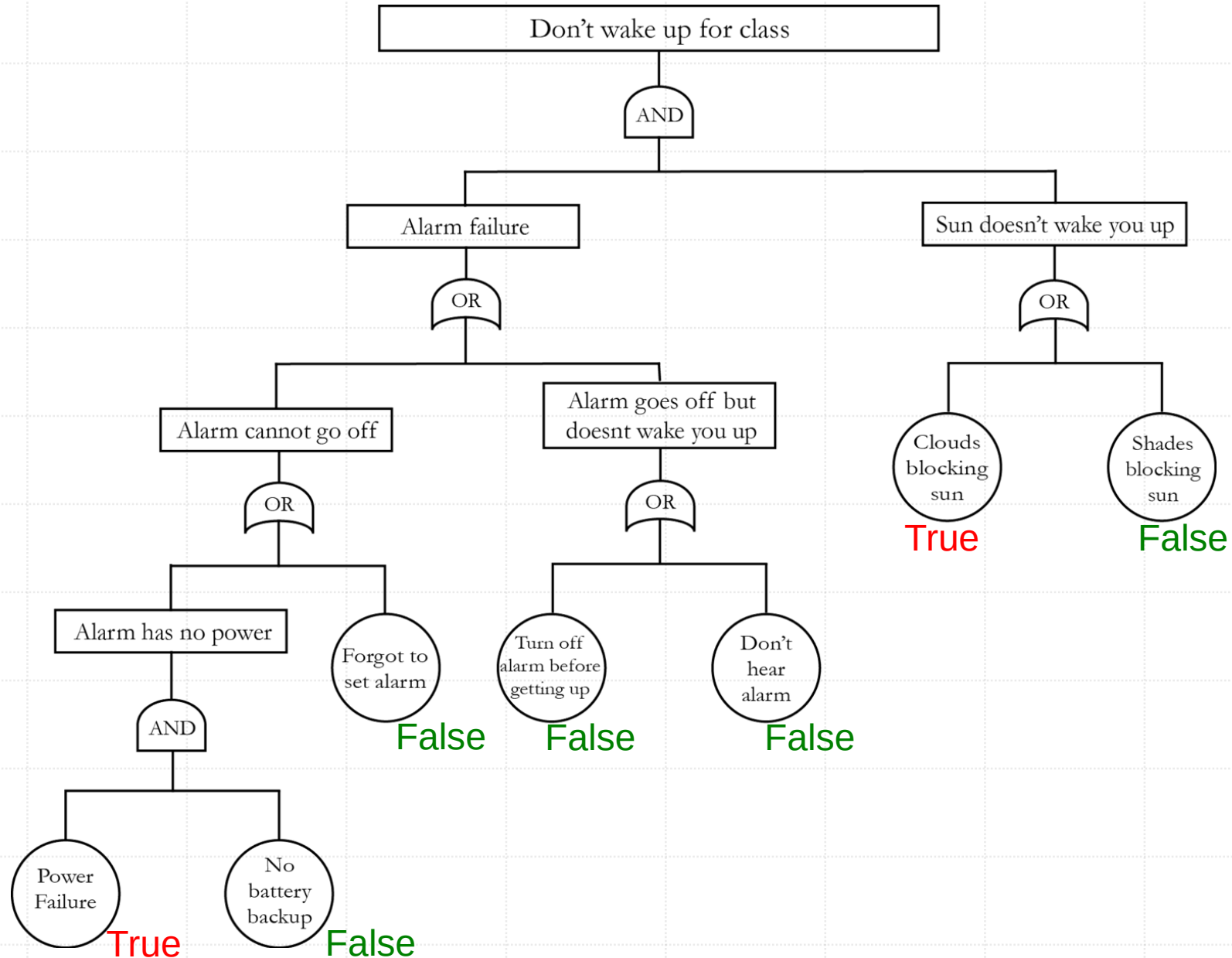
Create a Fault Tree

- Top level event: Don't wake up for class

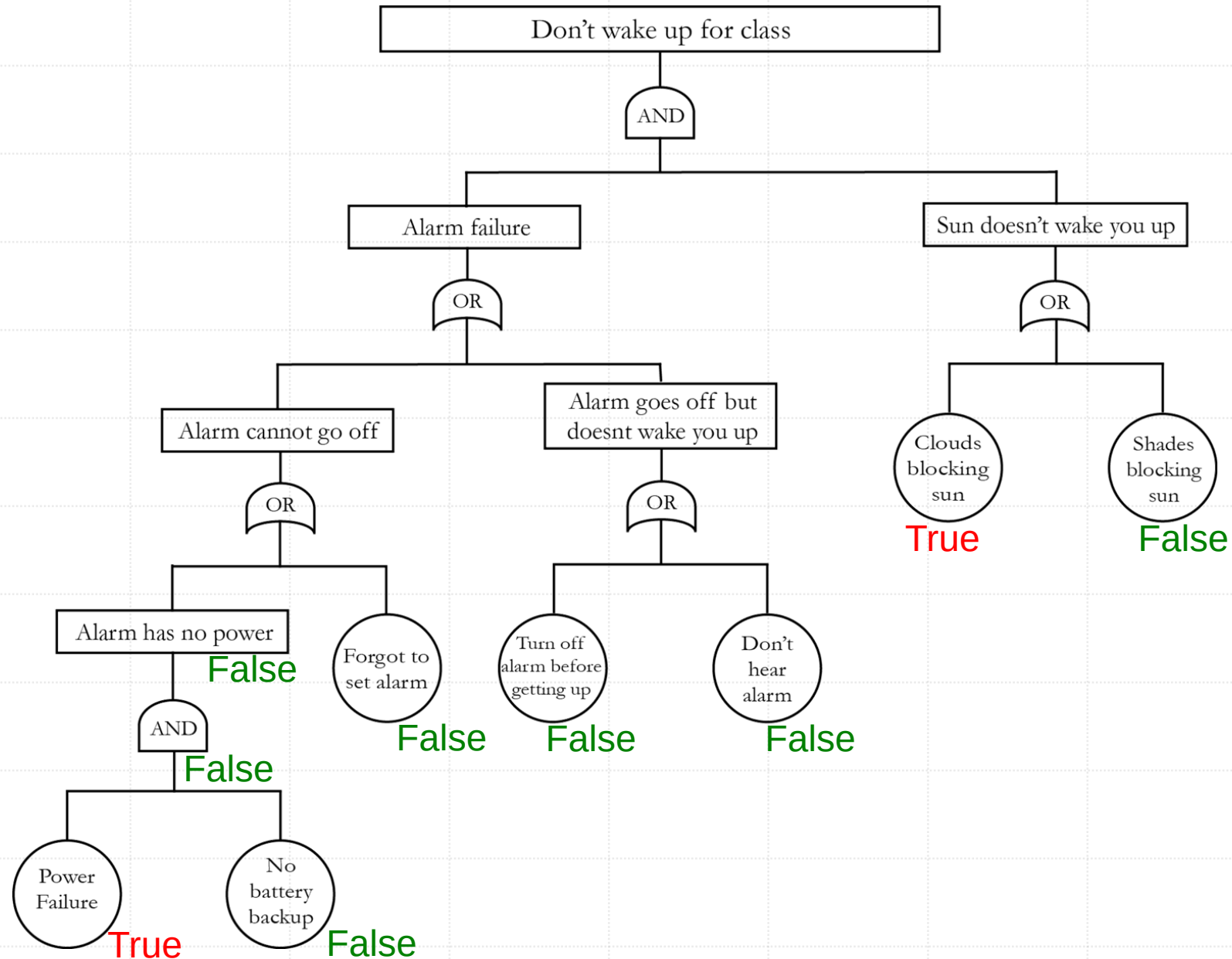
Deterministic Example



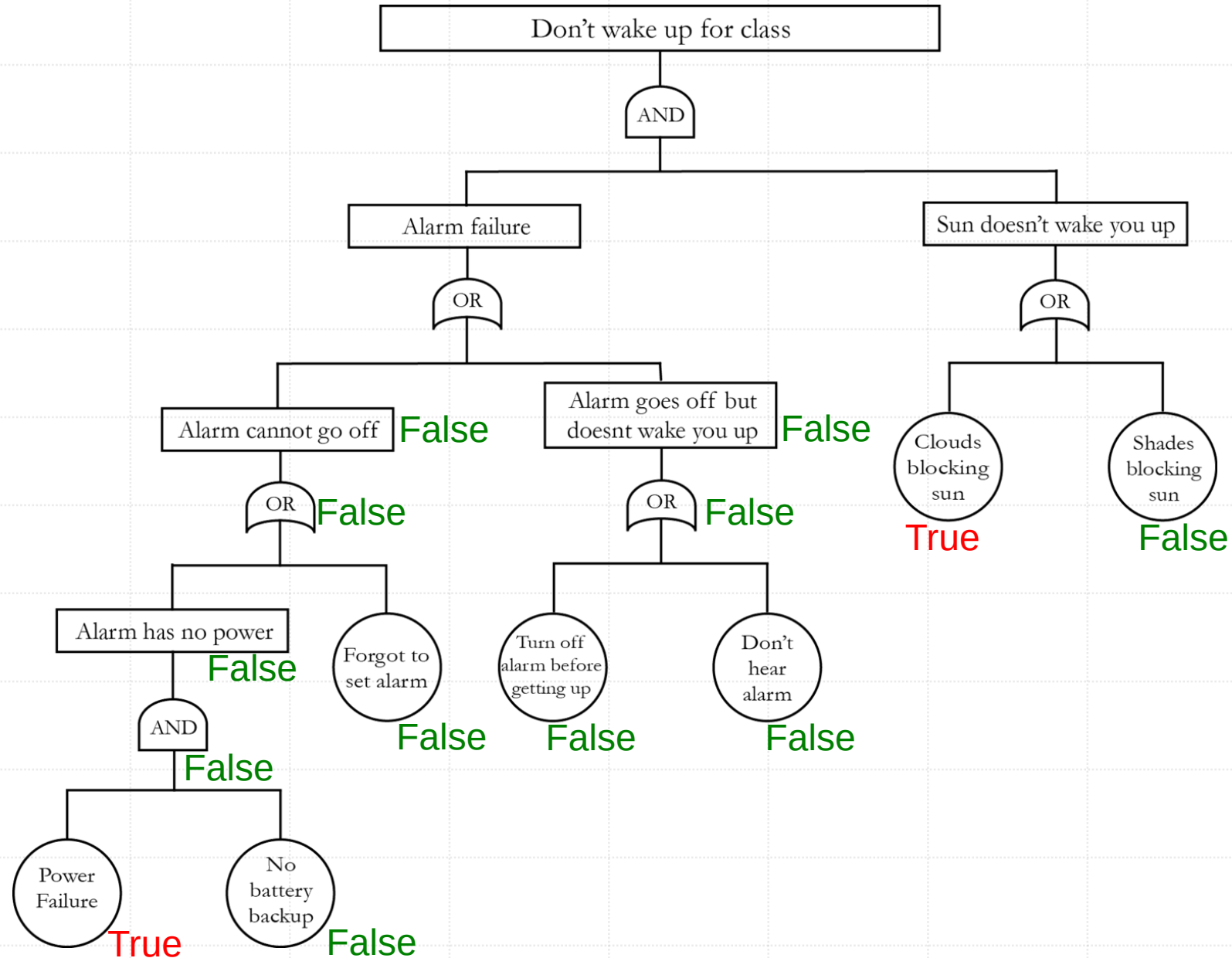
Deterministic Example



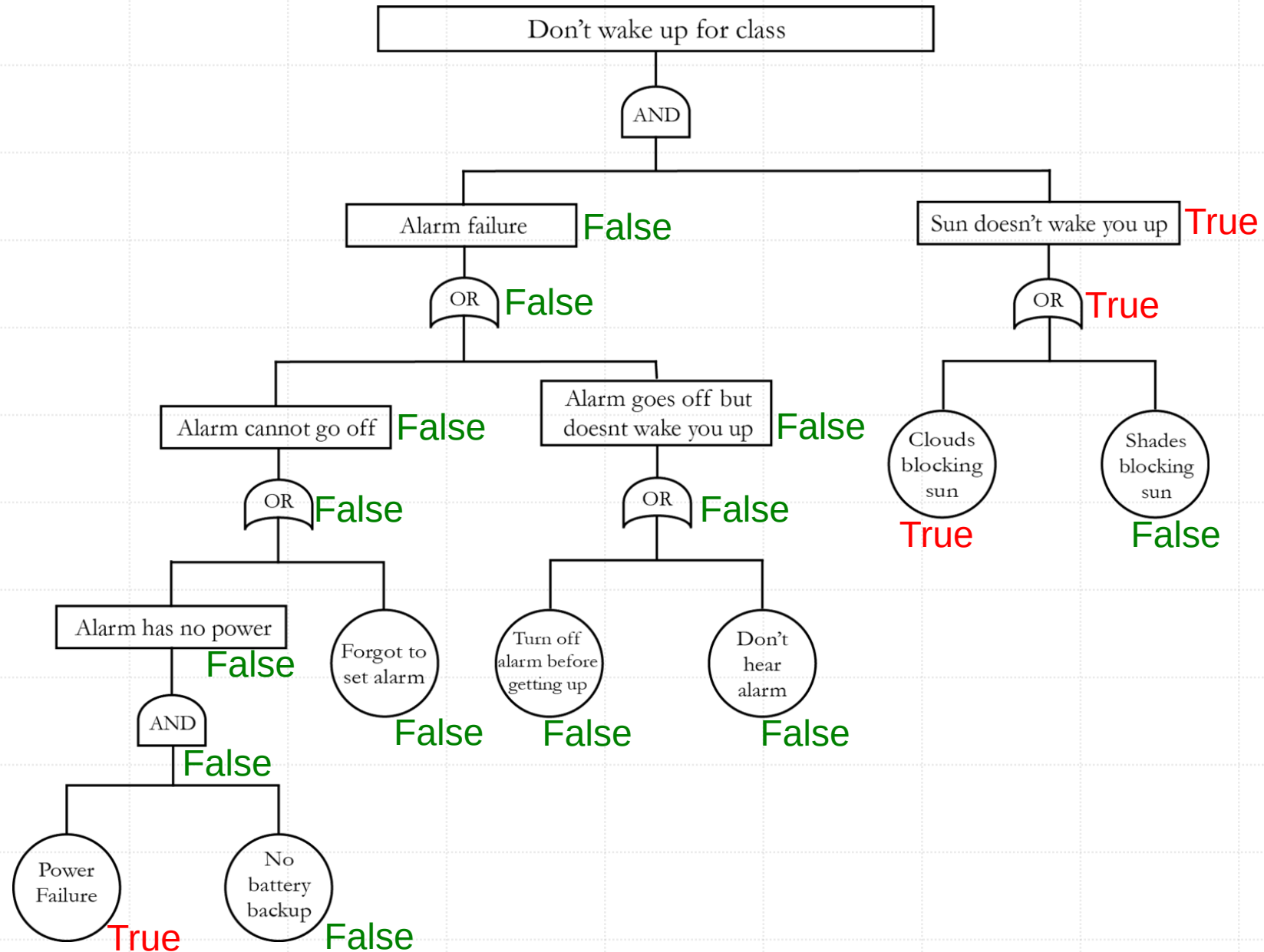
Deterministic Example



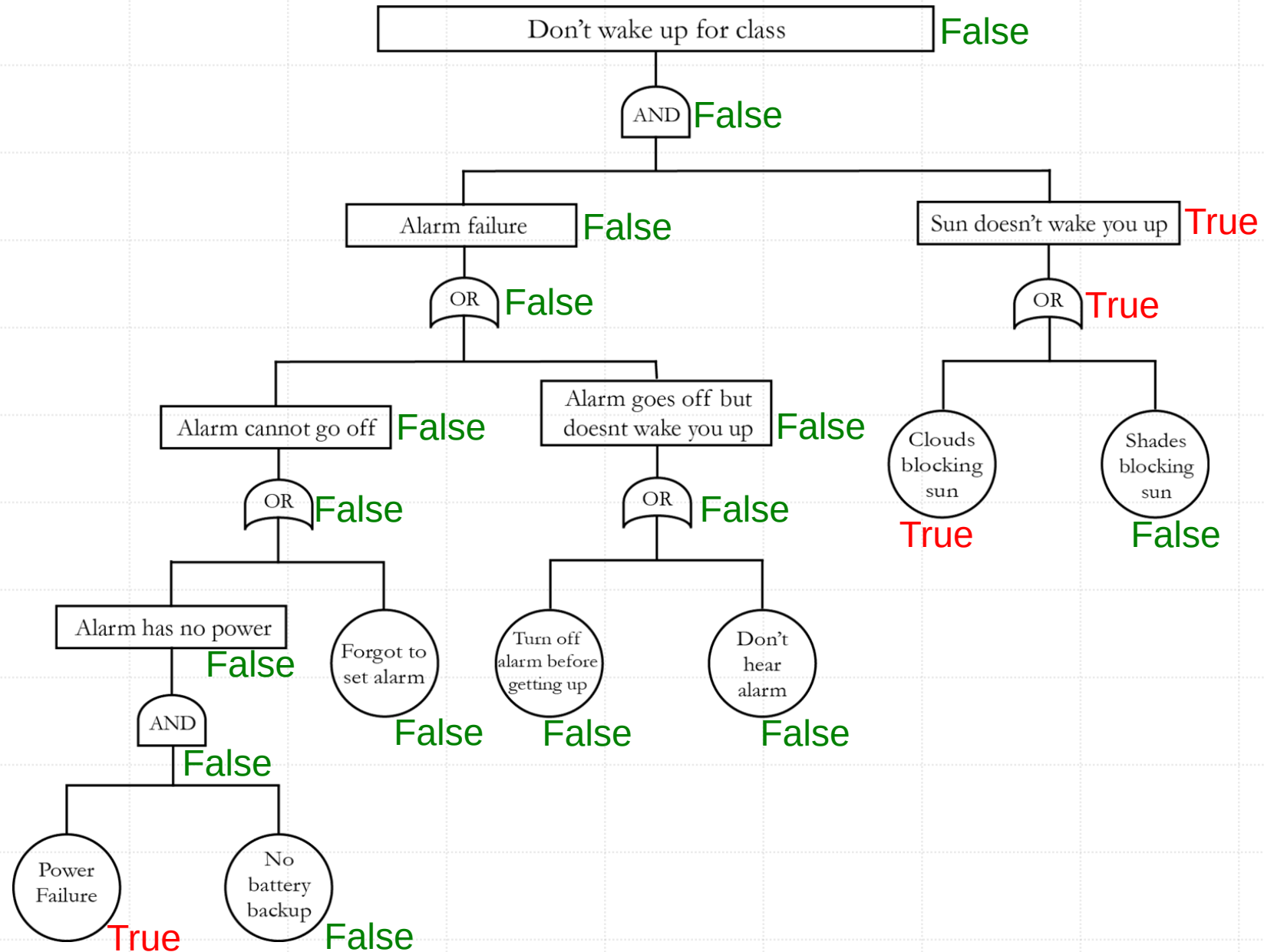
Deterministic Example



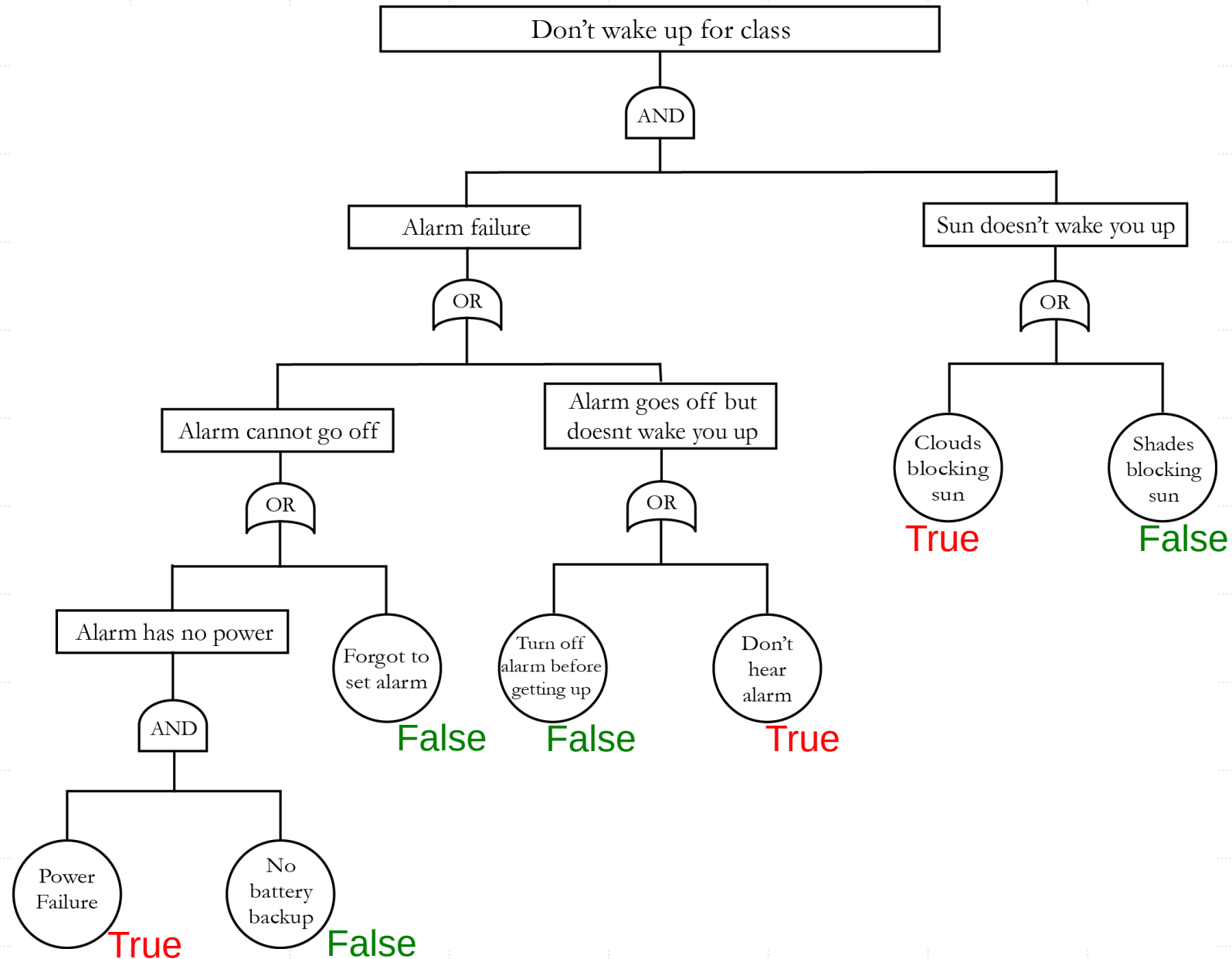
Deterministic Example



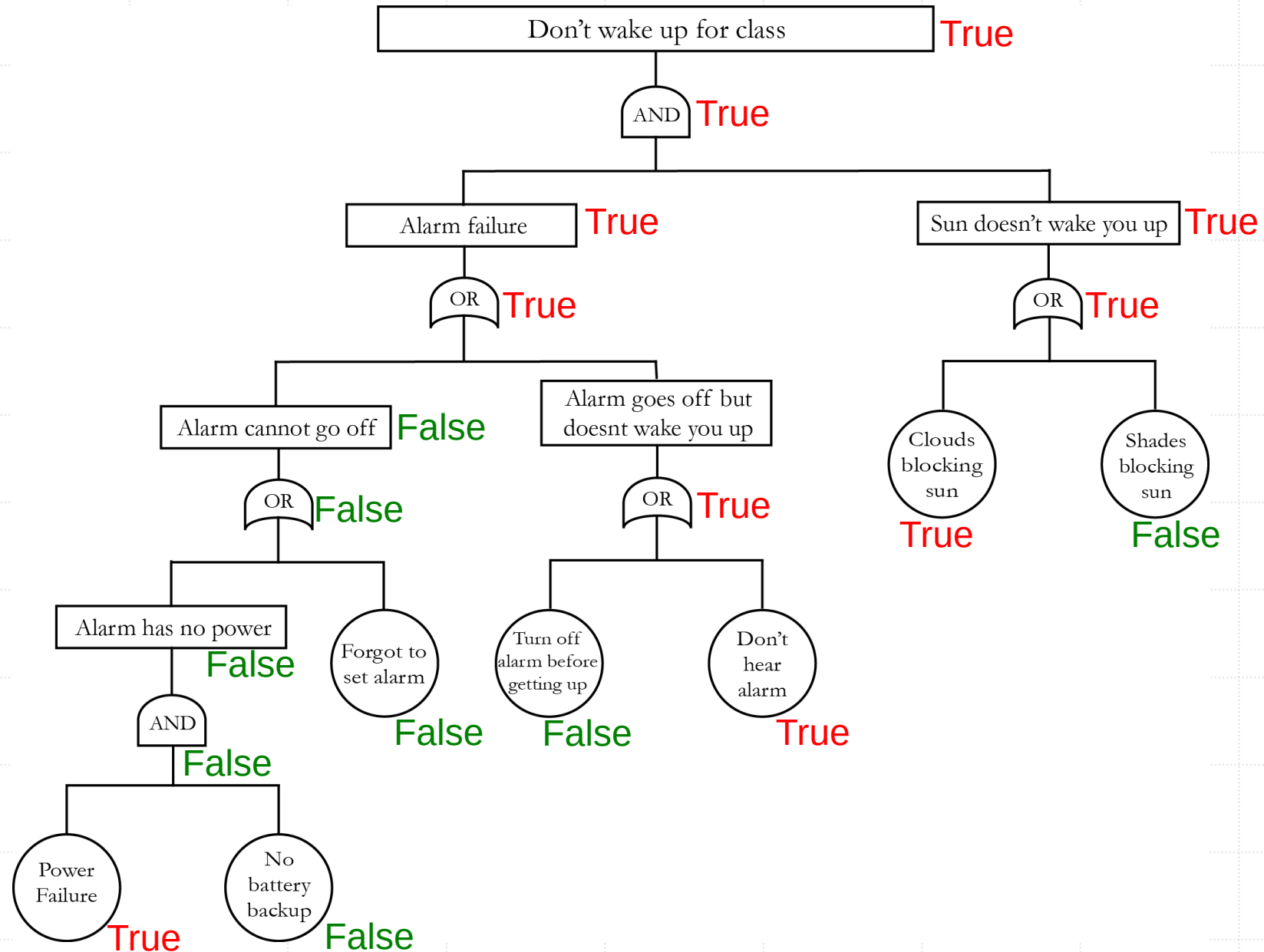
Deterministic Example



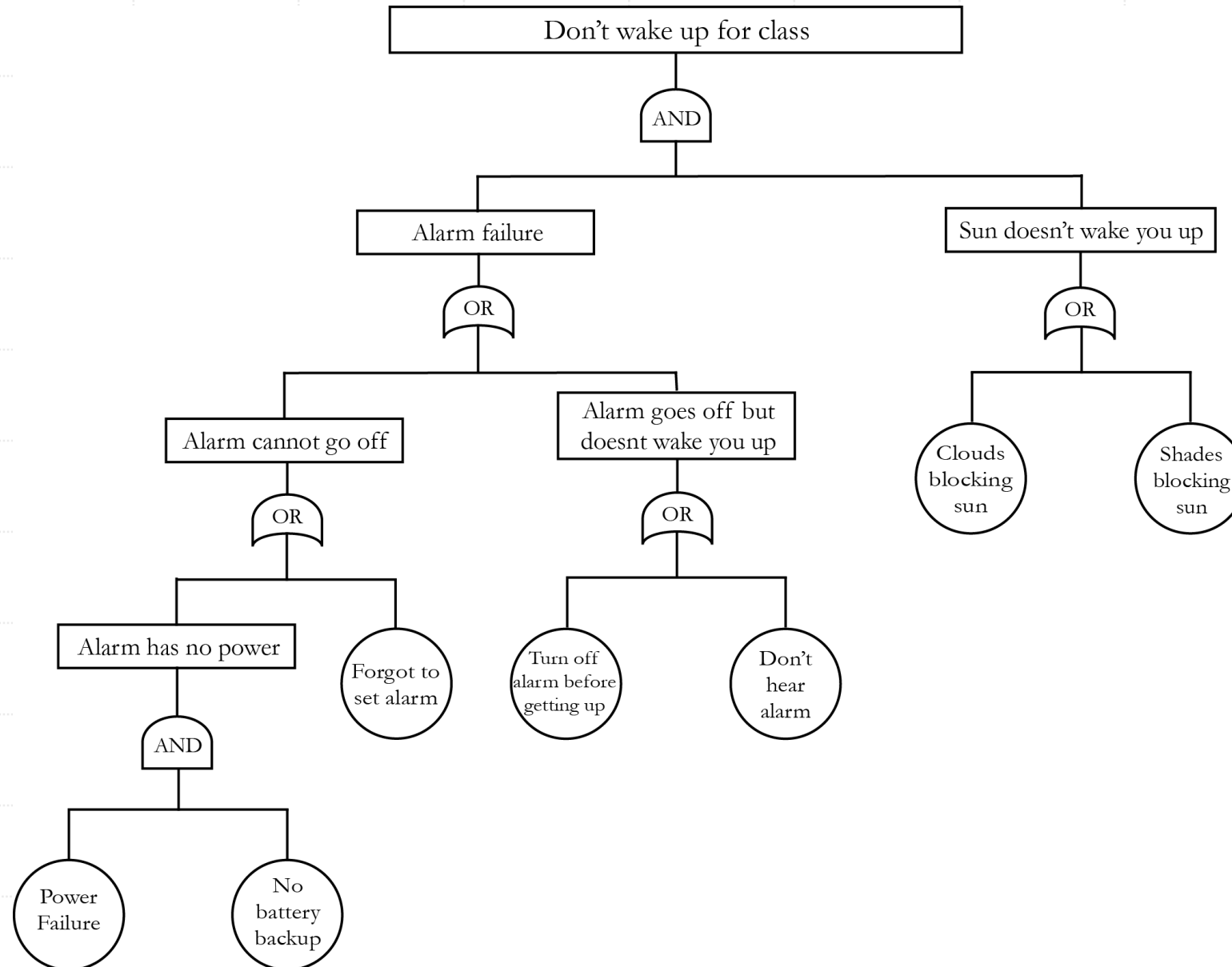
Deterministic Example



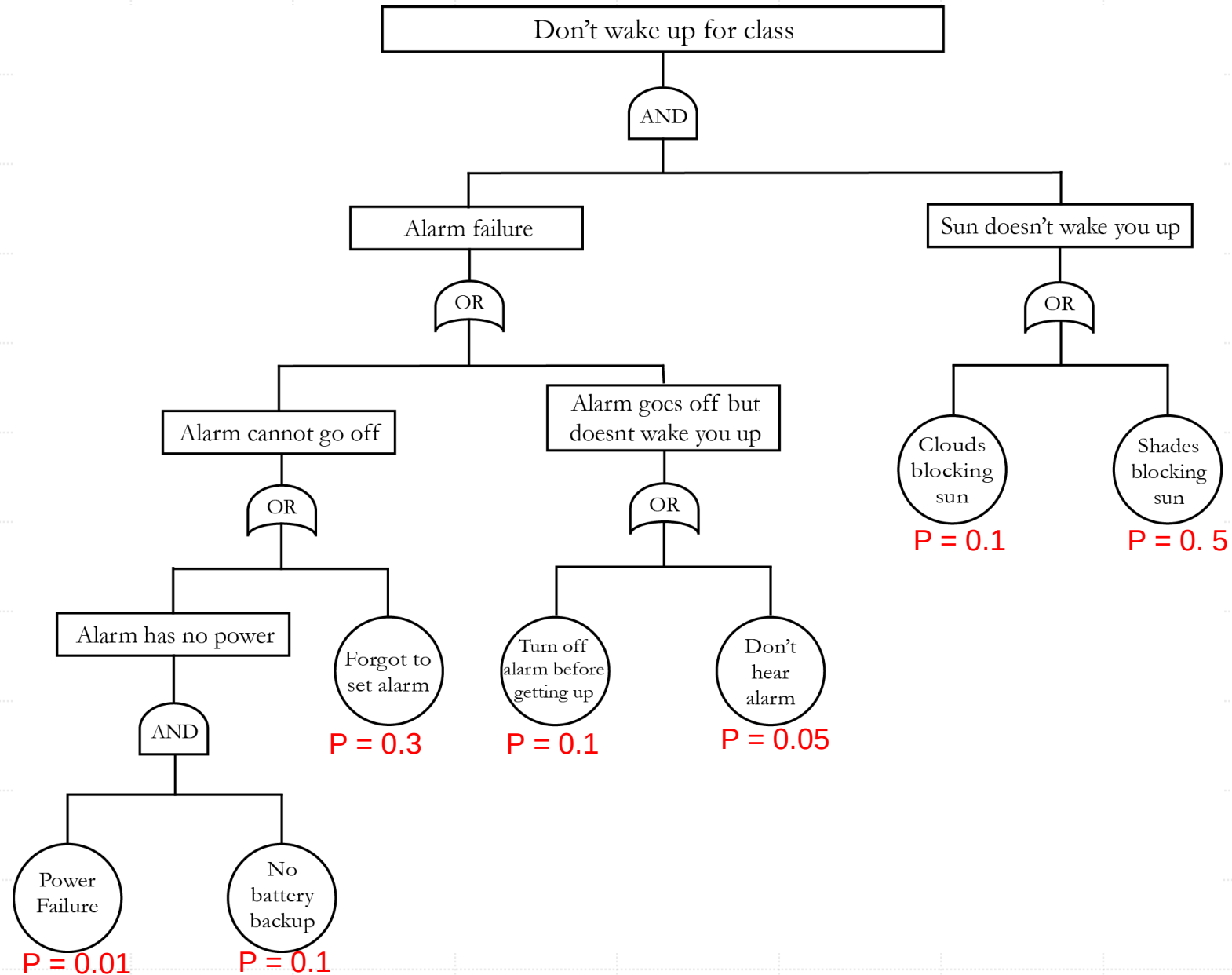
Deterministic Example



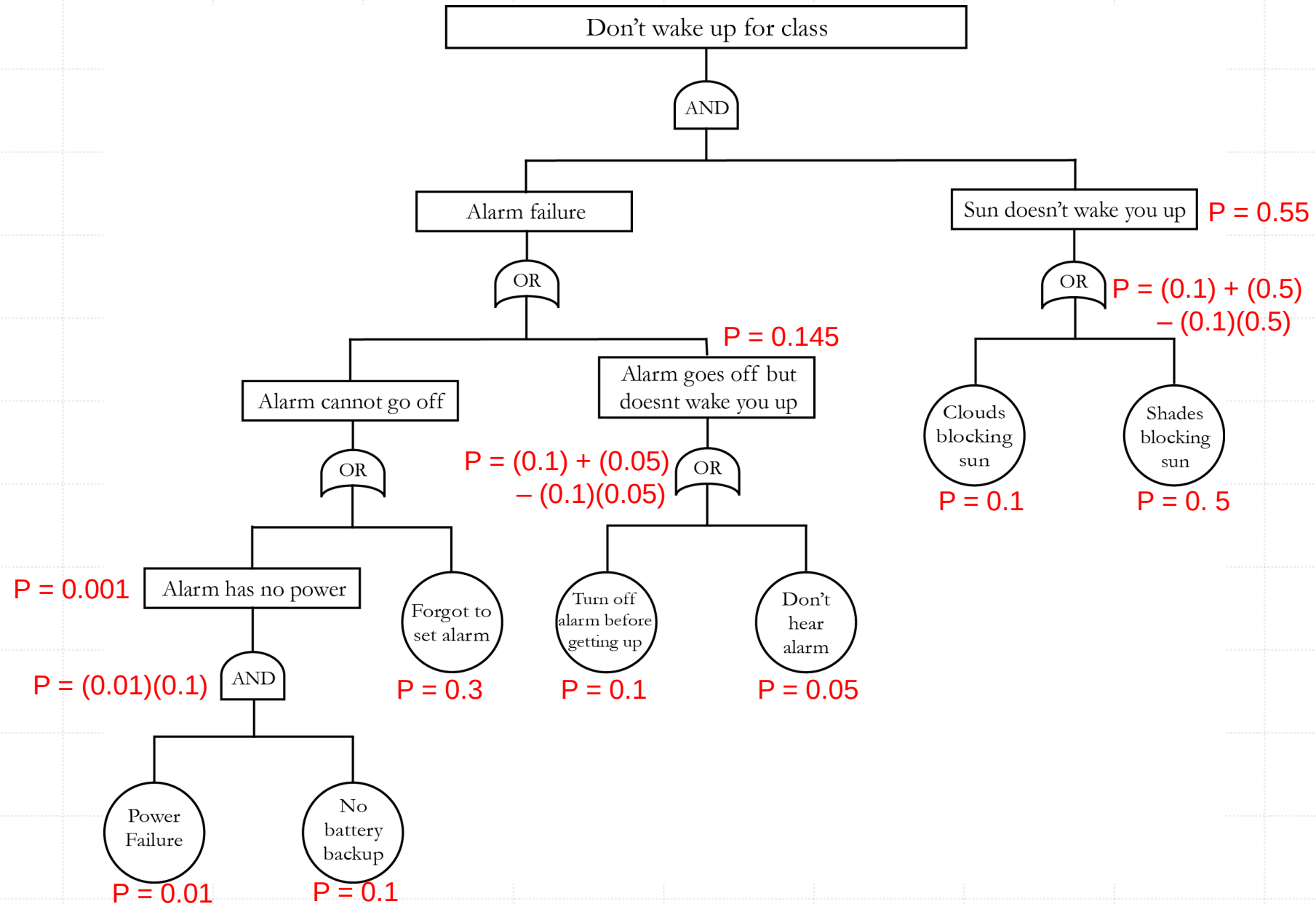
Probabilistic Example



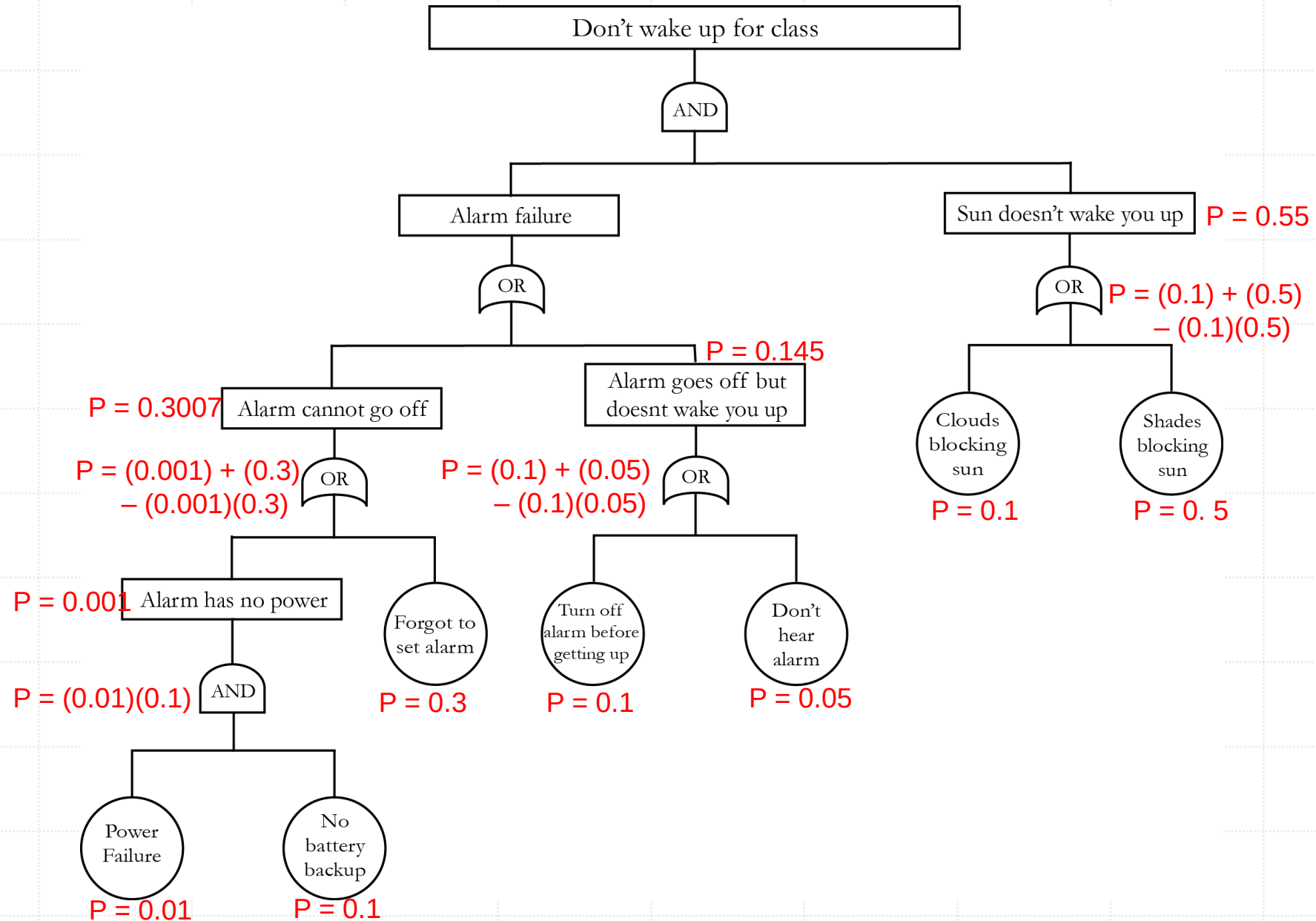
Probabilistic Example



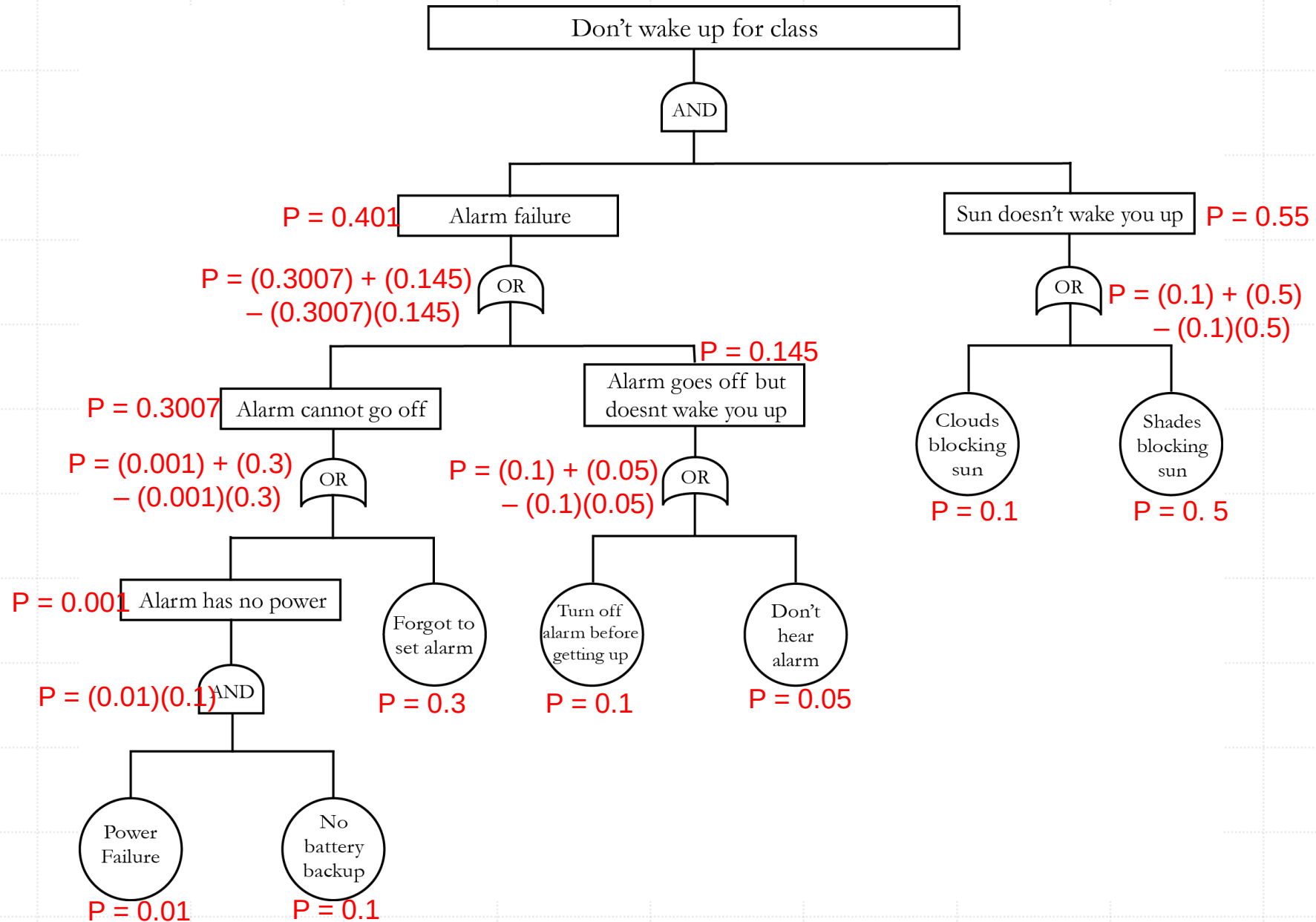
Probabilistic Example



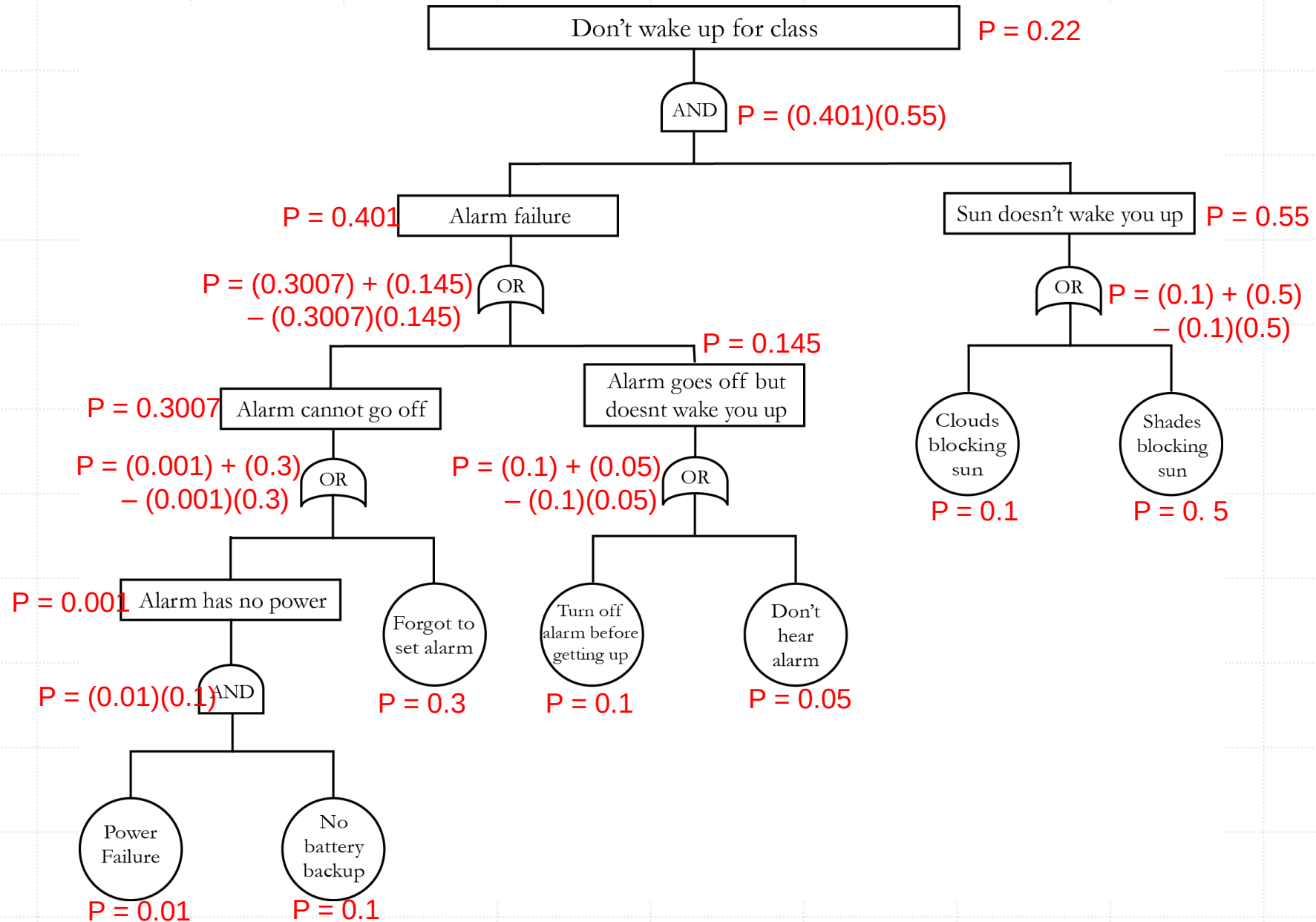
Probabilistic Example



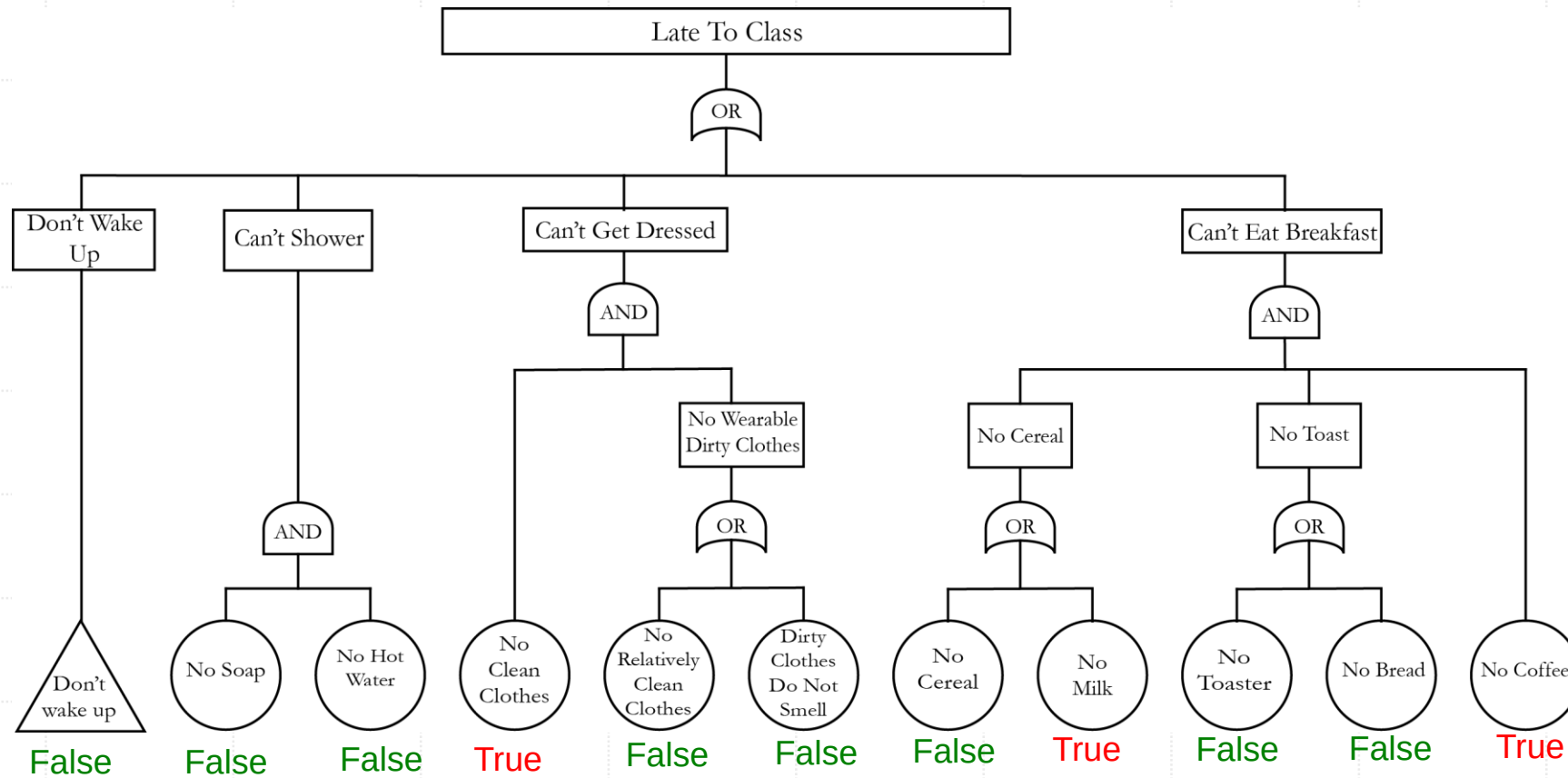
Probabilistic Example



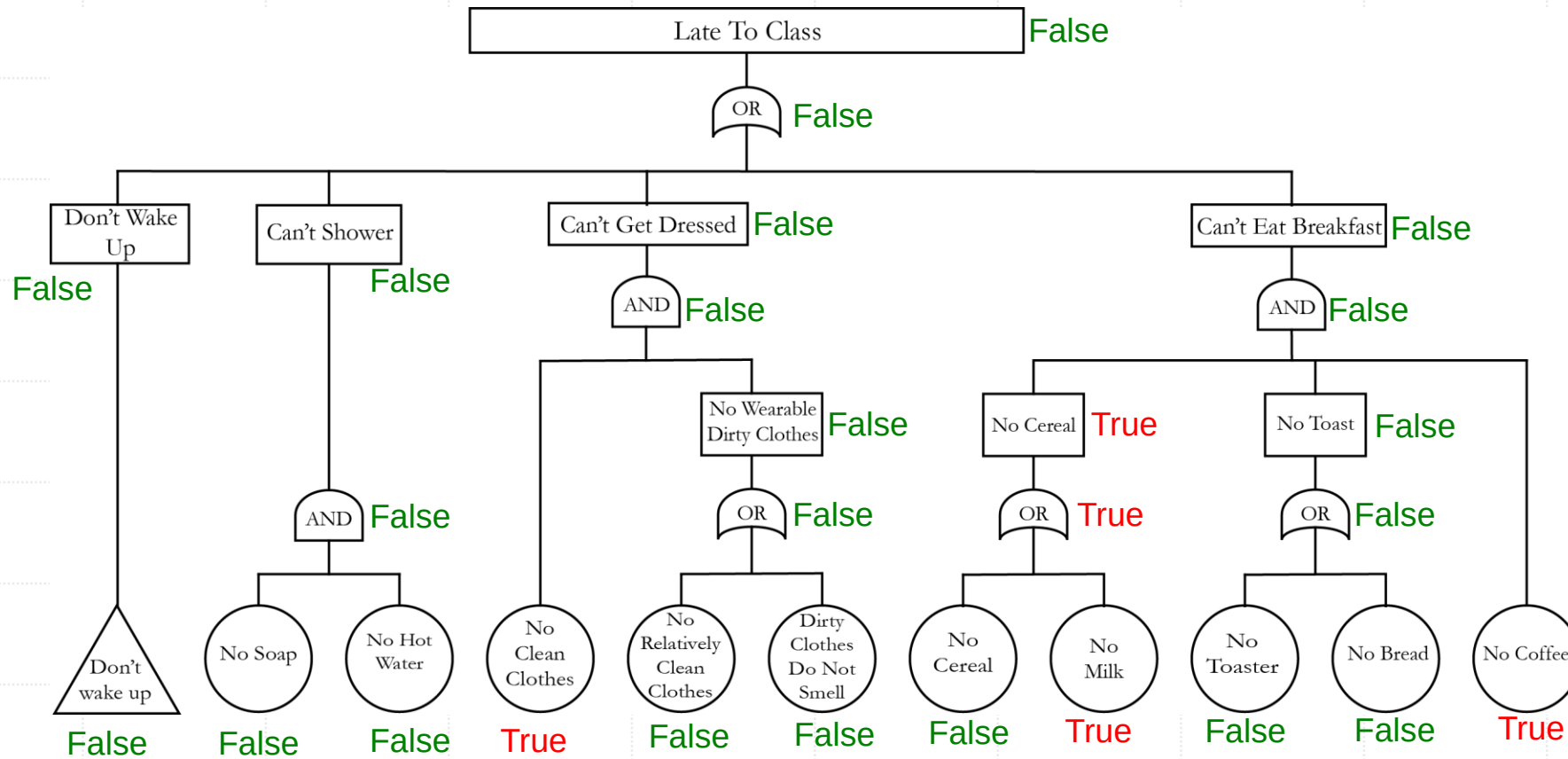
Probabilistic Example

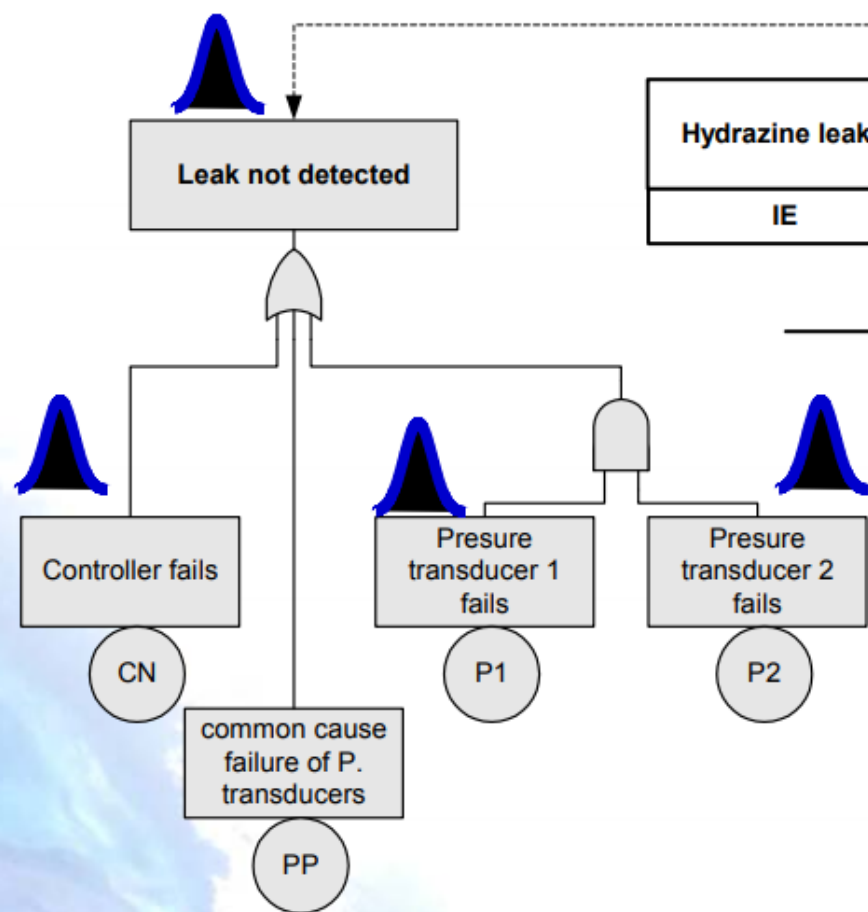


In Class Project



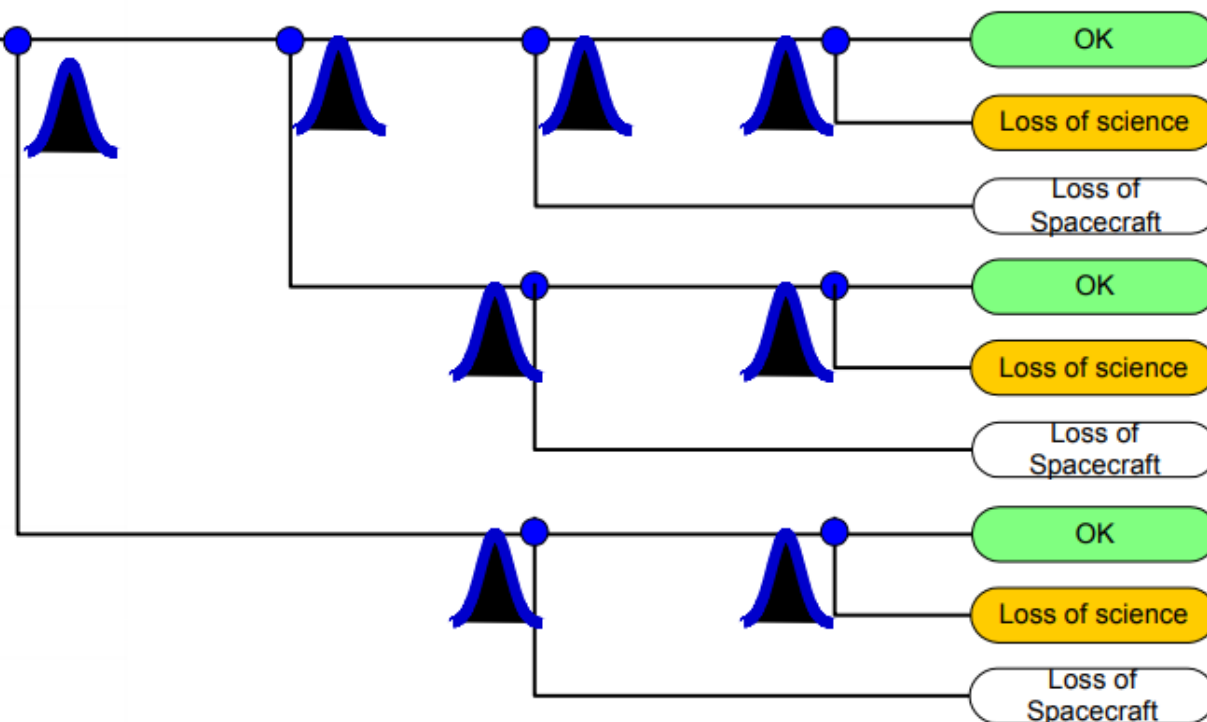
In Class Project Solution





Fault tree

Hydrazine leaks	Leak detected	Leak isolated	No damage to flight critical avionics	No damage to scientific equipment	End state
IE	LD	LI	A	S	



Event tree

■ Failure Modes and Effects

Risk Analysis Methods

Failure Modes and Effects

Process Step	Potential Failure Mode	Potential Failure Effect	SEV ¹	Potential Causes	OCC ²	Current Process Controls	DET ³	RPN ⁴	Action Recommended
What is the step?	In what ways can the step go wrong?	What is the impact on the customer if the failure mode is not prevented or corrected?	How severe is the effect on the customer?	What causes the step to go wrong (i.e., how could the failure mode occur)?	How frequently is the cause likely to occur?	What are the existing controls that either prevent the failure mode from occurring or detect it should it occur?	How probable is detection of the failure mode or its cause?	Risk priority number calculated as SEV x OCC x DET	What are the actions for reducing the occurrence of the cause or for improving its detection? Provide actions on all high RPNs and on severity ratings of 9 or 10.
ATM Pin Authentication	Unauthorized access	- Unauthorized cash withdrawal - Very dissatisfied customer	8	Lost or stolen ATM card	3	Block ATM card after three failed authentication attempts	3	72	
	Authentication failure	Annoyed customer	3	Network failure	5	Install load balancer to distribute work-load across network links	5	75	
Dispense Cash	Cash not disbursed	Dissatisfied customer	7	ATM out of cash	7	Internal alert of low cash in ATM	4	196	Increase minimum cash threshold limit of heavily used ATMs to prevent out-of-cash instances
	Account debited but no cash disbursed	Very dissatisfied customer	8	- Transaction failure - Network issue	3	Install load balancer to distribute work-load across network links	4	96	
	Extra cash dispensed	Bank loses money	8	- Bills stuck to each other - Bills stacked incorrectly	2	Verification while loading cash in ATM	3	48	

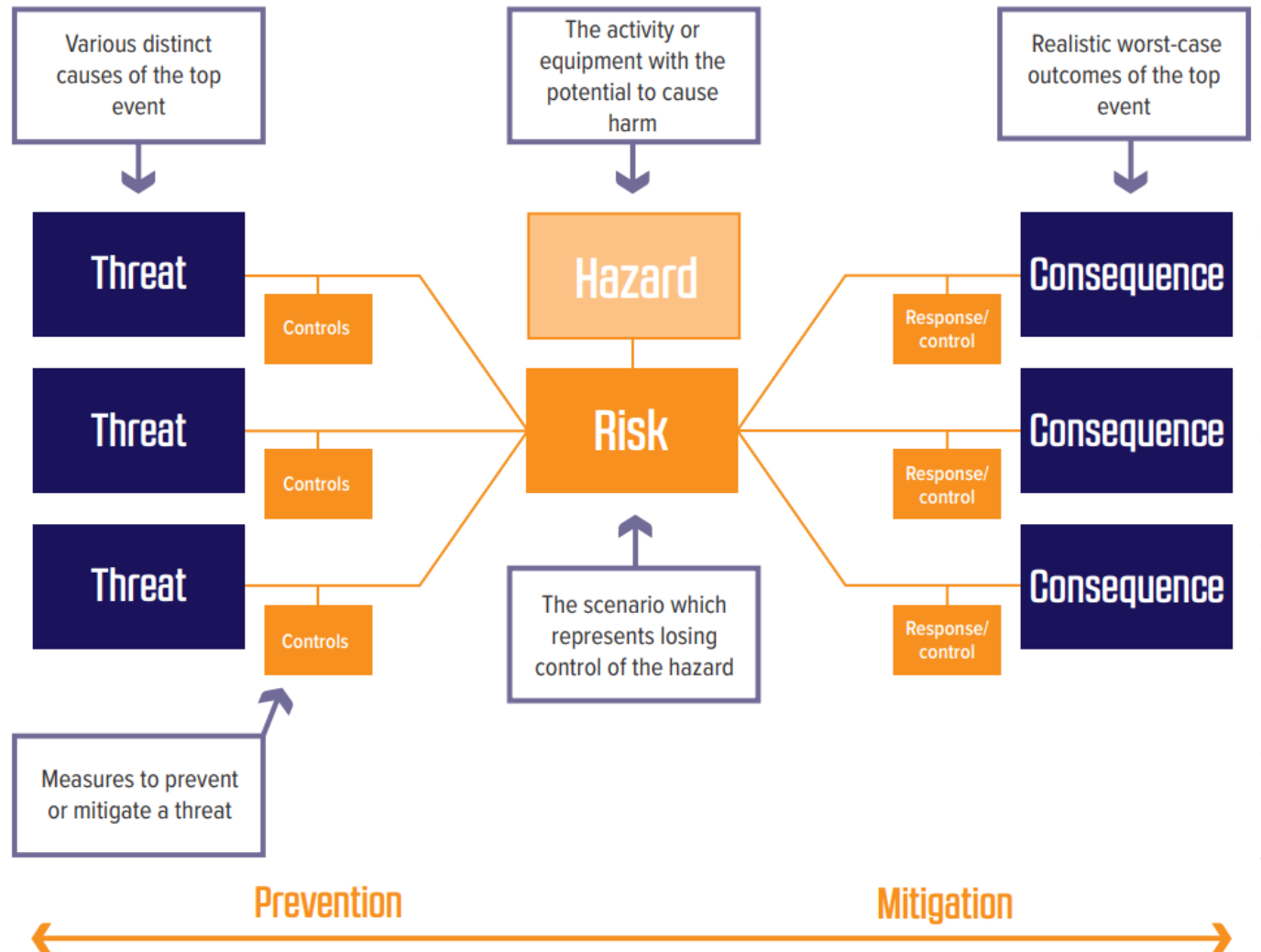
1. **Severity:** Severity of impact of failure event. It is scored on a scale of 1 to 10. A high score is assigned to high-impact events while a low score is assigned to low-impact events.
 2. **Occurrence:** Frequency of occurrence of failure event. It is scored on a scale of 1 to 10. A high score is assigned to frequently occurring events while events with low occurrence are assigned a low score.
 3. **Detection:** Ability of process control to detect the occurrence of failure events. It is scored on a scale of 1 to 10. A failure event that can be easily detected by the process control is assigned a low score while a high score is assigned to an inconspicuous event.
 4. **Risk priority number:** The overall risk score of an event. It is calculated by multiplying the scores for severity, occurrence and detection. An event with a high RPN demands immediate attention while events with lower RPNs are less risky.

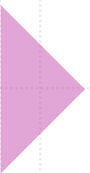
Risk Analysis M

- Bow Tie Analysis

Canterbury Flood - Hazard

BOW TIE ANALYSIS





Making decisions

Risk Matrix

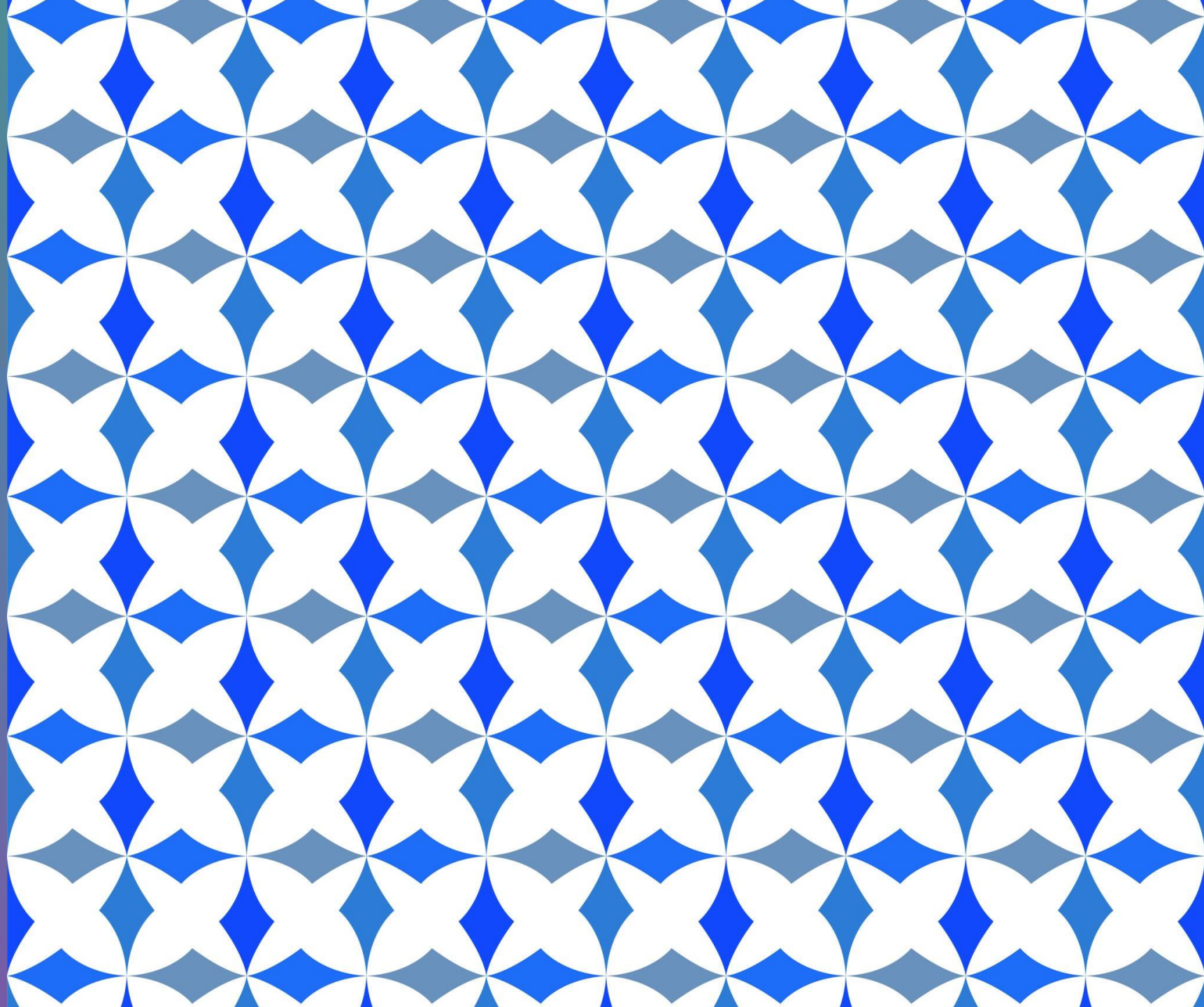
RISK ASSESSMENT MATRIX

		CONSIDER THE LIKELIHOOD OF A HAZARDOUS EVENT OCCURRING				
RISK ASSESSMENT MATRIX		Very unlikely to happen	Unlikely to happen	Possibly could happen	Likely to happen	Very likely to happen
	Catastrophic (e.g fatal)	Moderate	Moderate	High	Critical	Critical
	Major (e.g Permanent Disability)	Low	Moderate	Moderate	High	Critical
	Moderate (e.g Hospitalisation/Short or Long Term Disability)	Low	Moderate	Moderate	Moderate	High
	Minor (e.g First Aid)	Very Low	Low	Moderate	Moderate	Moderate
	Superficial (e.g No Treatment Required)	Very Low	Very Low	Low	Low	Moderate

ENGCV370

DR MEGAN BOSTON

RISK,
HAZARD, AND
RESILIENCE
ASSESSMENTS



LEARNING OBJECTIVES

OUTLINE

Definitions

Hazard Identification
& Vulnerability
Assessment
(Wednesday)

Risk Analysis and
Tools (Thursday)

Finding Sources
(Friday)

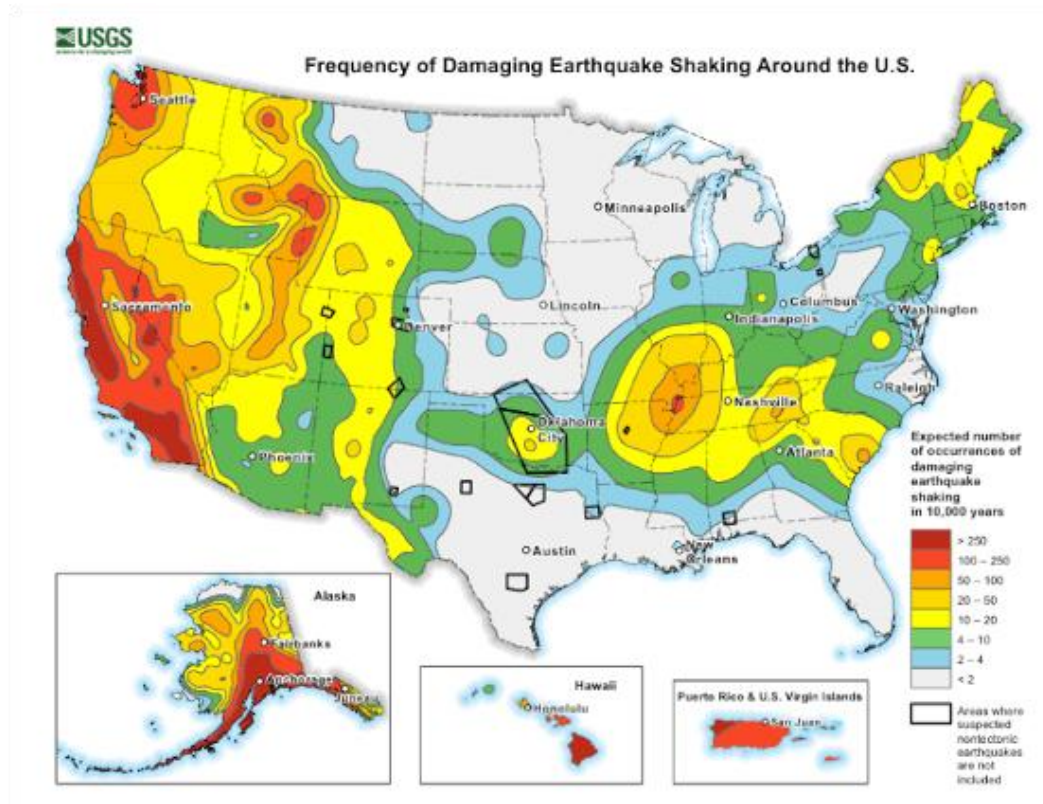
Resilience
(Wednesday)

Applying Resilience
(Thursday)

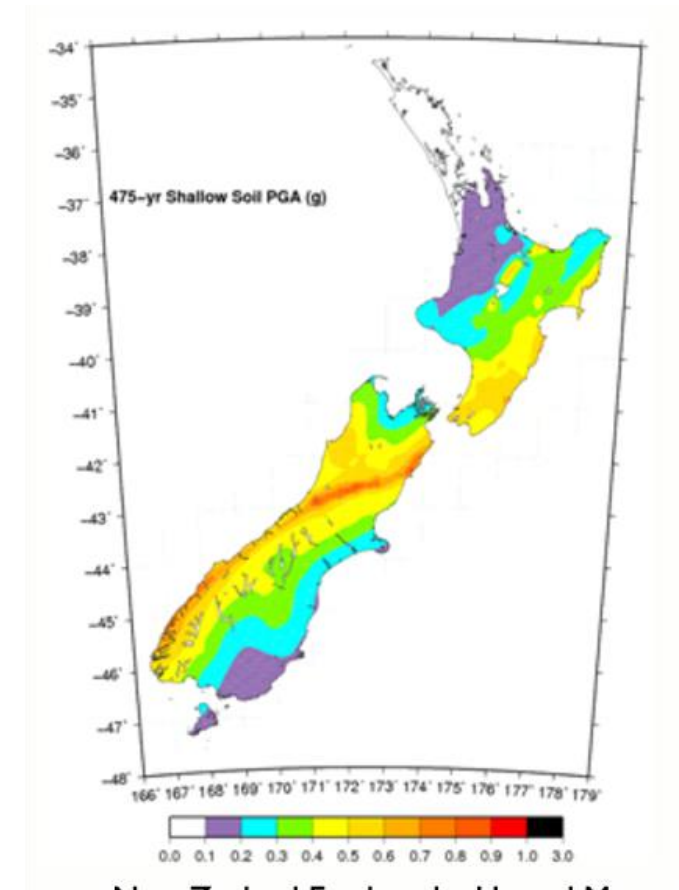
Resilience
Applications, Cultural
and Social (Friday)

Report help (Study
Break)

Hazard – A threat from a natural or manmade source that has the potential to cause property damage, loss of life, injury, social or economic disruption, or degrade the environment.



USGS Earthquake Hazard Map.
Source: USGS



New Zealand Earthquake Hazard Map.
Source: GNS

RISK

```
graph TD; RISK[RISK] --- Q1[What can go wrong? (scenarios combining hazards and outcomes)]; RISK --- Q2[How likely is it to go wrong? (probability of the scenario)]; RISK --- Q3[What are the consequences if it goes wrong?];
```

What can go wrong? (scenarios combining hazards and outcomes)

How likely is it to go wrong? (probability of the scenario)

What are the consequences if it goes wrong?

HAZARD

is the potential
to cause harm



when crossing a road,
cars are a **hazard**

RISK

is the likelihood of
harm taking place



when crossing
a highway, the risk of an
accident is **high**



when crossing
a country road, the risk of
an accident is **low**

HAZARD

in foods can be...



physical
for instance
pieces of bones
in fish products



biological
for instance
harmful bacteria,
viruses or parasites



chemical
for instance mercury
in fish or acrylamide
in starchy food

RISK

is determined by
the **exposure...**



how much

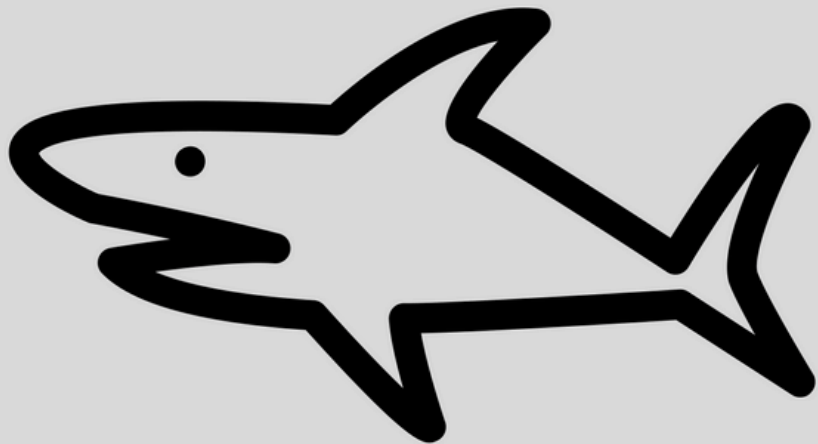
how long

how often

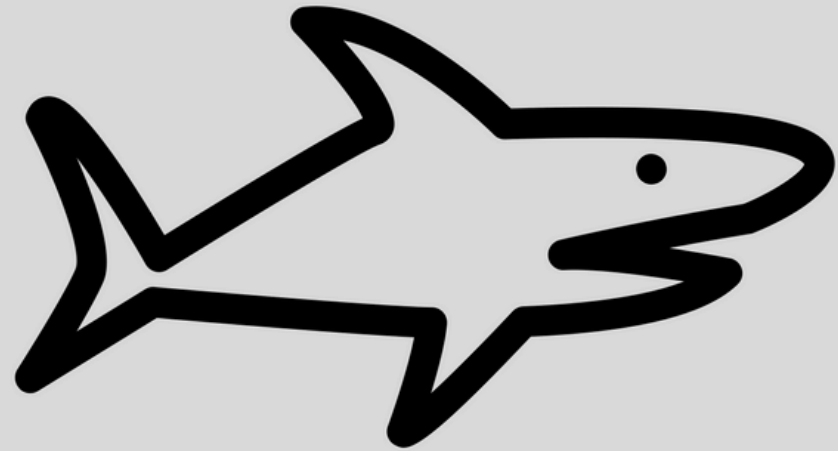
...to a hazard

Hazard

Something that can potentially cause harm



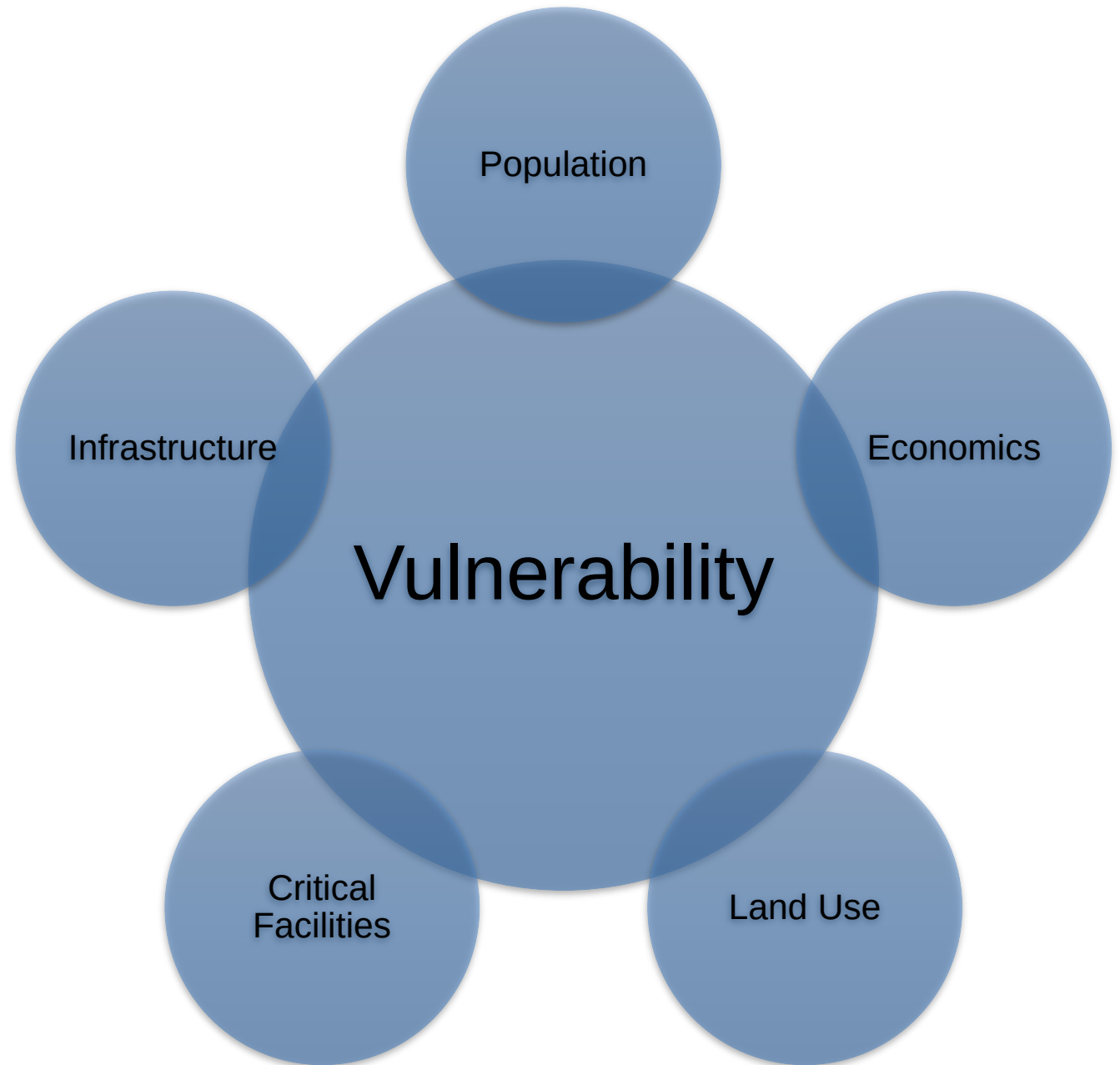
Risk



= hazard + exposure

VULNERABILITY

The diminished capacity of an individual or group to anticipate, cope with, resist and recover from the impact of a natural or man-made hazard



DISASTER

A serious disruption of the functioning of a community or a society at any scale due to hazardous events interacting with conditions of exposure, vulnerability and capacity, leading to one or more of the following: human, material, economic and environmental losses and impacts



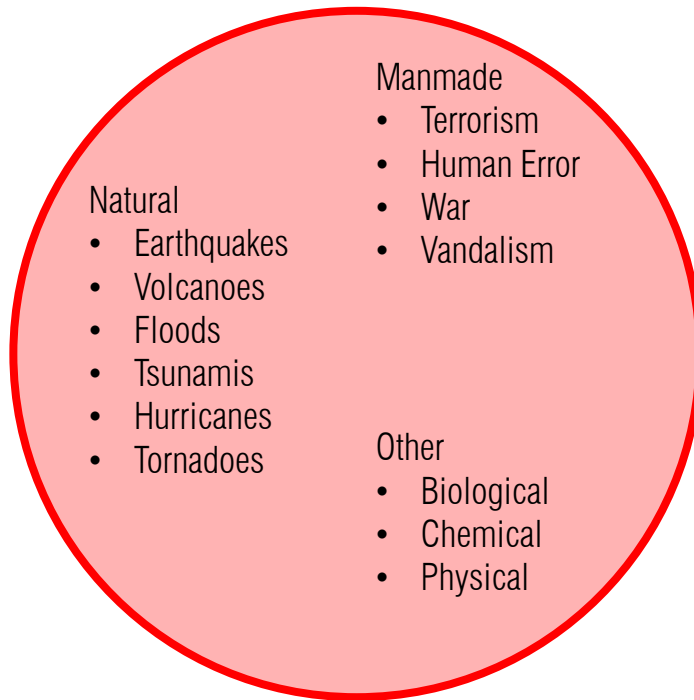
Downtown Christchurch after Feb 2011 Earthquake. Source: Wikimedia Commons

RESILIENCE

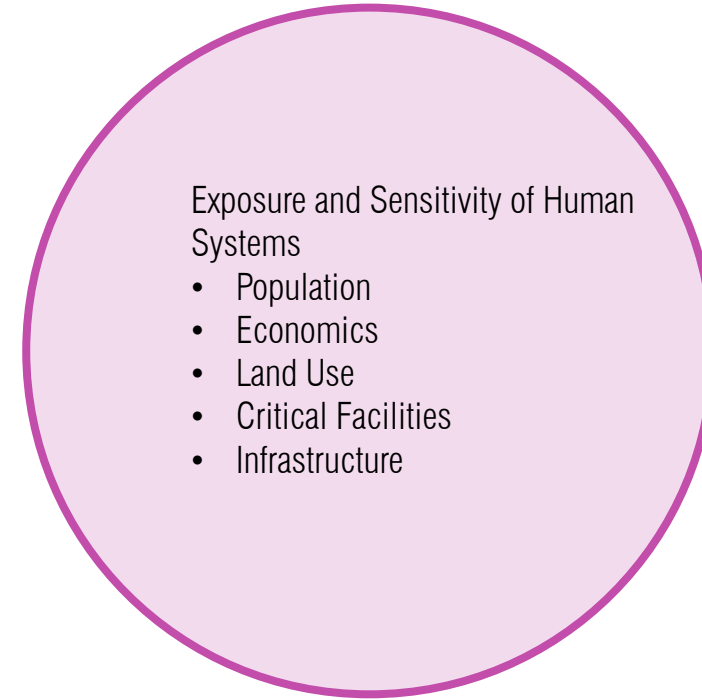
The capacity to absorb or mitigate the impact of hazard events while maintaining and restoring essential services

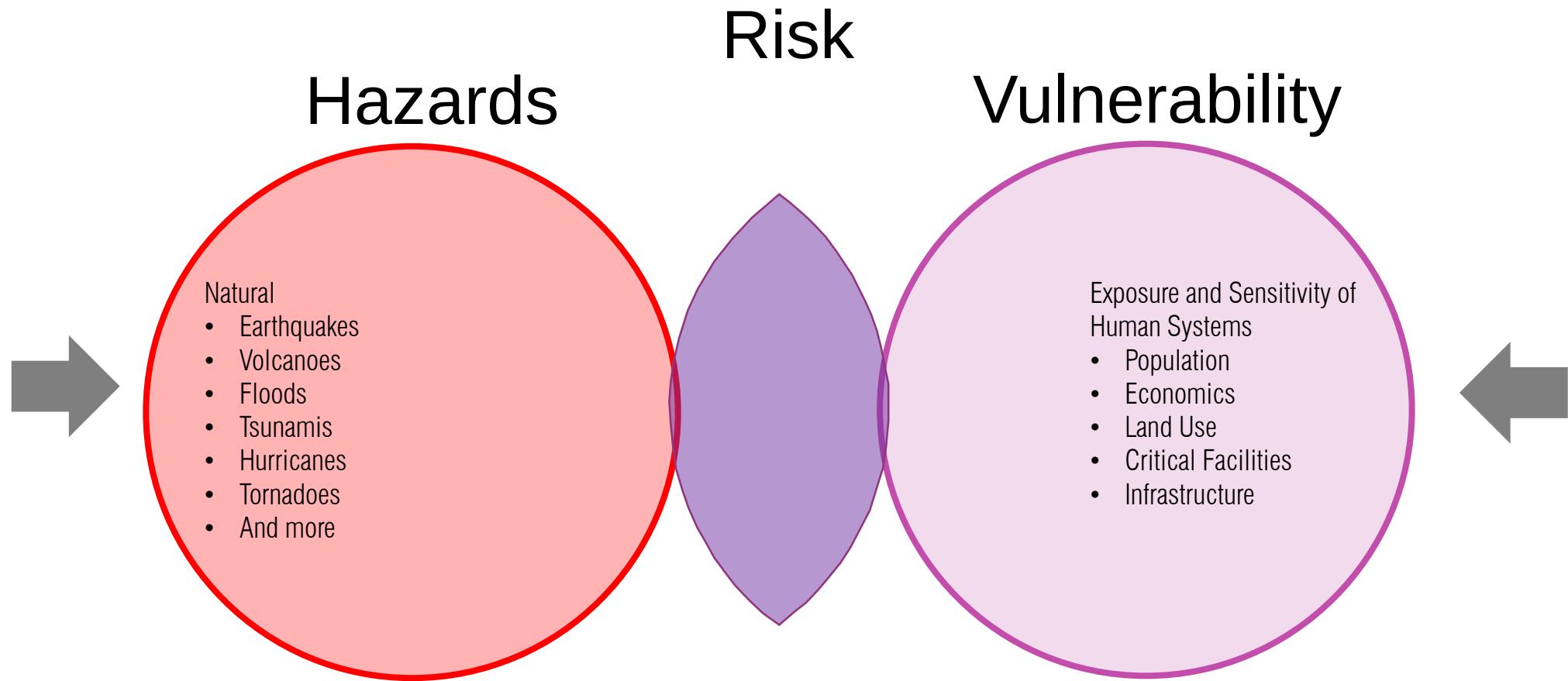


Hazards



Vulnerability





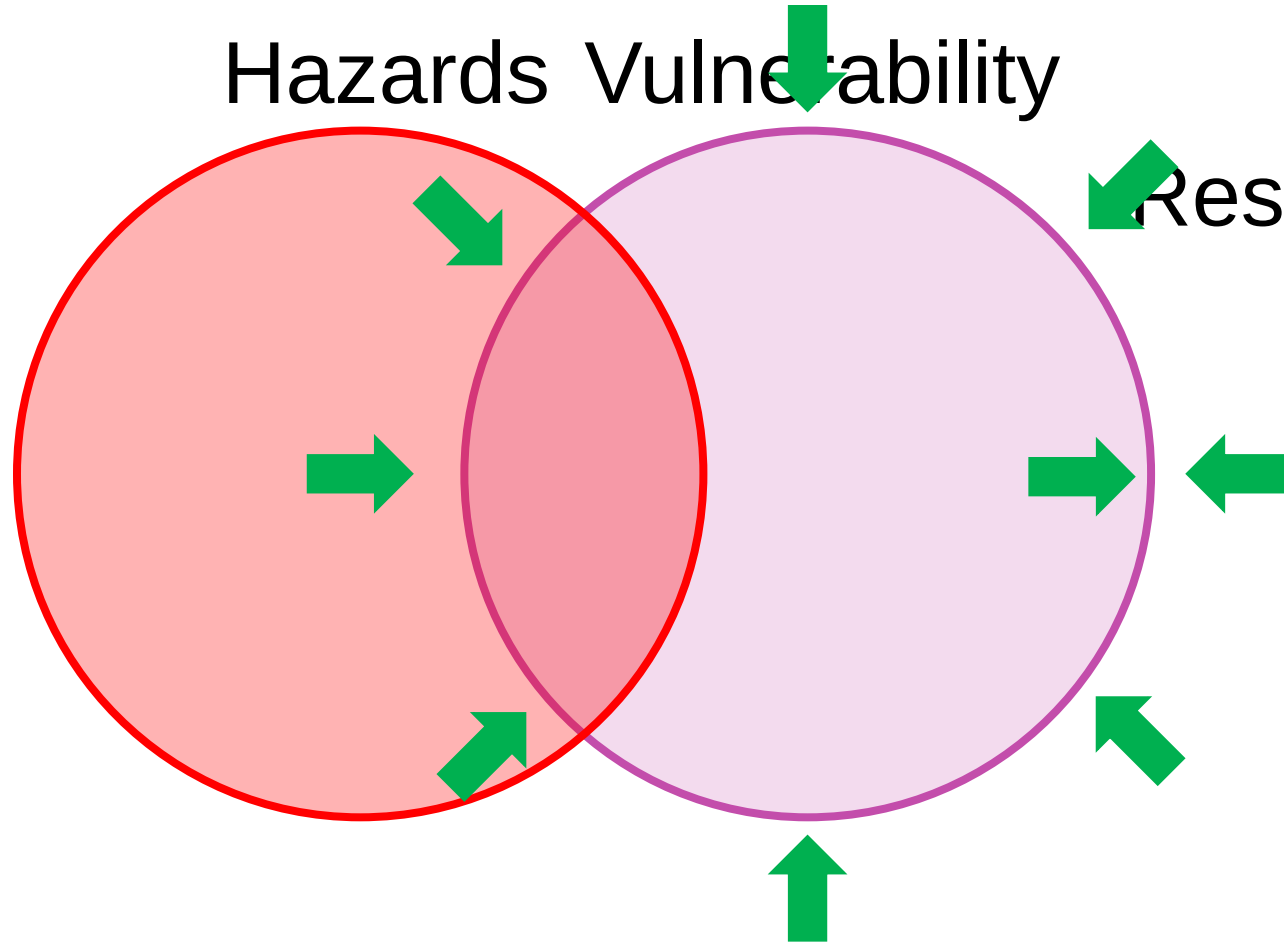


Volcanic eruption that is not near population centers or human infrastructure will have little to no impact on a community



Hazards Vulnerability

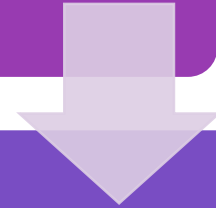
Resilience



RISK ASSESSMENT

HAZARD AND RISK ASSESSMENT

Methodical assessment of
the potential hazards



Determining potential
impacts from hazards



Determine the
vulnerabilities that are
making things susceptible
to hazards

HAZARD IDENTIFICATION

- What are the potential hazards for an engineering system?

TYPES OF HAZARDS

Natural

Anthropogenic

Technological

TYPES OF HAZARDS

UNDRR

Meteorological and Hydrological

Extraterrestrial

Geohazards

Environmental

Chemical

Biological

Technical

Societal

GH0003 / GEOHAZARDS / Seismogenic (Earthquakes)

Liquefaction (Earthquake Trigger)

Definition

Soil liquefaction occurs when soil is transformed from a solid to a liquid state as a result of increased pore pressure and reduced effective stress. It is typically caused by rapid loading of the soil during earthquake shaking (AGI, 2017).

Reference

AGI, 2017. Liquefaction [soil]. American Geosciences Institute (AGI). www.americangeosciences.org/word/liquefaction-soil Accessed 14 October 2002.

TL0019 / **TECHNOLOGICAL** / Cyber Hazard

Malware

Definition

Malware is a summary term for different forms of malevolent software designed to infiltrate and infect computers, typically without the knowledge of the owner (ITU, 2008).

Reference

ITU, 2008. ITU Study on the Financial Aspects of Network Security: Malware and Spam. International Telecommunication Union (ITU). www.itu.int/ITU-D/cyb/cybersecurity/docs/itu-study-financial-aspects-of-malware-and-spam.pdf Accessed 3 October 2020.

S00008 / **SOCIETAL / Economic**

Financial shock

Definition

A financial shock is an unexpected disturbance which originates from the financial sector and has a significant effect on an economy (e.g. national, regional, or global). The term is largely used to refer to events which have negative impacts (ECB, 2013).

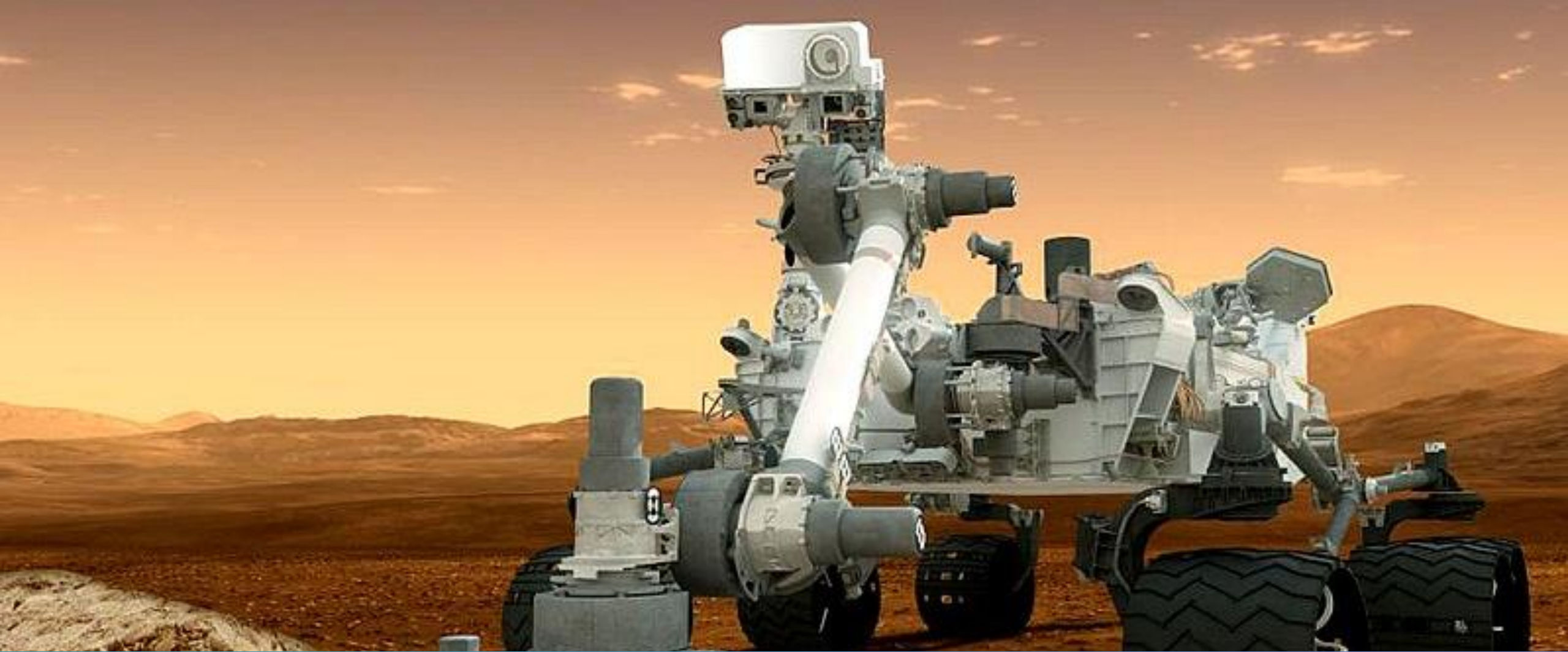
Reference

ECB, 2013. Financial Shocks and the Macroeconomy: Heterogeneity and Non-Linearities. European Central Bank (ECB), Occasional Paper Series No. 143. www.esm.europa.eu/sites/default/files/financialshocksandthemacroeconomy.pdf Accessed 12 September 2020.



HAZARD IDENTIFICATION: CASE STUDY

IMAGE: HAMILTON CITY COUNCIL

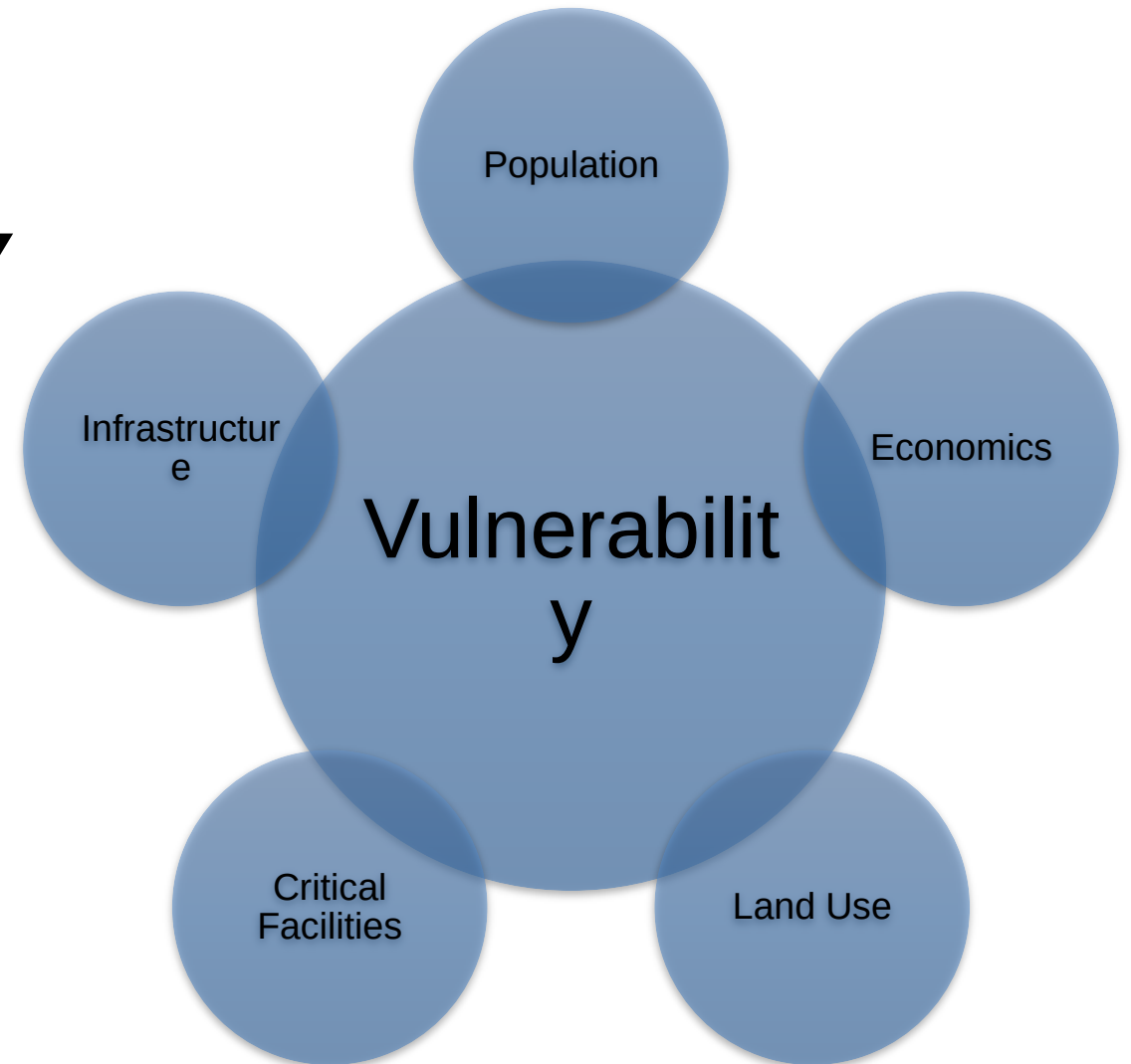


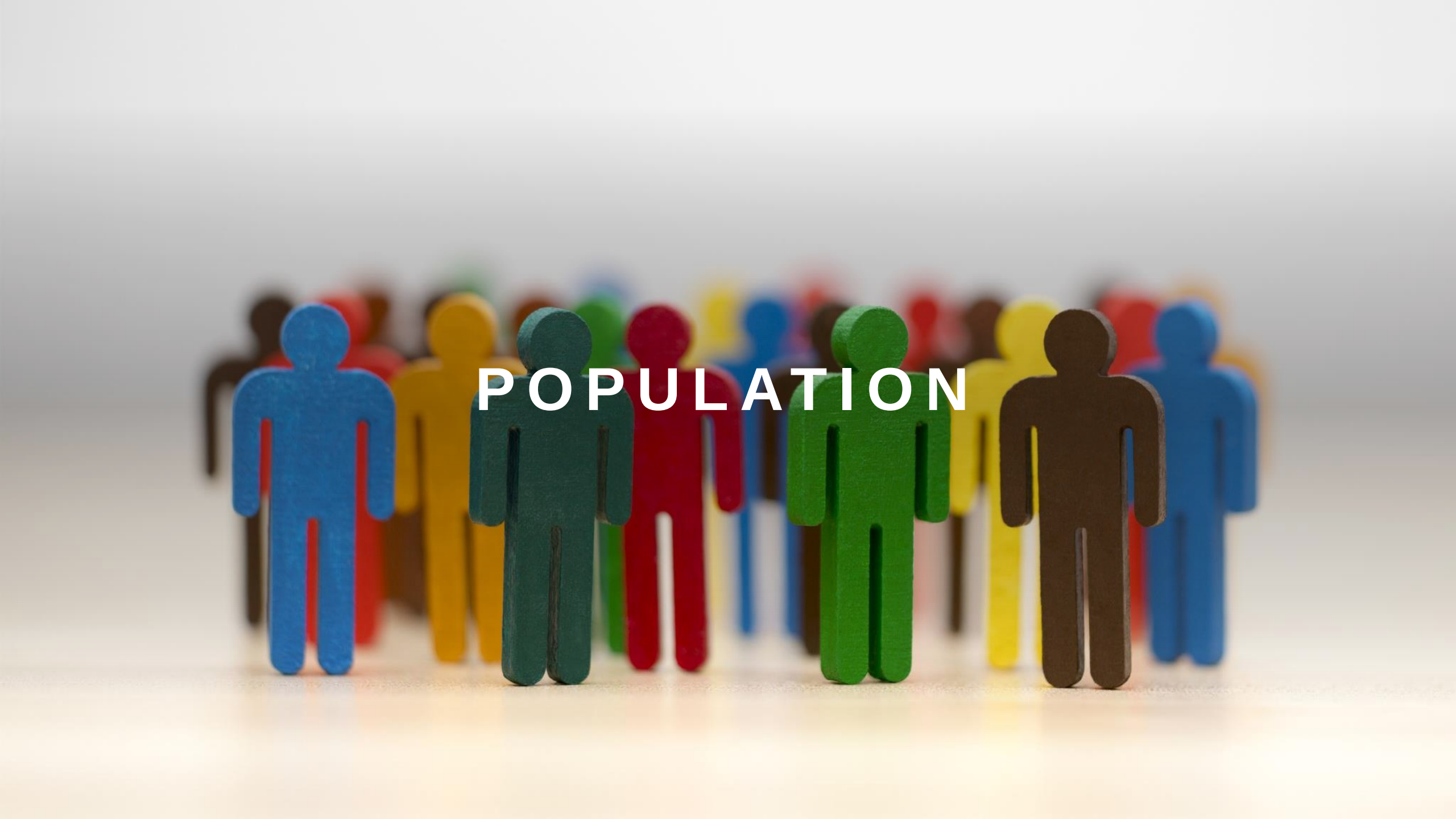
HAZARD IDENTIFICATION

VULNERABILITY ASSESSMENT

VULNERABILITY

- Identify all critical vulnerabilities
- Consider critical assets for the system



A group of colorful wooden human figures standing in a row, with the word "POPULATION" overlaid in white capital letters. The figures are in various colors including blue, yellow, red, green, and brown, and are arranged in a slightly staggered line. The background is a soft, out-of-focus light gray.

POPULATION



ECONOMICS

LAND USE





INFRASTRUCTURE